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**WEB-BASED FACULTY EVALUATION SYSTEM WITH AI-ENHANCED
FEEDBACK ANALYSIS AND CHED-COMPLIANT INSTRUMENTS FOR THE
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES**

An Undergraduate Capstone Project Presented to the Faculty of the
Institute of Computer Studies
NATIONAL AVIATION ACADEMY OF THE PHILIPPINES
Piccio Garden, Villamor, Pasay City

In Partial Fulfillment of the Requirements for the Degree of
BACHELOR OF SCIENCE IN INFORMATION SYSTEM
WITH SPECIALIZATION IN AVIATION INFORMATION SYSTEM

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MARCH 2026

CHAPTER 1

INTRODUCTION

Introduction

Faculty evaluation systems have become a critical component of quality assurance in higher education, supporting instructional improvement, accountability, and data-driven decision-making. Institutions worldwide are increasingly adopting web-based platforms, multi-source evaluation models, and artificial intelligence (AI) to improve the accuracy and efficiency of performance assessment. Existing studies highlight the benefits of combining student, peer, and supervisor inputs, as well as using AI to analyze large volumes of qualitative feedback. However, several gaps remain evident: many systems lack multi-source integration, limiting the completeness and reliability of evaluations, and the use of AI in qualitative feedback analysis is still limited, particularly in combining advanced language models with validation techniques. Few systems also undergo comprehensive evaluation using ISO/IEC 25010 standards, and real-world implementation assessments, user acceptance studies, and post-evaluation insights are often overlooked. These gaps highlight the need for a more integrated, intelligent, and standards-compliant faculty evaluation solution.

The National Aviation Academy of the Philippines (NAAP) is a specialized higher education institution dedicated to training future aviation professionals, such

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as pilots, aviation maintenance technicians, and other industry specialists. Because it prepares students for safety-critical careers, NAAP places strong emphasis on maintaining high instructional standards, regulatory compliance, and continuous quality improvement. Faculty evaluation is central to this effort, allowing the institution to monitor teaching effectiveness and ensure that instructors consistently meet the academic and industry standards expected of an aviation training institution. As NAAP continues to grow, however, its reliance on manual and fragmented evaluation methods has become increasingly inadequate, creating a clear need for a more integrated and technology-driven evaluation system.

These challenges are evident in NAAP's current evaluation practices. Existing systems are fragmented, relying on isolated data sources without a centralized platform for efficient data management. The absence of automated insight extraction requires time-consuming manual analysis, increasing administrative workload and delaying decision-making. Without standardized quality assessment frameworks, the institution cannot objectively evaluate the performance and reliability of its evaluation tools, and the level of user adoption and acceptance of existing systems remains unclear, further complicating any improvement efforts. Together, these limitations highlight the urgent need for an evaluation system that integrates multiple data sources, automates analysis, ensures quality compliance, and supports continuous optimization.

In response to these challenges, this study aims to design, implement, and evaluate a Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for NAAP. Specifically, it seeks to answer the following research questions:

1. How does the integration of multi-source faculty evaluation (student, peer, and supervisor inputs) affect the reliability and completeness of faculty performance assessment?
2. How effective is AI-assisted analysis in processing and interpreting qualitative faculty evaluation feedback?
3. How does the proposed system perform in terms of ISO/IEC 25010 software quality characteristics, particularly usability, functional suitability, performance efficiency, and reliability?
4. How does the system compare with the existing evaluation process in terms of operational efficiency, data accuracy, usability, and administrative workload?
5. What is the level of user acceptance in terms of perceived usefulness and ease of use?
6. What data-driven improvements can be proposed based on the evaluation results of the system?

These questions guide the development and evaluation of the proposed solution, ensuring alignment with the study's objectives of enhancing effectiveness, efficiency, and usability in faculty evaluation systems.

Project Context

The National Aviation Academy of the Philippines (NAAP) operates as a specialized higher education institution focused on aviation training and education, where maintaining high instructional standards is essential to ensuring student competence and operational safety. As an institution responsible for developing future aviation professionals, NAAP plays a critical role in upholding academic quality, regulatory compliance, and industry readiness. Within this environment, faculty evaluation serves as a key mechanism for assessing teaching effectiveness and supporting continuous improvement, involving multiple stakeholders such as students, faculty members, supervisors, administrators, deans, and Human Resources personnel.

At present, faculty evaluation at NAAP is conducted using a hybrid approach that combines paper-based forms and Google Forms. While these methods allow the institution to collect feedback, the overall process remains fragmented and largely manual: administrators must consolidate responses from different sources, verify the completion of evaluations, and generate reports without the support of a centralized system, and qualitative feedback is manually interpreted, making the

process time-consuming and dependent on subjective analysis. These practices reflect a semi-digital system that lacks integration, automation, and real-time processing capabilities.

This workflow presents several operational challenges. The absence of multi-source integration limits the ability to capture a holistic view of faculty performance from students, peers, and supervisors, while the lack of automated validation and analytics forces administrators to manually detect duplicate, incomplete, or unauthorized submissions. Without a centralized and secure data management system, it is also difficult to track historical records and generate comprehensive reports. These limitations increase administrative workload, cause processing delays, and expose the system to data security and confidentiality risks, ultimately affecting the reliability, accuracy, and usefulness of evaluation data for institutional decision-making.

Given these challenges, there is a clear need to improve the existing evaluation system at NAAP, as current practices are insufficient to support the efficient, secure, and data-driven evaluation processes required in a high-stakes aviation education environment. There is an opportunity to implement a centralized and secure web-based platform that integrates multi-stakeholder evaluation, incorporates AI-assisted techniques for qualitative feedback analysis, and automates validation and reporting. These improvements would significantly

reduce administrative burden while improving data accuracy, system reliability, and institutional decision-making.

Purpose and Description

Faculty and student evaluations play a vital role in higher education by improving teaching quality, ensuring institutional accountability, and supporting continuous academic development. However, existing manual and semi-digital evaluation methods often lead to inefficiencies, data inconsistencies, and limited analytical capabilities, reducing the overall effectiveness of evaluation outcomes. In response to these challenges, this study aims to design, develop, and evaluate an AI-enhanced, integrated, web-based faculty evaluation system that centralizes multi-source evaluation processes and improves data accuracy, administrative efficiency, and data-driven academic decision-making. The system is intended for use by students, faculty members, supervisors, administrators, deans, and Human Resources personnel, contributing to a more structured and reliable evaluation environment within the National Aviation Academy of the Philippines (NAAP). Furthermore, the system is developed in accordance with CHED standards and will be evaluated based on efficiency, accuracy, compliance with ISO/IEC 25010 software quality standards, and user acceptance to ensure its effectiveness and practical applicability.

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The proposed solution is a web-based faculty evaluation system that integrates multiple evaluation components into a unified platform. It is designed to replace fragmented and manual processes with a centralized, secure, and accessible system that supports real-time data processing and reporting. The system adopts a multi-source evaluation approach by combining inputs from students, peers, and supervisors, ensuring a more comprehensive and balanced assessment of faculty performance. In addition, it incorporates AI-assisted techniques to enhance the interpretation of qualitative feedback, providing deeper insights that are not easily achievable through manual analysis alone. By integrating evaluation, analysis, and reporting into a single platform, the system offers a cohesive and efficient solution aligned with institutional and regulatory requirements.

The system includes several key features that address the limitations of existing practices. It supports multi-source evaluation through structured instruments for students, peers, and supervisors, and applies AI-assisted analysis to process and summarize large volumes of qualitative feedback for faster, more consistent interpretation. A centralized database manages all evaluation data, allowing secure storage, easy retrieval, and efficient report generation, while built-in security safeguards protect user information and prevent unauthorized access. An analytics dashboard further provides real-time performance insights through charts, tables, and key indicators, supporting monitoring and evaluation activities.

Together, these features enable automation, real-time processing, and systematic data management within the system.

The expected outcomes of the proposed system include significant improvements in evaluation accuracy, efficiency, and security compared to existing methods. By automating data collection, validation, and analysis, the system reduces administrative workload and minimizes human errors. It enhances the reliability of evaluation results by integrating multiple data sources and applying consistent analytical methods to both quantitative and qualitative data. The system also strengthens data privacy and security, directly addressing the vulnerabilities present in existing evaluation practices. For stakeholders, the platform provides clearer, more actionable insights that support informed decision-making and continuous improvement of teaching performance. Ultimately, the study aims to demonstrate that the proposed system achieves acceptable levels of usability and user acceptance while offering a scalable and sustainable solution for institutional evaluation processes.

Objective of the Study

To develop a Web-Based Faculty Evaluation System with AI-Assisted Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines that ensures accurate, secure, accessible, and efficient faculty evaluation and student clearance processes. The system supports

structured feedback collection and enables HR personnel to perform data-driven evaluation and decision-making. Guided by the Agile Software Development Life Cycle, the study specifically aims to:

1. To investigate how the integration of multi-source faculty evaluation (student, peer, and supervisor inputs) influences the reliability and completeness of faculty performance assessment.
2. To examine the effectiveness of AI-assisted analysis in processing and interpreting qualitative faculty evaluation feedback, including the use of text and frequency analysis for improved insight extraction.
3. To empirically evaluate the proposed faculty evaluation system using the ISO/IEC 25010:2023 software quality characteristics, including functional suitability, reliability, usability, security, performance efficiency, maintainability, compatibility, portability, and accessibility.
4. To compare the proposed system with the existing evaluation process in terms of operational efficiency, data accuracy, usability, and administrative workload.
5. To assess the level of user acceptance of the proposed system, focusing on perceived usefulness, ease of use, and overall user experience.
6. To identify data-driven improvements and recommendations based on system evaluation results, user feedback, and performance analysis.

Scope and Limitations

Scope of the Study

This study focuses on the design, development, implementation, and evaluation of an AI-enhanced, CHED-compliant web-based faculty evaluation system that integrates multi-source evaluation from students, peers, and supervisors at the National Aviation Academy of the Philippines (NAAP), Villamor Campus. The system aims to improve the faculty evaluation process by incorporating AI-assisted qualitative feedback analysis while preserving the original CHED-aligned evaluation structure.

The study investigates the system's effectiveness in terms of reliability, fairness, efficiency, usability, system quality, and user acceptance. It specifically examines how multi-source evaluation contributes to a more comprehensive faculty assessment and how AI-assisted analysis supports the processing and summarization of qualitative feedback.

The respondents of the study consist of selected students, faculty members, supervisors, and Human Resource (HR) personnel who directly interact with the proposed system. A purposive sampling technique is employed to select participants who are knowledgeable and experienced with the faculty evaluation process.

The research adopts a developmental research design with a quantitative approach. Data are collected using structured survey questionnaires and system

evaluation instruments based on CHED standards. The evaluation of the proposed system is guided by the ISO/IEC 25010:2023 software quality characteristics, including functional suitability, reliability, usability, performance efficiency, security, compatibility, maintainability, and accessibility. The collected data are analyzed using descriptive statistical tools such as frequency distribution, mean, and weighted mean to determine the system's effectiveness and user acceptance.

The proposed system covers the core functionalities of faculty evaluation, including user authentication, evaluation submission, data processing, report generation, and AI-assisted summarization of qualitative feedback. The system operates within the institutional environment of NAAP–Villamor Campus and is accessible only to authorized users during the designated academic term for system implementation and evaluation.

Limitations of the Study

This study is limited to the implementation and evaluation of the proposed faculty evaluation system within the National Aviation Academy of the Philippines (NAAP), Villamor Campus. As such, the findings may not be generalizable to other campuses or higher education institutions with different organizational structures, policies, or evaluation practices.

The study employs a quantitative research approach using structured survey questionnaires and system evaluation instruments. Consequently, it does

not include qualitative methods such as interviews, focus group discussions, or thematic analysis that could provide more in-depth insights into users' experiences and perceptions.

A purposive sampling technique is used to select respondents who have direct interaction with the system. The relatively limited number of participants may affect the representativeness of the findings and may introduce sampling bias. In addition, the accuracy of the results depends on the respondents' honesty, understanding, and willingness to provide objective responses during the evaluation.

The proposed system is limited to essential functionalities, including evaluation submission, data processing, report generation, and AI-assisted summarization of qualitative feedback. It does not include integration with third-party platforms, deployment across multiple campuses, fully automated decision-making, or other advanced artificial intelligence capabilities.

The system operates in a controlled institutional environment and may be affected by technical factors such as internet connectivity, browser compatibility, device specifications, and hardware performance. Likewise, the AI-assisted feedback analysis may have limitations in accurately interpreting complex, ambiguous, or context-dependent qualitative responses. These technical and environmental factors may influence system performance and users' evaluation experience.

The study also does not modify the existing CHED-prescribed faculty evaluation criteria. Instead, it enhances the evaluation process by integrating multi-source assessment and AI-assisted analysis while maintaining compliance with established evaluation standards.

Conceptual Framework

The study is anchored on the Input–Process–Output (IPO) conceptual framework, a widely used model in systems and information science that illustrates how data are transformed into meaningful outputs through systematic processes. This framework is appropriate for the study as it provides a structured approach to analyzing how multi-source evaluation data are collected, processed, and converted into actionable results within an AI-enhanced faculty evaluation system. It is also supported by principles from information systems and software quality evaluation, particularly the ISO/IEC 25010:2023 standards, which guide the assessment of system performance and usability. The use of the IPO model enables a clear representation of system functionality and ensures that each stage of data handling contributes to achieving the study's objectives of improving evaluation accuracy, efficiency, and user acceptance.

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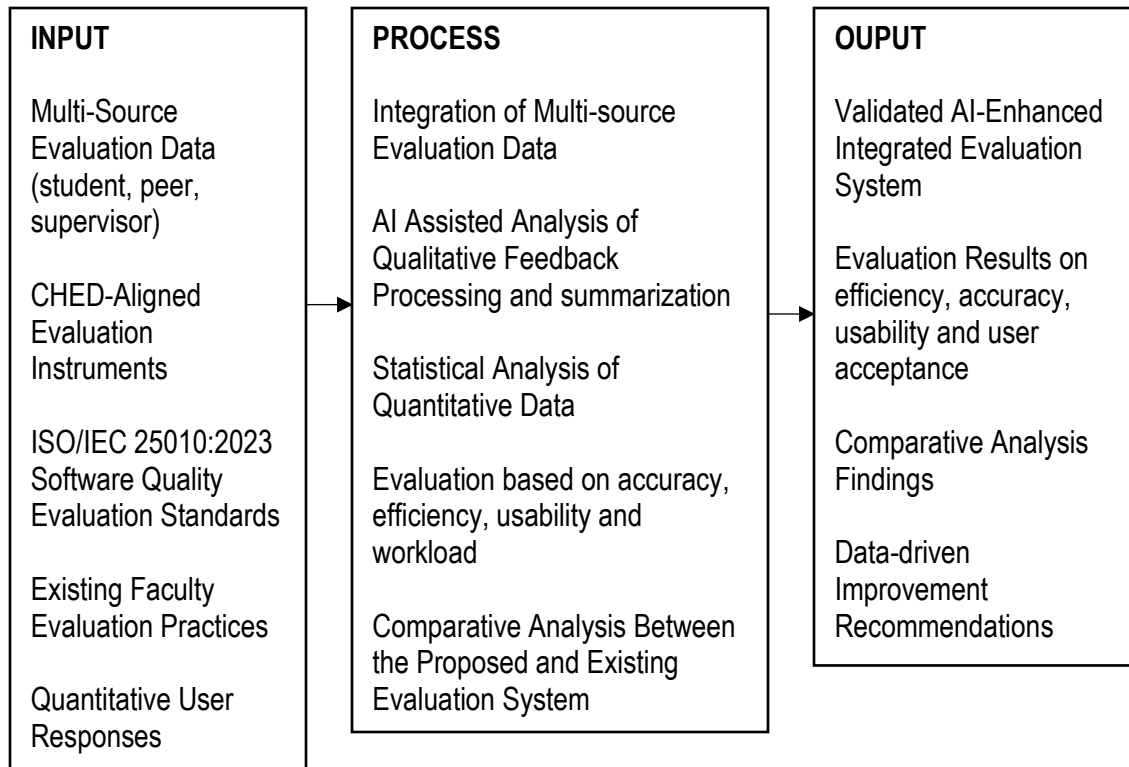


Figure 1: Input Process Output

INPUT

The input component consists of the resources and information required for the development and evaluation of the proposed system. These include multi-source evaluation data from students, peers, and supervisors, CHED-aligned evaluation instruments, ISO/IEC 25010:2023 software quality evaluation standards, existing faculty evaluation practices, and quantitative responses gathered from the respondents. These inputs serve as the foundation for assessing both faculty performance and the quality of the proposed evaluation system.

PROCESS

The process component involves the integration and analysis of the collected data within the proposed web-based faculty evaluation system. Multi-source evaluation data are consolidated into a unified assessment, while AI-assisted analysis processes and summarizes qualitative feedback. Quantitative responses are analyzed using appropriate statistical methods, and the system is evaluated based on software quality characteristics such as accuracy, efficiency, usability, and workload reduction. A comparative analysis is also conducted to determine the differences between the proposed system and the existing faculty evaluation process.

OUTPUT

The output component represents the expected results of the study. These include a validated AI-enhanced integrated faculty evaluation system, evaluation results regarding efficiency, accuracy, usability, and user acceptance, comparative analysis findings between the proposed and existing systems, and data-driven recommendations for improving the faculty evaluation process. These outputs demonstrate the effectiveness of the proposed system in providing a more reliable, efficient, and comprehensive faculty evaluation.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter reviews relevant local and foreign literature to establish the theoretical and empirical foundations of the proposed Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines (NAAP). Faculty and student evaluations are essential pillars of quality assurance in higher education, supporting teaching effectiveness, professional growth, and institutional accountability. In the Philippine context, these practices are guided by CHED Memorandum Order No. 19, Series of 2025, which promotes standardized and performance-based evaluation processes to ensure fairness and continuous improvement [45]. The importance of effective evaluation is especially critical in aviation education, where instructional quality directly influences student competence and safety. However, traditional paper-based and semi-digital evaluation methods remain inefficient and prone to inconsistent feedback, reporting delays, bias, and data integrity issues, limiting institutions' ability to generate meaningful insights and use evaluation results effectively for improvement and evidence-based decision-making.

AI Analysis

AI analysis in the context of faculty evaluation refers to the application of computational methods—including natural language processing (NLP), frequency analysis, and large language model (LLM)-based summarization—to transform large volumes of qualitative feedback into structured, actionable insights. The sheer volume of open-ended responses generated during evaluation cycles makes manual interpretation increasingly impractical, making AI-driven analysis a critical enabler for scalable and consistent feedback processing. However, a close examination of the literature reveals that existing implementations remain either broad in scope or limited to quantitative data streams, leaving qualitative open-ended comment analysis substantially underdeveloped in practice.

Studies widely agree that AI tools accelerate feedback interpretation while improving consistency. Sajja, Sermet, and Cwiertny demonstrated across two studies [1], [2] that integrating GPT-4-based learning analytics into educational evaluation enables descriptive analytics that surface patterns in engagement and cognition, with frequency analysis of instructional metrics supporting data-driven pedagogical decisions—yet both studies focus on broad theoretical frameworks rather than the systematic processing of faculty evaluation comments in context-specific deployments. Apetorgbor et al. [3] confirmed through a comprehensive review that AI technologies produce accurate and efficient evaluations with near-

instant feedback, substantially reducing turnaround time compared to traditional methods; however, their review does not propose or validate a hybrid method that pairs API-based AI summarization with non-API frequency validation, leaving the practical question of how to process qualitative faculty comments in resource-constrained environments unanswered. Ammari [4] analyzed Student Evaluations of Teaching (SETs) through generative AI and showed that such techniques yield richer faculty performance insights aligned with program improvement goals, but this analysis is restricted to student-generated text and does not extend to peer or supervisor qualitative inputs, confirming that multi-source qualitative comment integration remains an unaddressed dimension. Kumar [5] demonstrated that AI-driven automated assessment systems combining machine learning and NLP deliver real-time analysis alongside personalized recommendations, with frequency analysis proving especially effective for trend identification—yet the study does not operationalize this within a faculty evaluation system that simultaneously collects and processes qualitative responses from differentiated evaluator roles.

These findings confirm a consistent gap: while AI capability for qualitative text processing is well established in theory, no existing implementation combines API-based comment summarization with non-API frequency validation across student, peer, and supervisor inputs within a single operational faculty evaluation platform. This gap directly shapes the AI-assisted feedback component of the

proposed system, which employs OpenAI GPT-4.1 mini for comment summarization paired with non-API frequency validation—a hybrid approach designed to address both the analytical and resource constraints that existing studies have left unresolved.

Artificial Intelligence

Artificial intelligence (AI) in higher education evaluation encompasses a broader range of applications beyond feedback analysis, including automated scoring, performance prediction, and pattern detection in large datasets. Research consistently demonstrates that AI-based evaluation tools increase assessment consistency and reduce human bias. Almufarreh, Noaman, and Saeed [6] showed that machine learning frameworks applied to academic teaching quality evaluation produce more consistent performance scores than traditional rater-dependent methods. Qi et al. [7] and Gao [8] established that deep learning architectures detect complex patterns in large teaching evaluation datasets, enabling more nuanced distinctions among performance levels. Mradula et al. [9] and Bagunaid et al. [10] demonstrated that automated evaluation platforms with continuous monitoring capabilities provide ongoing recommendations supporting proactive faculty development. Darshini [11] supported the predictive dimension, showing that machine learning models analyzing academic data can forecast performance trajectories with measurable accuracy.

Where the literature reveals a critical gap is in its Philippine-specific application of advanced language models to qualitative evaluation data. Philippine-focused studies by Roaring et al. [12] and Gabayan et al. [13] prioritize practical text analysis and predictive classification within local institutional resource constraints. Roaring et al. [12] applied text mining to faculty teaching performance ratings at Pangasinan State University but relied on frequency-based methods without AI-assisted comment summarization. Gabayan et al. [13] conducted a systematic review of predictive classification and trend forecasting in faculty performance evaluation in Philippine higher education but found no study that had deployed advanced language models such as GPT-4 or equivalent systems for qualitative comment interpretation within a Philippine institutional context. This absence confirms that the integration of large language model capabilities into faculty evaluation systems remains effectively unrealized within local Philippine higher education practice—a gap that the proposed system addresses by applying GPT-4.1 mini within NAAP's operational environment, connecting the advanced analytical capabilities demonstrated in global research with the locally relevant institutional context, while the inclusion of frequency analysis reinforces the pattern detection strategies proven effective by Roaring et al. [12].

Assessment Instruments

Assessment instruments in faculty evaluation refer to the standardized tools, criteria, and measurement frameworks used to systematically capture faculty performance across defined competency areas. Well-designed instruments are the foundation of any credible evaluation system; without clearly defined criteria, evaluation results risk becoming subjective and inconsistent, undermining institutional accountability and faculty development goals. Multiple studies agree that structured, standards-aligned evaluation instruments significantly improve the validity, reliability, and clarity of faculty assessments. Vasilev et al. [14] emphasized that structured frameworks improve measurement clarity and validity, reducing ambiguity in how performance criteria are interpreted by different evaluators. Nausheen et al. [15] developed a novel tool using student evaluations to assess faculty development needs, combining descriptive and frequency analysis to identify critical improvement areas. Alenezi et al. [16] highlighted the importance of aligning instruments with modern digital practices, while Mounira et al. [17] stressed that standardized tools must remain responsive to changing institutional environments. Isaifan and Hasna [18] further demonstrated that assessment tools incorporating analytics capabilities enhance the overall measurement process.

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Philippine implementations reveal a more practice-oriented perspective: Daluyon and Zamora [19], Santillan [20], and Wenceslao [21] each integrated structured instruments into web-based systems for State Universities and Colleges (SUCs), demonstrating that digitized instruments improve data management and reporting efficiency. Most directly relevant is the work of Dueñas and Martin [22], who developed contextualized faculty evaluation criteria for technical instructors at the Philippine State College of Aeronautics (PhilSCA). However, a critical limitation runs across both international and Philippine studies: instrument design remains focused on quantitative rating scales, and none of the reviewed works integrate draft-enabled digital form delivery that pairs instrument completion with AI-assisted qualitative analysis of accompanying comments. Furthermore, while Dueñas and Martin [22] address the aviation-specific context, their instrument operates without digital workflow automation, real-time feedback processing, or compliance verification against the current CHED Memorandum Order No. 19, Series of 2025—requirements that did not exist at the time of their publication. This gap shapes the instrument design of the proposed system, which employs CHED-aligned and institution-approved standardized instruments delivered through draft-enabled digital forms, following Dueñas and Martin's aviation-specific approach [22] while meeting current CHED regulatory requirements.

Secure Authentication

Secure authentication refers to the mechanisms by which digital systems verify the identity of users before granting access, encompassing password hashing, one-time password (OTP) protocols, multi-factor verification, and role-based access control. In the context of faculty evaluation systems, strong authentication is not an optional security layer but a core functional requirement, as these platforms handle sensitive personal and professional information whose unauthorized access or manipulation can compromise institutional integrity and individual privacy. Research converges on the effectiveness of multi-layered authentication systems in protecting sensitive data. Madan Mohan [23] demonstrated that email-based OTP systems significantly enhance authentication security through advanced cryptographic techniques and industry-standard protocols that generate unique, time-sensitive tokens, with bcrypt password hashing and secure server communication directly addressing threats such as credential exposure and unauthorized account access—achieving both high security and user-friendliness that is critical for maintaining evaluator participation rates.

The limitation of Madan Mohan's framework [23], and of the available authentication literature more broadly, is that it addresses general web application security rather than the specific compliance and role-differentiated access

requirements of higher education evaluation systems operating under Philippine law. The Data Privacy Act of 2012 (Republic Act 10173) imposes legal obligations on institutions handling personal academic data that general cryptographic frameworks do not explicitly address, and no reviewed study proposes an authentication architecture that simultaneously satisfies multi-role access segregation (students, peers, supervisors, administrators), OTP-based anomaly detection for irregular login events, and compliance with RA 10173 within a faculty evaluation context. These considerations directly inform the security architecture of the proposed system, which develops secure authentication mechanisms validating evaluator identity through a targeted OTP protocol and bcrypt password hashing, Cloudflare firewall protection, automated backups, and audit trails—all in compliance with RA 10173—with role-based access control ensuring each evaluator role accesses only the instruments appropriate to their designated function.

Digital Transformation

Digital transformation in higher education refers to the organizational, cultural, and technological shift from paper-based and fragmented semi-digital processes to integrated institutional platforms that automate workflows, centralize data, and enable evidence-based decision-making. Studies broadly agree that successful digital transformation requires alignment across technology,

organization, and institutional culture. Vasilev et al. [14] explained that transformation involves technical, organizational, and cultural adjustments that must proceed together for systems to be effective and adopted. Mounira et al. [17] showed that digitization supports comprehensive quality assurance through enhanced operational efficiency, while Alenezi et al. [16] identified the need to balance innovation with accessibility and inclusivity. Isaifan and Hasna [18] demonstrated that embedding AI into quality assurance systems produces scalable solutions that outperform manual processes in both speed and consistency.

In the Philippine context, Cadiz et al. [24] documented increasing digital tool adoption in academic management, while Wagan et al. [25] showed that digital platforms improve documentation and reporting efficiency in institutional records management. Pardiñan et al. [26] emphasized digitization as a mechanism for quality assurance among SUCs pursuing accreditation. Malatsico and Manayan [27] identified training and organizational readiness as critical success factors—a finding that distinguishes local implementations, which must account for varying levels of technological familiarity among faculty and staff, from international frameworks that often assume baseline digital proficiency. Critically, however, none of the Philippine digital transformation studies examined operationalize transformation within a faculty evaluation system that simultaneously incorporates AI-assisted feedback analysis, multi-source data collection, and automated

administrative workload reduction within a single platform. The inefficiency and administrative workload problems documented across local implementations persist precisely because no existing digitization initiative has combined these capabilities cohesively. These insights shape the implementation strategy of the proposed system, which integrates stakeholder consultation, automated evaluation workflows, centralized databases, and audit trails to reduce administrative burden, while CHED standards and ISO/IEC 25010:2023 quality compliance ensure that digital transformation at NAAP strengthens rather than compromises institutional effectiveness.

Faculty Evaluation

Faculty evaluation systems are formalized mechanisms for assessing instructional performance, professional conduct, and institutional contributions, providing the evidence base needed for promotion, retention, and professional development decisions. Research consistently establishes that structured evaluation produces more objective and reliable performance assessments. Mangila [28] established a strong empirical link between evaluation outcomes and actual teaching effectiveness and work performance in Philippine higher education, underscoring the stakes of evaluation system design. Wenceslao [21] developed faculty evaluation software for Philippine SUCs that promotes uniform procedures and fair implementation, demonstrating the feasibility of digitizing

structured evaluation processes within local institutional constraints. Cinizan and Valdez [29] implemented analytics-enhanced evaluation platforms at Ilocos Sur Polytechnic State College that consolidate multi-source data and improve evaluation usability through descriptive and comparative analytics.

However, these implementations—while representing meaningful progress—share a defining structural limitation: they function as fragmented, single-function systems rather than integrated platforms. Wenceslao [21] automates data collection but does not provide AI-assisted qualitative analysis or real-time decision-support reporting. Cinizan and Valdez [29] consolidate multi-source data but rely on post-hoc descriptive analytics rather than real-time insight generation, and their system does not extract structured optimization recommendations from evaluation outcomes. Mangila [28] establishes the performance-evaluation relationship empirically but offers no technological framework for acting on that relationship systematically. International frameworks tend to be more analytically sophisticated but are rarely adapted to the regulatory and resource context of Philippine state universities. Across the reviewed literature, no existing faculty evaluation system in the Philippines integrates multi-source assessment, AI-powered qualitative analysis, real-time reporting, and structured post-evaluation recommendations into a single unified environment—leaving administrators without the data-driven optimization basis that evaluation data is theoretically capable of providing. The proposed system directly addresses

these limitations by designing a scalable, secure evaluation platform that delivers these capabilities cohesively, transforming faculty evaluation from a manual administrative burden into a strategic institutional decision-making tool.

Learning Analytics

Learning analytics refers to the measurement, collection, analysis, and reporting of data about learners and learning contexts for the purpose of understanding and optimizing learning environments and the decisions that affect them. Applied to faculty evaluation, learning analytics provides the conceptual and technical infrastructure for converting raw evaluation data into actionable insights. Studies widely agree that analytics-driven approaches improve the quality and utility of evaluation data. Xu et al. [30] and Chen and Yang [31] showed that big data frameworks integrating multiple information sources enable cross-departmental comparison and longitudinal trend tracking. Patacsil et al. [32] demonstrated through associative rule analysis that teaching behavior relationships become visible when evaluation data is processed systematically. Dinamling and Depaynos [33] applied exploratory factor analysis to organize student feedback into measurable instructional dimensions. Gonugunta and Leo [34] confirmed that data-driven decision-making frameworks inform institutional strategies more reliably than intuition-based approaches, while Cadiz et al. [24] highlighted the critical role of analytics in monitoring digital learning environments.

Philippine implementations, however, reveal a tendency toward retrospective descriptive analytics suited to institutional resource conditions rather than advanced, real-time processing. Cinizan and Valdez [29] implemented descriptive and comparative analytics at Ilocos Sur Polytechnic State College with demonstrable improvements in evaluation usability, yet this approach lacks real-time accessibility and interactive visualization that would enable administrators to act on evaluation data without the lag of separate analytical processing cycles. More fundamentally, none of the reviewed Philippine or international implementations generate structured post-evaluation insights—that is, explicit, data-derived recommendations for faculty development or institutional improvement based on the current evaluation cycle's results. Evaluation data is collected and summarized, but the translation of that data into an actionable optimization basis for the subsequent cycle remains unaddressed. These findings inform the analytics architecture of the proposed system, which provides a descriptive analytics dashboard delivering completion rates, average ratings, response rates, rankings, and historical trends in real time to professors, HR personnel, administrators, supervisors, and deans—extending local descriptive indicators through live visualization, interactive reporting, and structured post-evaluation recommendations without requiring specialized analytical expertise.

Multi-Source Evaluation

Multi-source evaluation, also referred to as 360-degree feedback, is an assessment approach that integrates performance data from multiple rater groups—typically students, peers, and supervisors—to produce a more comprehensive and balanced view of faculty performance than any single-rater design can achieve. Studies converge on the superiority of multi-source evaluation for producing balanced and reliable faculty performance assessments. Dalvi et al. [35] demonstrated that 360-degree feedback enhances employee development and performance by integrating diverse viewpoints. Hussain and Matore [36] confirmed that multi-evaluator approaches produce richer and more reliable instructional quality information than single-source designs. Tatari, Ebrahimi, and Tadayon [37] implemented an electronic 360-degree faculty evaluation method that assigned weighted contributions to student, peer, and supervisor assessments, using descriptive-analytical methods to measure satisfaction and performance levels. Cinizan and Valdez [29] further demonstrated that analytics-enhanced platforms consolidating multi-source data improve evaluation usability in Philippine SUC contexts.

Despite the convergence of evidence supporting multi-source evaluation, a critical implementation gap persists: no existing study or system has combined multi-source evaluation with AI-assisted qualitative analysis within a single unified

digital platform. Tatari et al. [37] integrate multiple rater groups but do not apply AI-powered analysis to the qualitative comments generated by those raters. Cinizan and Valdez [29] consolidate multi-source data within a Philippine SUC context but rely on descriptive analytics without AI-assisted insight extraction from open-ended responses. Hussain and Matore [36] and Dalvi et al. [35] establish the theoretical and empirical value of multi-source designs but do not address the computational infrastructure needed to process the qualitative data these designs generate at scale. This gap represents a significant missed opportunity: multi-source evaluation generates richer qualitative feedback precisely because it captures perspectives from different relational contexts, yet this richness remains unexploited without AI-assisted analysis. This theme directly informs the multi-source evaluation design of the proposed system, which investigates how the integration of student, peer, and supervisor inputs—processed through AI-assisted analysis and real-time reporting within a role-based access-controlled centralized platform—influences the reliability, completeness, and actionability of faculty performance assessment.

Peer Evaluation

Peer evaluation of teaching refers to the structured process by which faculty colleagues assess each other's professional conduct, instructional practices, collaborative engagement, and institutional citizenship—dimensions that students

are not positioned to evaluate. Research consistently demonstrates that well-designed peer assessment frameworks improve both the quality of feedback and professional collaboration among faculty. Cutroni and Paladino [38] found through systematic review that peer feedback encourages shared responsibility, constructive communication, and collective commitment to academic standards. Chovriy et al. [39] demonstrated that structured peer interactions improve cooperation, knowledge sharing, and institutional engagement. Darvishi et al. [40], [41] and Topping et al. [42] showed that AI-assisted peer assessment platforms enhance feedback clarity, fairness, and transparency, establishing that technological mediation can reduce the inconsistency that often undermines informal peer review. Alipour et al. [43] further demonstrated that decentralized collaborative peer assessment standardizes judgments and reduces interpersonal bias through organized digital workflows.

Philippine implementations of peer evaluation remain in relatively early stages of formalization. Gutierrez [44] documented peer assessment becoming a regular faculty review component in Manila higher education institutions but noted that it occurs largely through informal or semi-structured processes without digital workflow support. Cinizan and Valdez [29] implemented systematic recording of peer feedback but did not apply AI-powered pattern analysis to the qualitative content of peer comments. This creates a dual gap: locally, peer evaluation lacks the digital infrastructure to ensure consistency and reduce bias; and even in

international implementations where AI-assisted tools exist, they have not been deployed within integrated multi-source faculty evaluation systems that simultaneously collect student and supervisor inputs. The proposed system bridges both dimensions by using organized digital workflows to standardize peer assessment and reduce interpersonal bias, extending this through AI-powered pattern analysis of peer feedback, and employing anonymized response mechanisms and role-based access control to promote the honest, constructive communication demonstrated effective in prior research.

Policy Frameworks

Policy frameworks in the context of Philippine higher education faculty evaluation refer to the national regulatory instruments, compliance mandates, and governance structures that define evaluation criteria, roles, reporting requirements, and accountability mechanisms. Research widely agrees that strong policy frameworks are prerequisites for sustainable evaluation system implementation. Vasilev et al. [14] emphasized that governance structures supporting continuous assessment and institutional accountability are among the strongest predictors of evaluation system effectiveness. Isaifan and Hasna [18] demonstrated that integrating emerging technologies into quality assurance systems requires explicit ethical standards and oversight mechanisms. Malatsico and Manayan [27] provided Philippine-specific evidence that supportive

institutional policies encourage technology adoption and modernization, while their absence or ambiguity is a major barrier to consistent system application.

At the national policy level, CHED Memorandum Order No. 19, Series of 2025 [45] establishes revised requirements that align faculty evaluation with national quality standards, and DBM-CHED Joint Circular No. 3, Series of 2022 [46] institutionalizes evaluation procedures through the NBC 461 reclassification cycle, reinforcing standardized performance-based frameworks across public higher education institutions. A significant gap in the existing literature is that no reviewed system demonstrates full alignment with these current Philippine regulatory instruments within a web-based evaluation platform that simultaneously incorporates AI-assisted analysis. International research, while rich in governance theory, does not directly address the specific compliance requirements that Philippine SUCs and specialized aviation HEIs must satisfy. Philippine studies that do address policy compliance—such as Wenceslao [21] and Cinizan and Valdez [29]—predate or do not explicitly align with CHED Memorandum Order No. 19, Series of 2025 in their system design. Policy alignment is therefore woven into the foundational design of the proposed system, governing everything from instrument construction to data handling and workflow authorization, ensuring that the system achieves innovation through AI and web-based technologies while maintaining full regulatory compliance.

Student Evaluation

Student evaluation of teaching refers to the systematic collection of student perceptions and ratings regarding instructor performance, course delivery, communication effectiveness, and classroom engagement. Research consistently validates the importance of well-structured student evaluation instruments collected through accessible digital platforms. Zhao et al. [47] conducted a comprehensive literature review establishing that carefully structured instruments ensure meaningful and consistent student feedback, while Yang [48] recommended incorporating contextual and experiential factors to improve response quality and reduce superficial ratings. Ahmed, Hamim, and Nandi [49] demonstrated that web-based platforms efficiently gather student evaluations in online and blended learning environments, reducing administrative burden while increasing participation rates. Katende and Namutebi [50] confirmed that student perceptions provide valuable insights into instructional communication and engagement that other evaluators cannot as easily observe.

Philippine implementations reflect both the potential and the current limitations of student evaluation practice. Reyes et al. [51] implemented online faculty evaluation platforms promoting consistency and organized data management across PUP campuses. Dela Fuente [52] and Cinizan and Valdez [29] applied data-driven models to generate performance indicators and

summarize evaluation trends. Lopez, Eguia, and Simon [53] efficiently documented teaching outcomes through systematic student input at Baguio Central University, and Gutierrez [44] documented increasing reliance on web-based tools incorporating student feedback into formal faculty reviews. However, the limitation across all local and most international implementations is twofold: first, qualitative student comments are collected but not systematically analyzed through AI-assisted methods to extract structured insights; and second, evaluation results are summarized after the fact rather than made available in real time for immediate administrative decision-making. Student feedback therefore remains an underleveraged resource—data is gathered but its qualitative dimension is processed minimally or not at all. The proposed system addresses this by implementing structured, CHED-aligned student evaluation instruments delivered through draft-enabled digital forms with anonymized response mechanisms, with AI-assisted analysis using GPT-4.1 mini transforming qualitative comments into structured insights and real-time reporting providing immediate evaluation summaries.

System Quality

System quality in the context of digital evaluation platforms refers to the degree to which a software system meets defined quality characteristics encompassing functional correctness, reliability, usability, security, performance

efficiency, maintainability, compatibility, portability, and accessibility. The effectiveness of any digital evaluation platform ultimately depends on its software quality, as systems that perform unreliably or are difficult to use undermine the evaluation data they collect regardless of how well their assessment instruments are designed. The ISO/IEC 25010:2023 standard is widely recognized as the comprehensive international framework for measuring these characteristics objectively. Sarwosri et al. [54] demonstrated that ISO/IEC 25010 evaluates functional suitability, performance efficiency, and reliability with objective, verifiable benchmarks, making it superior to informal or ad hoc quality assessments. Vasilev et al. [14] established that system quality directly influences evaluation outcome accuracy and usefulness. Isaifan and Hasna [18] showed that technology embedded into quality assurance systems must itself meet high quality standards to support scalable, sustainable evaluation.

Philippine implementations of ISO/IEC 25010 in evaluation systems confirm its applicability in local SUC contexts: Daluyon and Zamora [19] and Wenceslao [21] integrated quality considerations into Philippine SUC system design. Santillan [20] explicitly applied ISO/IEC 25010 standards to validate software reliability—but applied only a subset of quality characteristics rather than the full complement. Cinizan and Valdez [29] emphasized user-friendly and flexible interfaces as determinants of system adoption, and Gutierrez [44] demonstrated that well-designed systems promote transparency and consistent reporting. The common

and significant limitation across all Philippine studies is the tendency to apply only a partial selection of ISO/IEC 25010 characteristics—typically functional suitability, reliability, and usability—while omitting maintainability, compatibility, portability, and accessibility from formal evaluation. No Philippine study has evaluated a faculty evaluation system against the complete ISO/IEC 25010:2023 quality model, leaving a gap in the evidence base regarding whether these systems meet international standards comprehensively. The proposed system addresses this by empirically evaluating all ISO/IEC 25010:2023 software quality characteristics—functional suitability, reliability, usability, security, performance efficiency, maintainability, compatibility, portability, and accessibility—ensuring both international standards compliance and practical operability within NAAP's administrative environment.

Web-Based Systems

Web-based evaluation systems are digital platforms delivered through internet browsers that automate the collection, processing, and reporting of faculty evaluation data, eliminating the delays and inconsistencies characteristic of paper-based administration and enabling real-time access to evaluation results by authorized stakeholders. Studies broadly agree that web-based evaluation systems improve accessibility, transparency, operational efficiency, and reporting reliability compared to manual approaches. Tatari, Ebrahimi, and Tadayon [37]

demonstrated that electronic 360-degree systems integrating multiple perspectives into single platforms capture holistic performance data more efficiently than fragmented paper processes. Abhang and Gaike [55] emphasized that digital evaluation frameworks improve institutional effectiveness and accountability through automation. Jia et al. [56] validated reliable online instruments in teaching evaluation contexts, confirming that digital platforms produce psychometrically sound data when instruments are properly designed. Ahmed, Hamim, and Nandi [49] showed that digital platforms effectively adapt evaluations to both on-campus and remote instruction contexts, while Katende and Namutebi [50] confirmed that structured online feedback contributes meaningfully to teaching quality improvement.

Philippine implementations demonstrate contextual adaptations: Daluyon and Zamora [19] developed customized platforms for the University of Rizal System addressing specific operational needs, Santillan [20] applied ISO/IEC 25010 quality standards partially to reinforce software reliability, and Lopez et al. [53] and Gutierrez [44] documented growing institutional reliance on digital tools for transparent and consistent evaluation reporting. However, the gap in existing Philippine web-based systems is the absence of integrated AI analysis, real-time multi-source dashboards, structured post-evaluation recommendations, and formal user acceptance assessment within a single unified platform. While these systems automate data collection, they do not transform that data into structured

improvement insights, nor do they assess whether the platform itself achieves sustained user adoption—a critical factor for long-term institutional effectiveness that remains unexamined in existing local implementations. The proposed system addresses these gaps by customizing its architecture for NAAP's aviation education context, following ISO/IEC 25010:2023 quality standards, and formally assessing user acceptance in terms of perceived usefulness, ease of use, and overall user experience—providing the adoption evidence that existing systems have not generated.

Synthesis of Literature

A comparative analysis of the reviewed literature across the thirteen thematic areas reveals a coherent and directional story: faculty evaluation systems are evolving from isolated administrative tasks into integrated mechanisms for quality assurance, instructional improvement, and evidence-based institutional decision-making. Common trends across both international and Philippine studies confirm that web-based platforms have established themselves as the practical standard for scalable evaluation, with digital automation of collection, processing, and reporting dramatically outperforming manual approaches in speed, consistency, and accessibility. Multi-source evaluation—combining student, peer, and supervisor inputs—produces measurably more balanced and reliable faculty performance data than any single-rater design [35], [36], [37], [29]. Artificial

intelligence, particularly advanced language models such as GPT-4 and its successors, has demonstrated substantial capacity to transform qualitative feedback into structured, actionable insights through frequency analysis, comment summarization, and pattern detection [1], [2], [3], [4], [5]. Policy frameworks—particularly CHED Memorandum Order No. 19, Series of 2025 [45] and DBM-CHED Joint Circular No. 3, Series of 2022 [46]—provide the regulatory structure within which technological innovation must operate, while strong authentication mechanisms and compliance with the Data Privacy Act of 2012 are non-negotiable requirements for systems handling sensitive faculty performance records. Software quality, assessed through ISO/IEC 25010:2023, ensures that evaluation platforms are reliably performant, secure, maintainable, and accessible.

The strengths of existing research lie in its breadth: international studies offer methodologically sophisticated frameworks and AI integration models, while Philippine studies provide contextually grounded implementations demonstrating the feasibility of digitized evaluation within local SUC resource constraints. The weaknesses, however, are consistent and cross-cutting. First, multi-source evaluation and AI-assisted qualitative analysis have never been combined within a single operational system: systems that collect multi-source data do not apply AI to its qualitative components, and systems that apply AI do not integrate differentiated rater groups. Second, no Philippine evaluation system has been assessed against the complete ISO/IEC 25010:2023 quality framework;

assessments remain partial and selective, leaving comprehensive quality compliance undemonstrated. Third, no reviewed system contextualizes its design and compliance within current Philippine aviation higher education requirements, combining CHED Memorandum Order No. 19, Series of 2025 alignment with aviation-specific instrument adaptation. Fourth, post-evaluation structured insights—explicit, data-derived recommendations for faculty development and institutional improvement—are absent from existing implementations; evaluation data is collected and reported but not translated into an actionable optimization basis. Fifth, user acceptance has not been formally and systematically assessed in existing Philippine faculty evaluation systems, leaving the adoption level and its determinants unknown. These five convergent gaps—lack of multi-source and AI integration, no full ISO/IEC 25010 evaluation, absence of post-evaluation insights, and limited acceptance evidence—constitute the research problem that the proposed system is designed to resolve.

Conceptual Link to the Current Study

The reviewed literature collectively establishes the theoretical and empirical foundation upon which the proposed system is built and positions the current study within—and beyond—the existing body of knowledge. The AI analysis and artificial intelligence themes [1]–[13] directly support the system's AI-enhanced feedback component, justifying the adoption of GPT-4.1 mini and frequency analysis as

validated approaches for processing qualitative evaluation data at scale—while simultaneously confirming that this combination has not been operationalized within a Philippine faculty evaluation context. The assessment instruments literature [14]–[22] grounds the instrument design in standards-aligned measurement theory and aviation-specific precedents established by Dueñas and Martin [22], ensuring CHED compliance and contextual appropriateness, and confirms the absence of current CMO No. 19, Series of 2025 alignment in existing digital instrument implementations. The secure authentication research [23] informs the targeted OTP protocol and bcrypt-based architecture that protects sensitive faculty data in compliance with RA 10173.

The digital transformation and policy frameworks literature [14], [16], [17], [24]–[27], [45], [46] justifies the systemic, stakeholder-inclusive implementation approach and CHED regulatory alignment, confirming that technological innovation must be accompanied by organizational readiness and policy compliance—while also revealing that no existing Philippine system has achieved this combination within a faculty evaluation platform. The faculty evaluation, multi-source evaluation, peer evaluation, student evaluation, and learning analytics themes [28]–[53] collectively validate the multi-source assessment architecture and analytics dashboard, demonstrating that integrating student, peer, and supervisor inputs within a real-time analytics environment produces evaluation outcomes superior to single-source, manual approaches—while confirming that no

existing system has paired this integration with AI-assisted qualitative analysis and structured post-evaluation recommendations. Finally, the system quality literature [14], [18], [19], [20], [21], [29], [54] establishes ISO/IEC 25010:2023 as the appropriate evaluation standard and confirms its applicability in Philippine SUC contexts, while also confirming that no existing study has applied the full quality framework to a faculty evaluation system.

The proposed system addresses the identified gaps by consolidating web-based infrastructure, multi-source evaluation mechanisms, GPT-4.1 mini-powered analytical tools, layered security safeguards, regulatory compliance, full ISO/IEC 25010:2023 quality evaluation, formal user acceptance assessment, and structured post-evaluation recommendations into a unified platform—delivering an evaluation environment that is more efficient than manual processes, more comprehensive than single-source approaches, more analytically insightful than traditional reporting methods, and more secure and compliant than fragmented implementations. This study thus advances the literature by providing the first integrated, AI-enhanced, CHED-compliant faculty evaluation system designed specifically for the operational and accreditation requirements of Philippine aviation higher education, establishing a replicable model for similar institutions navigating the same convergence of technological opportunity, regulatory obligation, and institutional improvement imperative.

CHAPTER 3

SYSTEM ANALYSIS, DESIGN, AND METHODOLOGY

Research and Development Methodology

This chapter discusses the technical side of our Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines, focusing on how each part of the system functions behind the scenes. Starting from the input, followed by the process, and ending with the output, the section explains how evaluation data is collected, analyzed, and transformed into accurate and secure results. The chapter presents the overall system outline, showing the main components and features of the platform, as well as the different levels of access granted to authorized users such as students, faculty members, supervisors, HR personnel, and the Office of Student Affairs. Overall, this section aims to provide a clear understanding of how the system operates and how it was developed to support a secure and efficient evaluation process.

Software Development Life Cycle Model

This section presents the Software Development Life Cycle (SDLC) adopted in the design and development of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments

for the National Aviation Academy of the Philippines. The study employed the Agile Software Development Life Cycle using a hybrid Scrum-inspired framework to support iterative development, continuous improvement, and structured stakeholder involvement throughout the system development process.

The Agile approach was selected to accommodate evolving system requirements, user feedback, and institutional policies, particularly those aligned with the evaluation framework of the Commission on Higher Education (CHED). Through incremental development, the system was implemented in two structured development cycles (sprints) to deliver functional components early, validate system features with users, and refine the system based on testing results and stakeholder feedback.

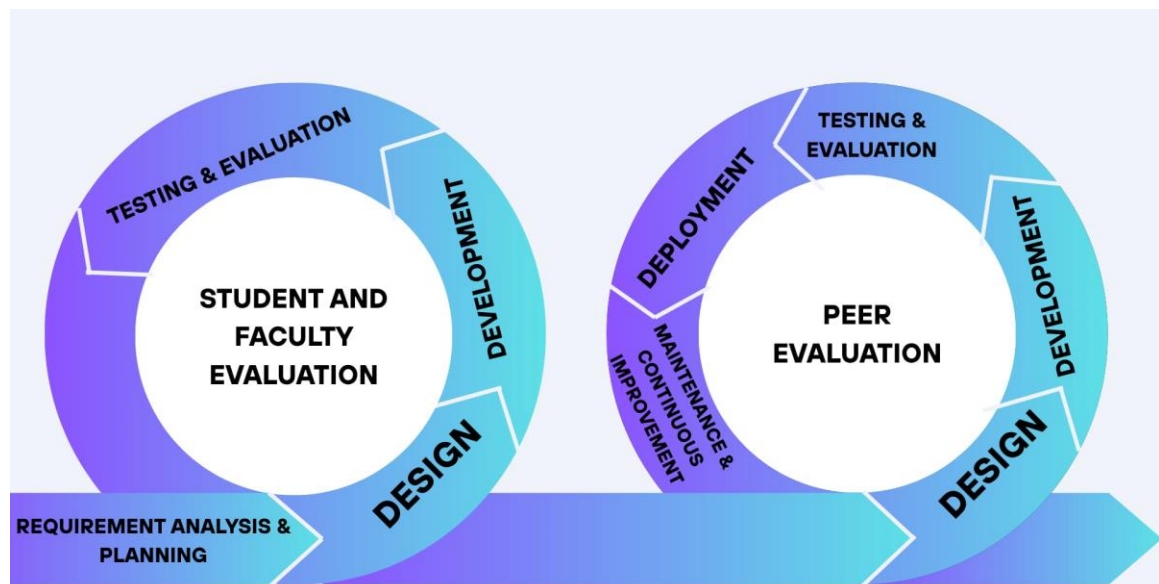


Figure 1. Agile Software Development Life Cycle Workflow for the Development of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments.

Figure 1 illustrates the Agile development workflow adopted in this study, demonstrating the iterative progression of system development phases supported by continuous stakeholder feedback and improvement cycles.

Agile Development Phases

The Agile development process implemented in this study consists of six major iterative phases, each producing specific deliverables that guided subsequent development activities.

Requirements Analysis and Planning

The requirements analysis and planning phase was conducted through structured consultations with key stakeholders, including students, faculty members, supervisors, the Vice President for Academic Affairs (VPAA), Office of Student Affairs (OSA), Human Resource personnel, and system administrators. Data were gathered using interviews and direct observation of existing processes to identify both functional and non-functional system requirements in accordance with CHED Memorandum Order No. 19, Series of 2025, as well as institutional evaluation policies. The main outcomes of this phase included comprehensive documentation of functional and non-functional requirements, clear definitions of stakeholder roles and responsibilities, a detailed system requirement specification aligned with ISO/IEC 29148 standards and identification of system performance, reliability, and security requirements. This phase provided a solid foundation for

the design of the system architecture and ensured that the proposed solution effectively addressed existing operational challenges.

Design

The design phase involved translating the documented requirements of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments into detailed technical blueprints. This process encompassed the creation of system architecture diagrams, database models, data flow diagrams, user interface prototypes, and security specifications. The system architecture was developed using a web-based client-server model to support centralized management of evaluation data, while AI-assisted analysis modules were integrated to enable automated summarization and frequency analysis of qualitative feedback, ensuring both data transparency and the authenticity of evaluator responses. Additionally, the design phase included defining user interface layouts for students, faculty members, supervisors, the Vice President for Academic Affairs (VPAA), Office of Student Affairs (OSA), Human Resource personnel, and system administrators, as well as specifying security and access control mechanisms to safeguard sensitive data. These design activities established a comprehensive framework for the subsequent development, testing, and implementation of the system.

Development

During the development phase, the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments was implemented using web-based programming technologies, database management systems, and client-side interface frameworks. Development followed iterative sprint cycles, allowing incremental implementation of features and integration testing to ensure seamless operation across system modules. The system incorporated AI-assisted qualitative feedback analysis through both frequency analysis and AI-assisted summarization techniques such as the OpenAI GPT-4.1 mini and natural language processing (NLP) techniques to identify recurring themes and patterns in open-ended evaluation responses. This AI component functions solely as an analytical support tool and does not modify original evaluator responses or numerical ratings, thereby upholding principles of transparency and fairness. In addition, the system integrated fully functional evaluation modules, role-based authentication and authorization mechanisms, automated reporting and dashboard features, and a clearance verification subsystem, providing a complete framework for managing evaluations, monitoring completion, and supporting data-driven decision-making by students, faculty members, supervisors, the Vice President for Academic Affairs (VPAA), and administrative personnel.

Testing and Evaluation

The testing and evaluation phase focused on verifying the functionality, usability, performance efficiency, reliability, and security of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments. The process followed structured testing protocols, including functional testing, integration testing, and performance validation, to ensure that all system components operated as intended. System quality evaluation was conducted in accordance with the ISO/IEC 25010:2023 software quality model, providing a standardized framework for assessing functional suitability, reliability, usability, security, and performance efficiency. Testing procedures also ensured compliance with institutional data privacy requirements under the Data Privacy Act of 2012 (Republic Act No. 10173). The outcomes of this phase included comprehensive documentation of functional and performance testing, security and authentication validation, usability assessment, and overall software quality evaluation, providing assurance that the system met defined criteria and was ready for pilot implementation and full operational deployment.

Deployment and Maintenance

Following successful testing and validation, the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments was prepared for institutional deployment within the National Aviation

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Academy of the Philippines' server infrastructure. The deployment process encompassed system configuration, user orientation and training, and planning for data migration to ensure seamless integration with existing institutional records. The maintenance phase facilitates continuous system improvement through iterative feedback collected from users, including students, faculty members, supervisors, the Vice President for Academic Affairs (VPAA), Office of Student Affairs (OSA), Human Resource personnel and administrative personnel. Leveraging the Agile methodology, the system undergoes periodic updates, security enhancements, and performance optimizations, ensuring long-term sustainability, adaptability to institutional needs, and continued alignment with operational requirements. This phase also includes comprehensive documentation of deployment procedures, user training materials, maintenance protocols, and plans for feedback-driven system enhancements.

Agile Methodology Justification

The selection of the Agile methodology for this study is supported by its demonstrated effectiveness in modern software development contexts, particularly in enhancing adaptability, stakeholder collaboration, and software quality. Recent research indicates that Agile approaches improve organizational responsiveness to changing requirements, enable frequent feedback loops with stakeholders, and support iterative development practices that can lead to higher quality outcomes

in complex projects. Studies show that Agile methodology enhances collaboration among team members and stakeholders, improves adaptability to evolving requirements, and contributes to enhanced software quality through continuous testing and refinement (Punitha & Kumanan, 2026). The iterative nature of Agile development aligns with the needs of educational system development, allowing the proposed system to respond effectively to evolving institutional policies, emerging CHED guidelines, and technological requirements, thereby ensuring long-term relevance and scalability.

Integration of AI-Assisted Components within the Agile Framework

Artificial intelligence modules were incorporated throughout the iterative development cycles to facilitate the analysis of qualitative feedback. The AI system employs frequency-based text summarization and thematic clustering to assist administrators in efficiently interpreting large volumes of evaluation comments. To ensure ethical transparency and compliance with regulatory standards, AI functionality is restricted to analytical support and does not modify original evaluation responses. All outputs generated by the AI undergo administrative review before being used in institutional reporting, ensuring accuracy, fairness, and accountability in decision-making.

System Development Framework

The system development framework was structured using a layered architecture comprising the presentation, application, and data management layers. The development process encompassed requirement gathering, iterative module creation, integration testing, and system validation. The system was implemented using modern web technologies, including server-side scripting, relational database management systems, and client-side interface frameworks. AI-assisted modules were incorporated to support the analysis of qualitative feedback through frequency-based text summarization techniques. These AI components are intended to aid administrative interpretation of feedback while maintaining the integrity and authenticity of user responses.

Analysis of the Existing System

This section examines the current state of faculty evaluation at the National Aviation Academy of the Philippines (NAAP), identifying the operational methods in place, the limitations they produce, and the resulting gaps in institutional capability. The analysis also evaluates the alternative approaches considered before arriving at the development of a custom web-based system as the most appropriate solution.

Current Evaluation Procedures

Faculty evaluation at NAAP is presently conducted through a combination of printed forms and general-purpose online survey tools that were not specifically designed for academic performance assessment. At the start of each evaluation period, administrative staff either distribute physical forms to students or share externally hosted survey links for completion. Supervisory evaluations follow a similar manual process, with department heads and supervisors filling out separate institutional instruments that are then submitted to the Human Resource office for encoding.

The consolidation of evaluation data is performed by HR staff, who manually encode responses from collected forms into spreadsheet files, aggregate numerical scores, and compile summary reports for distribution to department-level administrators. Open-ended comments written by students are reviewed individually by HR personnel, a process that is time-intensive and susceptible to subjective interpretation. No automated mechanism exists for synthesizing qualitative feedback across a faculty member's full evaluation set.

Clearance processing at the end of each semester requires the Office of Student Affairs (OSA) to manually verify, for each student, whether all required evaluations have been submitted. This cross-referencing is performed against the

HR office's encoded records and the course enrollment records, with no direct system linkage between the three data sources.

Limitations and Problem Analysis

The existing evaluation process exhibits several significant limitations that impair the institution's capacity to conduct reliable, efficient, and compliant faculty assessments. Manual data encoding introduces transcription errors and inconsistencies that reduce the accuracy of reported evaluation results. The absence of a centralized, role-differentiated digital platform means that data from different evaluation types — student, supervisor, dean, and peer — must be managed through separate processes, increasing the administrative burden and creating fragmented records.

The absence of any authentication mechanism linked to official enrollment data means that the existing survey tools cannot enforce submission eligibility. Students may submit evaluations for faculty members they were not taught by, submit duplicate responses, or fail to complete evaluations without any automated detection. General-purpose survey platforms do not support the CHED-mandated questionnaire format, preventing the institution from generating evaluation reports that conform to the specifications of CHED Memorandum Order No. 19, Series of 2025.

The manual review of qualitative, open-ended feedback is inherently time-consuming and produces inconsistent interpretations across different HR staff members. The lack of structured data prevents any longitudinal analysis of faculty performance trends across evaluation periods.

These limitations collectively impede the institution's capacity to use faculty evaluation data as a meaningful tool for academic quality management, faculty development, and regulatory compliance reporting.

Requirements Gap Analysis

A requirements gap analysis was undertaken to identify the specific functional and operational capabilities that the existing system fails to provide, and which are necessary for NAAP to meet its institutional, regulatory, and data governance obligations.

The most critical gap identified is the complete absence of a secure, enrollment-linked evaluation submission system. The institution currently has no mechanism for automatically verifying that a student is enrolled in a course before allowing them to evaluate the corresponding faculty member. This creates a structural vulnerability in the integrity of evaluation data that cannot be addressed through improvements to the existing tools.

A second major gap concerns the processing of qualitative feedback. The manual review process cannot scale to the volume of open-ended comments generated across all evaluation periods, and provides no objective basis for cross-faculty or longitudinal comparisons. An AI-assisted summarization capability would address this gap directly.

The gap analysis also identified the absence of role-differentiated dashboards, automated clearance verification, CHED-compliant reporting, a non-editable audit trail, and multi-campus data management as additional capability deficits that must be addressed in the proposed system.

Alternative Solutions and Feasibility Analysis

Before committing to the development of a custom system, three alternative approaches were evaluated against NAAP's requirements.

Enhancing the existing survey tools was considered first due to its low initial cost. However, the general-purpose survey platforms currently in use cannot be configured to enforce enrollment-linked access control, generate CHED-compliant PDF reports, or support role-based access across seven distinct user types. The fundamental architectural limitations of these tools make them unsuitable as a foundation for the proposed system's requirements.

Commercial evaluation platforms were evaluated as a second alternative. These products typically offer standardized evaluation workflows and reporting features, but none of the platforms reviewed supported the specific CHED-mandated questionnaire formats and impose ongoing licensing and per-user subscription costs that would exceed NAAP's long-term operational budget.

Proposed Technology Alternatives

The development of a custom web-based evaluation system was selected as the most appropriate solution based on its capacity to fully satisfy NAAP's institutional, regulatory, and operational requirements. A custom system allows direct integration with the institution's official enrollment and faculty assignment records, ensuring that evaluation access is automatically governed by current course data rather than manual list management. It also permits the full implementation of CHED-compliant questionnaire instruments, including the specific rating scale configurations and open-ended comment requirements mandated by CHED Memorandum Order No. 19, Series of 2025.

A custom solution enables the incorporation of AI-assisted feedback analysis — using the OpenAI GPT-4.1 mini API in combination with local frequency analysis — a capability that is not available in general-purpose survey tools and is absent from most commercial evaluation platforms. It also supports the

implementation of multi-layered security architecture, including Bcrypt password hashing, email-based one-time password (OTP) authentication, Cloudflare firewall protection, and role-based access control across seven distinct user roles.

The custom system further supports multi-campus deployment, automated clearance verification, a non-editable audit trail, soft-delete data preservation, and real-time monitoring dashboards — all features that are either unavailable or prohibitively expensive to configure in off-the-shelf alternatives. While custom development requires a higher upfront investment than the other options considered, the long-term operational savings, regulatory compliance assurance, and institutional scalability it provides justify the investment, as confirmed by the cost-benefit analysis presented in the subsequent section.

Feasibility Study

The feasibility study assesses the viability of developing and deploying the proposed AI-Enhanced CHED-Compliant Web-Based Faculty and Student Evaluation System across four critical dimensions: technical, operational, economic, and schedule. Each dimension was evaluated with specific reference to the conditions and resources of NAAP in order to arrive at a well-grounded determination of project viability. Together, these assessments establish that the

proposed system is both achievable within NAAP's current environment and capable of delivering meaningful institutional value.

Technical Feasibility

NAAP's existing information technology infrastructure is capable of supporting the proposed system without requiring major hardware upgrades. The system is designed for deployment on a standard web server environment compatible with the institution's current server hardware. The recommended server configuration — a multi-core processor, 16 GB or more of RAM, and SSD-based storage — falls within the capacity of modern institutional hardware available at accessible market prices, as detailed in the hardware requirements section of this chapter.

Network connectivity at NAAP is stable and sufficiently provisioned to support concurrent web-based access by students, faculty, and administrative staff during evaluation windows. The AI-assisted feedback analysis component is implemented through API integration with OpenAI GPT-4.1 mini rather than through locally hosted machine learning infrastructure, which ensures that AI processing capability is available without placing a sustained computational burden on institutional servers. This design choice significantly reduces technical risk and hardware cost while preserving full analytical functionality.

The security components of the proposed system — including Bcrypt password hashing, risk-adaptive email-based OTP authentication, Cloudflare web application firewall integration, and role-based access control — are all implemented using widely supported technologies that integrate smoothly with standard web server and relational database environments. The primary technical risk is elevated server load during peak evaluation periods. This risk is mitigated through database query optimization, efficient session management, and connection pooling strategies applied during the development phase. The modular architecture of the system also allows individual components to be scaled or updated independently as institutional requirements evolve. On the basis of these considerations, the proposed system is assessed as technically feasible within NAAP's existing environment.

Operational Feasibility

The proposed system is designed to align with and automate NAAP's existing evaluation workflows rather than replace established institutional procedures. Students will continue to evaluate faculty members during designated evaluation periods; the change is that submissions will occur through a secure, role-specific web portal linked to official enrollment data rather than through paper forms or general-purpose survey links. The system automatically presents each

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student with only the evaluation forms relevant to their enrolled courses, eliminating the risk of students rating faculty members they were not taught by.

Faculty members, deans, HR personnel, OSA officers, the VPAA, and system administrators each interact with a role-specific dashboard tailored to their responsibilities. Faculty gain timely access to their performance data and AI-generated feedback summaries. Deans can monitor department-level evaluation statistics, manage peer evaluation pairings, and conduct supervisor evaluations within the same platform. HR staff manage the full evaluation lifecycle — from setting evaluation periods and configuring questionnaires to publishing feedback summaries and generating CHED-compliant reports — without any manual data consolidation. OSA officers gain a focused clearance verification view that eliminates the need for manual cross-referencing.

User adoption is anticipated to be smooth given that the majority of NAAP's students and faculty already interact with web-based academic platforms as part of their regular routines. The role-specific interface design minimizes the learning curve by presenting each user group with only the functions relevant to their work. Orientation sessions and in-system guidance will support users during the initial deployment period. The proposed system is therefore assessed as operationally feasible.

Economic Feasibility

The economic feasibility of the proposed system is established through the cost-benefit analysis presented in the following section, which demonstrates a positive return on investment over a three-year lifecycle. The total project cost across all six Agile development phases is ₱446,310, covering requirements analysis, system design, core and advanced module development, quality assurance testing, deployment, evaluation, and long-term maintenance. Against these costs, the system is projected to generate ₱250,000 in annual tangible savings through the elimination of manual evaluation administration, reduction in paper and printing costs, and automation of clearance verification.

The three-year net benefit of ₱303,690 and a return on investment of 68.04 percent confirm that the proposed system is economically sound. Beyond measurable savings, the system delivers significant intangible value: full compliance with CHED CMO No. 19, Series of 2025, and RA 10173; improved data accuracy and institutional transparency; and a scalable platform that can serve all NAAP campuses without proportional increases in cost. The use of open-source web development frameworks and existing institutional hardware infrastructure keeps licensing and infrastructure costs low, further strengthening the economic case for development. The proposed system is therefore assessed as economically feasible.

Schedule Feasibility

Development of the proposed system follows an iterative Agile framework organized into structured sprint cycles. This approach enables the team to deliver functional modules incrementally, with each sprint producing testable, stakeholder-reviewable outputs before subsequent development proceeds. Sprint planning sessions at the project's outset establish a prioritized feature backlog, ensuring that core functionality — user authentication, role-based access, and evaluation submission — is delivered and validated early in the timeline before more complex modules such as AI integration and reporting are completed.

The implementation schedule is designed so that the fully tested and configured system is ready for deployment before the start of a scheduled evaluation period, giving administrative staff and faculty members adequate time to familiarize themselves with the platform before it is activated for institutional use. User training sessions are incorporated into the schedule as a formal activity, not an afterthought, ensuring all role groups are prepared before go-live.

Potential scheduling risks include delays from stakeholder availability during requirement consultations and the time required to validate the AI integration and security configuration. These risks are managed through flexible sprint scheduling, early and sustained stakeholder involvement, and a phased deployment strategy that releases completed modules progressively. Given

NAAP's academic calendar and the development team's capacity under the Agile model, the proposed system is assessed as schedule feasible.

Cost-Benefit Analysis

This Cost-Benefit Analysis presents a comprehensive financial and operational evaluation of the development and implementation of the proposed AI-Enhanced CHED-Compliant Web-Based Faculty and Student Evaluation System for NAAP. The analysis is structured across six Agile development phases, with each phase broken down into key activities, role-based labor cost computations, and summary totals. Following the phase-by-phase cost breakdown, a benefit analysis quantifies the projected institutional savings and returns over a three-year operational lifecycle.

Labor cost rates applied throughout this analysis are based on current Philippine market salary data for IT professionals as of 2025, sourced from salary benchmarking platforms including SalaryExpert, JobStreet, and industry compensation surveys. Hourly rates reflect mid-level professionals appropriate for an institutional academic system development project.

Formula and Computation Explanations

The following formulas govern all labor cost computations, total cost aggregations, and benefit calculations used throughout this analysis. Each formula is applied consistently across all phases and activities.

Labor Cost Formula

Each line-item labor cost in the activity tables is computed using the following formula:

$$\text{Cost (₱)} = \text{Number of Persons} \times \text{Hourly Rate (₱)} \times \text{Number of Hours}$$

Example: For 2 Developers at ₱480/hr working 10 hours:

$$\text{Cost} = 2 \times \text{₱480} \times 10 = \text{₱9,600}$$

This formula multiplies the number of personnel assigned to an activity by their respective hourly rate and the estimated hours required to complete that activity. Where a fixed estimated cost is listed (e.g., for tools or materials), this represents a lump-sum estimate not computed through the labor formula.

Phase Total Cost Formula

The total cost for each phase is the sum of all sub-total costs from the activities within that phase:

$$\text{Phase Total Cost} = \Sigma (\text{Sub-Total of each activity in the phase})$$

Overall Total Project Cost Formula

The overall total project cost sums all six phase totals:

$$\begin{aligned} \text{Total Project Cost} &= \text{Phase 1} + \text{Phase 2} + \text{Phase 3} + \text{Phase 4} + \text{Phase 5} + \text{Phase 6} \\ &= \text{P}52,675 + \text{P}61,485 + \text{P}181,030 + \text{P}61,465 + \text{P}45,965 + \text{P}44,690 = \text{P}446,310 + \\ &\quad \text{P}44,690 \\ &= \text{P}446,310 \end{aligned}$$

Total Tangible Benefits Formula

Total three-year tangible benefits are computed by multiplying the annual tangible savings by the number of operational years evaluated:

$$\begin{aligned} \text{Total Benefits (3 Years)} &= \text{Annual Tangible Savings} \times 3 \\ &= \text{P}250,000 \times 3 = \text{P}750,000 \end{aligned}$$

Net Benefit Formula

The net benefit represents the financial gain after deducting the total project cost from total benefits:

$$\begin{aligned}\text{Net Benefit} &= \text{Total Tangible Benefits (3 Years)} - \text{Total Project Cost} \\ &= \text{₱750,000} - \text{₱446,310} = \text{₱303,690}\end{aligned}$$

Return on Investment (ROI) Formula

ROI expresses the net benefit as a percentage of total investment, measuring efficiency of the project expenditure:

$$\begin{aligned}\text{ROI (\%)} &= (\text{Net Benefit} \div \text{Total Project Cost}) \times 100 \\ &= (\text{₱303,690} \div \text{₱446,310}) \times 100 = 68.04\%\end{aligned}$$

An ROI of 68.04% indicates that for every peso invested in the system, the institution recovers the investment plus an additional ₱0.68 in net savings over the three-year period.

Hourly Rate Basis (Philippine Market Rates, 2025)

The following hourly rates are applied uniformly across all phases. Rates are based on mid-level professional compensation benchmarks sourced from SalaryExpert, JobStreet, and industry surveys as of 2025.

Table 3.1. Labor Cost Hourly Rate Basis — Philippine IT Market, 2025

Role	Hourly Rate (P)
Project Manager	₱725
System Analyst	₱550
Developer	₱480
UI/UX Designer	₱430
QA / Test Specialist	₱380
Stakeholder Representative	₱350
Technical Support Staff	₱300

Phase 1: Requirements Analysis and Planning

The Requirements Analysis and Planning phase establishes the foundation of the entire system development effort. It focuses on identifying and engaging institutional stakeholders across all seven user roles, eliciting and documenting functional and non-functional system requirements, translating requirements into prioritized user stories, and organizing these into a product backlog ready for sprint execution. Given the complexity of the proposed system — which integrates AI feedback analysis, multi-role access controls, CHED compliance, and multi-campus support — thorough requirements documentation at this stage is essential to prevent costly rework in later phases.

Key activities include:

- Stakeholder Identification and Engagement

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- Requirement Elicitation
- User Story Development
- Product Backlog Creation and Prioritization
- Sprint Planning

A. Cost Analysis

Assumptions — Labor Cost Estimates

Table 3.2. Phase 1 — Hourly Rate Assumptions

Role	Hourly Rate (₱)
Project Manager	₱725
System Analyst	₱550
Developer	₱480
UI/UX Designer	₱430
Stakeholder Representative	₱350

Number of stakeholders involved: 5

1. Stakeholder Identification and Engagement Cost

Table 3.3. Phase 1 — Stakeholder Identification and Engagement Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Identify Stakeholders	Project Manager	1	₱725	3	$1 \times 725 \times 3$	₱2,175
Identify Stakeholders	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200

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Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Stakeholder Meetings	Project Manager	1	₱725	4	$1 \times 725 \times 4$	₱2,900
Stakeholder Meetings	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Stakeholder Meetings	Stakeholders	5	₱350	4	$5 \times 350 \times 4$	₱7,000
Requirement Documentation	System Analyst	1	₱550	5	$1 \times 550 \times 5$	₱2,750
Requirement Documentation	Developer	1	₱480	2	$1 \times 480 \times 2$	₱960
Communication Tools	Internet / Tools	—	—	—	Estimated	₱600
Documentation Materials	Cloud Storage	—	—	—	Estimated	₱350

Sub-Total: ₱21,135

2. Requirement Elicitation Cost

Table 3.4. Phase 1 — Requirement Elicitation Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Requirement Interviews	System Analyst	1	₱550	6	$1 \times 550 \times 6$	₱3,300
Workshops	Project Manager	1	₱725	3	$1 \times 725 \times 3$	₱2,175
Workshops	Stakeholders	5	₱350	3	$5 \times 350 \times 3$	₱5,250
Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱12,375

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3. User Story Development Cost

Table 3.5. Phase 1 — User Story Development Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
User Story Writing	System Analyst	1	₱550	5	$1 \times 550 \times 5$	₱2,750
Feature Discussion	Project Manager	1	₱725	3	$1 \times 725 \times 3$	₱2,175
UI Concept Planning	UI/UX Designer	1	₱430	4	$1 \times 430 \times 4$	₱1,720

Sub-Total: ₱6,645

4. Product Backlog Creation and Prioritization Cost

Table 3.6. Phase 1 — Product Backlog Creation Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Backlog Organization	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Feature Prioritization	Project Manager	1	₱725	3	$1 \times 725 \times 3$	₱2,175
Technical Review	Developer	1	₱480	3	$1 \times 480 \times 3$	₱1,440

Sub-Total: ₱5,815

5. Sprint Planning Cost

Table 3.7. Phase 1 — Sprint Planning Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Sprint Planning Meeting	Project Manager	1	₱725	3	$1 \times 725 \times 3$	₱2,175
Sprint Planning Meeting	Developer	2	₱480	3	$2 \times 480 \times 3$	₱2,880

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Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Sprint Planning Meeting	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱6,705

B. Total Cost of Phase 1

Table 3.8. Phase 1 — Total Cost Summary

Activity	Cost (P)
Stakeholder Identification and Engagement	₱21,135
Requirement Elicitation	₱12,375
User Story Development	₱6,645
Product Backlog Creation and Prioritization	₱5,815
Sprint Planning	₱6,705
TOTAL COST — PHASE 1	₱52,675

Phase 2: System Design and Prototype Development

The System Design and Prototype Development phase translates the validated requirements and prioritized backlog into concrete technical designs and an interactive working prototype. This phase covers the full architectural blueprint of the system — including its three-tier client-server structure, relational database schema, role-based access control model, and AI analysis module integration — alongside UI/UX wireframes for each of the seven role-specific portals. A prototype is developed to allow stakeholder validation before full-scale coding begins, reducing the risk of costly interface or architectural revisions in Phase 3.

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Key activities include:

- System Architecture Design
- Database Design and Data Modeling
- UI/UX Wireframing and Interface Design
- Prototype Development
- Design Review and Validation

A. Cost Analysis

Table 3.9. Phase 2 — Hourly Rate Assumptions

Role	Hourly Rate (₱)
Project Manager	₱725
System Analyst	₱550
Developer	₱480
UI/UX Designer	₱430
QA / Test Specialist	₱380

1. System Architecture Design Cost

Table 3.10. Phase 2 — System Architecture Design Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
System Architecture Planning	Project Manager	1	₱725	4	$1 \times 725 \times 4$	₱2,900

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Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Technical Architecture Design	System Analyst	1	₱550	6	$1 \times 550 \times 6$	₱3,300
Component and Module Mapping	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
Infrastructure Planning	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Architecture Documentation	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Collaboration Tools	Internet / Platforms	—	—	—	Estimated	₱600

Sub-Total: ₱12,570

2. Database Design and Data Modeling Cost

Table 3.11. Phase 2 — Database Design and Data Modeling Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Data Requirement Analysis	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Database Schema Design	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
Entity Relationship Diagram (ERD)	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Data Normalization (3NF)	Developer	1	₱480	3	$1 \times 480 \times 3$	₱1,440
Database Design Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱10,370

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3. UI/UX Wireframing and Interface Design Cost

Table 3.12. Phase 2 — UI/UX Wireframing Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
User Interface Planning	UI/UX Designer	1	₱430	4	$1 \times 430 \times 4$	₱1,720
Wireframe Design (8 Portals)	UI/UX Designer	1	₱430	8	$1 \times 430 \times 8$	₱3,440
User Flow Mapping	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
UX Review Session	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
UI Prototype Mockups	UI/UX Designer	1	₱430	4	$1 \times 430 \times 4$	₱1,720
Design Collaboration Tools	Design Platforms	—	—	—	Estimated	₱500

Sub-Total: ₱11,030

4. Prototype Development Cost

Table 3.13. Phase 2 — Prototype Development Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Prototype Coding	Developer	2	₱480	8	$2 \times 480 \times 8$	₱7,680
Front-End Interface Build	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
API Mock Development	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
Prototype Integration	Developer	1	₱480	3	$1 \times 480 \times 3$	₱1,440
Technical Supervision	Project Manager	1	₱725	3	$1 \times 725 \times 3$	₱2,175
Prototype Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱17,745

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5. Design Review and Validation Cost

Table 3.14. Phase 2 — Design Review and Validation Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Prototype Review Meeting	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
Design Evaluation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Usability Testing	QA Specialist	1	₱380	4	$1 \times 380 \times 4$	₱1,520
Stakeholder Validation	Stakeholders	5	₱350	2	$5 \times 350 \times 2$	₱3,500
Feedback Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱9,770

B. Total Cost of Phase 2

Table 3.15. Phase 2 — Total Cost Summary

Activity	Cost (₱)
System Architecture Design	₱12,570
Database Design and Data Modeling	₱10,370
UI/UX Wireframing and Interface Design	₱11,030
Prototype Development	₱17,745
Design Review and Validation	₱9,770
TOTAL COST — PHASE 2	₱61,485

Phase 3: Core System Module Development (Sprint 1)

Sprint 1 covers the development of the system's foundational layer: user authentication with email-based OTP, role-based access control for all seven user roles, evaluation form submission and draft saving, enrollment-linked form

assignment, automated clearance verification, and the basic HR and student dashboards. These modules form the backbone on which all subsequent features depend and must be fully functional before Sprint 2 begins.

Key activities include:

- Sprint 1 Feature Coding — Core Modules
- Front-End and Back-End Integration
- Continuous Testing and Debugging
- Sprint 1 Review and Demonstration
- Sprint 1 Retrospective and Improvement Planning

A. Cost Analysis — Sprint 1

Table 3.16. Phase 3 (Sprint 1) — Hourly Rate Assumptions

Role	Hourly Rate (₱)
Project Manager	₱725
System Analyst	₱550
Developer	₱480
UI/UX Designer	₱430
QA / Test Specialist	₱380

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1. Sprint 1 Feature Coding — Core Modules

Table 3.17. Phase 3 (Sprint 1) — Core Module Feature Coding Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Authentication + OTP Module	Developer	2	₱480	10	$2 \times 480 \times 10$	₱9,600
RBAC for 8 User Roles	Developer	2	₱480	12	$2 \times 480 \times 12$	₱11,520
Evaluation Form + Draft Saving	Developer	1	₱480	10	$1 \times 480 \times 10$	₱4,800
Enrollment-Linked Form Assignment	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
Clearance Verification Module	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
Core Dashboards (Student + HR)	Developer	1	₱480	8	$1 \times 480 \times 8$	₱3,840
Code Review	System Analyst	1	₱550	5	$1 \times 550 \times 5$	₱2,750
Sprint 1 Coordination	Project Manager	1	₱725	5	$1 \times 725 \times 5$	₱3,625
Development Tools and Platforms	Software License	—	—	—	Estimated	₱1,200

Sub-Total: Sub-Total: ₱43,095

2. Front-End and Back-End Integration — Sprint 1

Table 3.18. Phase 3 (Sprint 1) — Front-End and Back-End Integration Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
API Integration (Auth + Forms)	Developer	1	₱480	8	$1 \times 480 \times 8$	₱3,840
Front-End Interface Integration	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880

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Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Interface Refinement	UI/UX Designer	1	₱430	5	$1 \times 430 \times 5$	₱2,150
System Logic Verification	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Integration Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: Sub-Total: ₱12,720

3. Continuous Testing and Debugging — Sprint 1

Table 3.19. Phase 3 (Sprint 1) — Continuous Testing and Debugging Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Unit Testing	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
System Testing	QA Specialist	1	₱380	8	$1 \times 380 \times 8$	₱3,040
Bug Fixing	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
Test Case Documentation	QA Specialist	1	₱380	4	$1 \times 380 \times 4$	₱1,520
Testing Supervision	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450

Sub-Total: Sub-Total: ₱11,770

4. Sprint 1 Review and Demonstration

Table 3.20. Phase 3 (Sprint 1) — Sprint 1 Review and Demonstration Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Sprint 1 Review Meeting	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
System Demonstration	Developer	1	₱480	3	$1 \times 480 \times 3$	₱1,440

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Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Feature Validation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Stakeholder Review	Stakeholders	5	₱350	2	$5 \times 350 \times 2$	₱3,500
Feedback Documentation	System Analyst	1	₱550	2	$1 \times 550 \times 2$	₱1,100

Sub-Total: Sub-Total: ₱9,140

5. Sprint 1 Retrospective and Improvement Planning

Table 3.21. Phase 3 (Sprint 1) — Retrospective and Improvement Planning Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Retrospective Meeting	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
Process Evaluation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Development Improvement Planning	Developer	2	₱480	3	$2 \times 480 \times 3$	₱2,880
QA Feedback Analysis	QA Specialist	1	₱380	3	$1 \times 380 \times 3$	₱1,140
Retrospective Documentation	System Analyst	1	₱550	2	$1 \times 550 \times 2$	₱1,100

Sub-Total: Sub-Total: ₱8,220

B. Total Cost — Sprint 1

Table 3.22. Phase 3 (Sprint 1) — Total Cost Summary

Activity	Cost (₱)
Sprint 1 Feature Coding — Core Modules	₱43,095
Front-End and Back-End Integration	₱12,720

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Activity	Cost (P)
Continuous Testing and Debugging	₱11,770
Sprint 1 Review and Demonstration	₱9,140
Sprint 1 Retrospective and Improvement Planning	₱8,220
TOTAL COST — SPRINT 1	₱84,945

Phase 3: Advanced Module Development (Sprint 2)

Sprint 2 builds upon the validated core system from Sprint 1 and develops the advanced features that differentiate this system from generic evaluation tools: AI-assisted feedback summarization using the OpenAI GPT-4.1 mini API, local frequency analysis, CHED-compliant PDF report generation, supervisor-to-faculty and peer-to-peer evaluation modules, the full analytics and ranking dashboards for HR and the VPAA, the non-editable audit trail, Bcrypt password hashing implementation, automated email notification system, multi-campus data partitioning, and the questionnaire management module. Sprint 2 also addresses all refinements identified during Sprint 1 review.

Key activities include:

- Sprint 2 Feature Coding — Advanced Modules
- AI Module and Security Integration
- Continuous Testing and Debugging
- Sprint 2 Review and Demonstration

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- Sprint 2 Retrospective and Improvement Planning

A. Cost Analysis — Sprint 2

Table 3.23. Phase 3 (Sprint 2) — Hourly Rate Assumptions

Role	Hourly Rate (₱)
Project Manager	₱725
System Analyst	₱550
Developer	₱480
UI/UX Designer	₱430
QA / Test Specialist	₱380

1. Sprint 2 Feature Coding — Advanced Modules

Table 3.24. Phase 3 (Sprint 2) — Advanced Module Feature Coding Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
AI Summarization Module (GPT-4.1 mini API)	Developer	2	₱480	12	$2 \times 480 \times 12$	₱11,520
Frequency Analysis Module	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
CHED-Compliant PDF Report Generator	Developer	1	₱480	8	$1 \times 480 \times 8$	₱3,840
Supervisor + Peer Evaluation Modules	Developer	1	₱480	8	$1 \times 480 \times 8$	₱3,840
Analytics Dashboards (HR, Dean, VPAA)	Developer	2	₱480	10	$2 \times 480 \times 10$	₱9,600
Audit Trail Module	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880

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Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Email Notification System	Developer	1	₱480	5	1 × 480 × 5	₱2,400
Questionnaire Management Module	Developer	1	₱480	6	1 × 480 × 6	₱2,880
Multi-Campus Data Partitioning	Developer	1	₱480	5	1 × 480 × 5	₱2,400
Sprint 2 Coordination	Project Manager	1	₱725	5	1 × 725 × 5	₱3,625
Code Review	System Analyst	1	₱550	5	1 × 550 × 5	₱2,750
Development Tools and Licenses	Software License	—	—	—	Estimated	₱1,500

Sub-Total: ₱50,115

2. AI Module and Security Integration

Table 3.25. Phase 3 (Sprint 2) — AI Module and Security Integration Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Bcrypt Hashing Implementation	Developer	1	₱480	6	1 × 480 × 6	₱2,880
OTP and Hashing Integration	Developer	1	₱480	4	1 × 480 × 4	₱1,920
Cloudflare Firewall Configuration	Developer	1	₱480	3	1 × 480 × 3	₱1,440
OpenAI API Integration and Prompting	Developer	1	₱480	6	1 × 480 × 6	₱2,880
Security Integration Verification	System Analyst	1	₱550	4	1 × 550 × 4	₱2,200
Interface Adjustments (Sprint 2)	UI/UX Designer	1	₱430	5	1 × 430 × 5	₱2,150

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Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Integration Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱15,120

3. Continuous Testing and Debugging — Sprint 2

Table 3.26. Phase 3 (Sprint 2) — Continuous Testing and Debugging Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Unit Testing	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
Security and AI Testing	QA Specialist	1	₱380	10	$1 \times 380 \times 10$	₱3,800
Bug Fixing	Developer	1	₱480	8	$1 \times 480 \times 8$	₱3,840
Test Case Documentation	QA Specialist	1	₱380	4	$1 \times 380 \times 4$	₱1,520
Testing Supervision	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450

Sub-Total: ₱13,490

4. Sprint 2 Review and Demonstration

Table 3.27. Phase 3 (Sprint 2) — Sprint 2 Review and Demonstration Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Sprint 2 Review Meeting	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
Full System Demonstration	Developer	1	₱480	3	$1 \times 480 \times 3$	₱1,440
Feature Validation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Stakeholder Review	Stakeholders	5	₱350	2	$5 \times 350 \times 2$	₱3,500

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Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Feedback Documentation	System Analyst	1	₱550	2	$1 \times 550 \times 2$	₱1,100

Sub-Total: ₱9,140

5. Sprint 2 Retrospective and Improvement Planning

Table 3.28. Phase 3 (Sprint 2) — Retrospective and Improvement Planning Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Retrospective Meeting	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
Process Evaluation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Development Improvement Planning	Developer	2	₱480	3	$2 \times 480 \times 3$	₱2,880
QA Feedback Analysis	QA Specialist	1	₱380	3	$1 \times 380 \times 3$	₱1,140
Retrospective Documentation	System Analyst	1	₱550	2	$1 \times 550 \times 2$	₱1,100

Sub-Total: ₱8,220

B. Total Cost — Sprint 2

Table 3.29. Phase 3 (Sprint 2) — Total Cost Summary

Activity	Cost (P)
Sprint 2 Feature Coding — Advanced Modules	₱50,115
AI Module and Security Integration	₱15,120
Continuous Testing and Debugging	₱13,490
Sprint 2 Review and Demonstration	₱9,140

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Activity	Cost (P)
Sprint 2 Retrospective and Improvement Planning	₱8,220
TOTAL COST — SPRINT 2	₱96,085

C. Total Cost of Phase 3 (Sprint 1 + Sprint 2)

Table 3.30. Phase 3 — Combined Sprint Total Cost

Activity	Cost (P)
Sprint 1 — Core System Modules	₱84,945
Sprint 2 — Advanced Modules and Completion	₱96,085
TOTAL COST — PHASE 3	₱181,030

Phase 4: System Testing, Deployment, and Maintenance

The System Testing, Deployment, and Maintenance phase ensures that the fully developed system is rigorously validated before institutional go-live. Comprehensive testing covers functional correctness, security, performance under peak load, and user acceptance across all seven role groups. Following successful testing, the system is deployed on NAAP's institutional server infrastructure with Cloudflare firewall integration and database security configuration applied. User training sessions are conducted for each role group before the system is activated for live use. Initial system maintenance covers the first operational period, addressing any post-deployment issues discovered in the production environment.

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Key activities include:

- System Testing and Quality Assurance
- User Acceptance Testing (UAT)
- System Deployment and Configuration
- User Training and Documentation
- Initial System Maintenance and Support

A. Cost Analysis

Table 3.31. Phase 4 — Hourly Rate Assumptions

Role	Hourly Rate (P)
Project Manager	₱725
System Analyst	₱550
Developer	₱480
UI/UX Designer	₱430
QA / Test Specialist	₱380
Stakeholder Representative	₱350

1. System Testing and Quality Assurance Cost

Table 3.32. Phase 4 — System Testing and Quality Assurance Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Test Plan Preparation	QA Specialist	1	₱380	4	$1 \times 380 \times 4$	₱1,520
Functional Testing	QA Specialist	1	₱380	10	$1 \times 380 \times 10$	₱3,800

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Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Security and Encryption Testing	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
Performance and Load Testing	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
Bug Fix Implementation	Developer	1	₱480	5	$1 \times 480 \times 5$	₱2,400
Test Result Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱14,170

2. User Acceptance Testing (UAT) Cost

Table 3.33. Phase 4 — User Acceptance Testing Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
UAT Planning	Project Manager	1	₱725	3	$1 \times 725 \times 3$	₱2,175
UAT Coordination	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
UAT Session (All Roles)	Stakeholders	5	₱350	4	$5 \times 350 \times 4$	₱7,000
Issue Identification	QA Specialist	1	₱380	3	$1 \times 380 \times 3$	₱1,140
UAT Report Preparation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱14,165

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3. System Deployment and Configuration Cost

Table 3.34. Phase 4 — System Deployment and Configuration Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Deployment Planning	Project Manager	1	₱725	3	$1 \times 725 \times 3$	₱2,175
Server Configuration	Developer	1	₱480	5	$1 \times 480 \times 5$	₱2,400
Database Deployment	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
Cloudflare Firewall Setup	Developer	1	₱480	3	$1 \times 480 \times 3$	₱1,440
Deployment Verification	QA Specialist	1	₱380	3	$1 \times 380 \times 3$	₱1,140
Cloud / Hosting Setup	Cloud Services	—	—	—	Estimated	₱1,200

Sub-Total: ₱10,275

4. User Training and Documentation Cost

Table 3.35. Phase 4 — User Training and Documentation Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Training Preparation	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Training Materials Creation	UI/UX Designer	1	₱430	4	$1 \times 430 \times 4$	₱1,720
User Training Sessions	Project Manager	1	₱725	3	$1 \times 725 \times 3$	₱2,175
User Training Sessions	Stakeholders	5	₱350	3	$5 \times 350 \times 3$	₱5,250

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Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
User Manual Documentation	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200

Sub-Total: ₱13,545

5. Initial System Maintenance and Support Cost

Table 3.36. Phase 4 — Initial System Maintenance and Support Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
System Monitoring	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
Bug Fix and Patch Updates	Developer	1	₱480	5	$1 \times 480 \times 5$	₱2,400
User Support Assistance	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Performance Monitoring	QA Specialist	1	₱380	3	$1 \times 380 \times 3$	₱1,140
Maintenance Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱9,310

B. Total Cost of Phase 4

Table 3.37. Phase 4 — Total Cost Summary

Activity	Cost (₱)
System Testing and Quality Assurance	₱14,170
User Acceptance Testing (UAT)	₱14,165
System Deployment and Configuration	₱10,275
User Training and Documentation	₱13,545
Initial System Maintenance and Support	₱9,310

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Activity	Cost (₱)
TOTAL COST — PHASE 4	₱61,465

Phase 5: System Evaluation and Continuous Improvement

Following initial deployment, Phase 5 focuses on systematically assessing the live system's performance, collecting structured feedback from all user groups, and planning and implementing targeted improvements based on real operational data. For the proposed system, this phase is especially important given the complexity of the AI-assisted feedback module and the multi-role dashboard architecture — both of which may require calibration based on actual usage patterns and HR workflow preferences. Security and performance are also formally reviewed to ensure that the system remains compliant with data privacy requirements and responsive under growing data volumes.

Key activities include:

- System Performance Evaluation
- User Feedback Collection and Analysis
- System Enhancement Planning
- Security and Performance Review
- Continuous Improvement Implementation

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A. Cost Analysis

Table 3.38. Phase 5 — Hourly Rate Assumptions

Role	Hourly Rate (₱)
Project Manager	₱725
System Analyst	₱550
Developer	₱480
UI/UX Designer	₱430
QA / Test Specialist	₱380
Stakeholder Representative	₱350

1. System Performance Evaluation Cost

Table 3.39. Phase 5 — System Performance Evaluation Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Performance Monitoring	Developer	1	₱480	5	$1 \times 480 \times 5$	₱2,400
System Usage Analysis	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Performance Testing	QA Specialist	1	₱380	4	$1 \times 380 \times 4$	₱1,520
Performance Report Preparation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Evaluation Meeting	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450

Sub-Total: ₱9,220

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2. User Feedback Collection and Analysis Cost

Table 3.40. Phase 5 — User Feedback Collection and Analysis Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Feedback Survey Design	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Feedback Collection	Stakeholders	5	₱350	3	$5 \times 350 \times 3$	₱5,250
Feedback Analysis	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Feedback Report	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Feedback Review Meeting	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
Survey Tools and Materials	Tools	—	—	—	Estimated	₱250

Sub-Total: ₱12,450

3. System Enhancement Planning Cost

Table 3.41. Phase 5 — System Enhancement Planning Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Enhancement Requirement Review	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Development Planning	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
Enhancement Prioritization	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
UX Enhancement Planning	UI/UX Designer	1	₱430	3	$1 \times 430 \times 3$	₱1,290

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Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Enhancement Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Enhancement Review Meeting	Project Manager	1	₱725	1	$1 \times 725 \times 1$	₱725

Sub-Total: ₱9,235

4. Security and Performance Review Cost

Table 3.42. Phase 5 — Security and Performance Review Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Security Assessment	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
Vulnerability Testing	QA Specialist	1	₱380	4	$1 \times 380 \times 4$	₱1,520
System Audit Review	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Security Review Meeting	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
Security Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱8,190

5. Continuous Improvement Implementation Cost

Table 3.43. Phase 5 — Continuous Improvement Implementation Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Minor System Enhancements	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880

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Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
UI/UX Improvements	UI/UX Designer	1	₱430	4	1 × 430 × 4	₱1,720
Regression Testing	QA Specialist	1	₱380	4	1 × 380 × 4	₱1,520
Enhancement Review	System Analyst	1	₱550	3	1 × 550 × 3	₱1,650
Improvement Documentation	System Analyst	1	₱550	2	1 × 550 × 2	₱1,100

Sub-Total: ₱8,870

B. Total Cost of Phase 5

Table 3.44. Phase 5 — Total Cost Summary

Activity	Cost (₱)
System Performance Evaluation	₱9,220
User Feedback Collection and Analysis	₱10,450
System Enhancement Planning	₱9,235
Security and Performance Review	₱8,190
Continuous Improvement Implementation	₱8,870
TOTAL COST — PHASE 5	₱45,965

Phase 6: Long-Term Maintenance and System Sustainability

Phase 6 covers the ongoing operational maintenance and sustainability of the deployed system over its active institutional life. Activities in this phase ensure that the system continues to meet evolving institutional requirements, remains secure against emerging threats, and operates reliably as the volume of evaluation data and registered users grows over time. Regular database maintenance,

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security monitoring, feature enhancement cycles, and knowledge transfer procedures are conducted on a scheduled basis. This phase is essential for protecting the institution's investment and ensuring that the system delivers consistent value beyond the initial deployment period.

Key activities include:

- System Monitoring and Maintenance
- Technical Support and Issue Resolution
- System Updates and Feature Enhancements
- Security Monitoring and Risk Management
- System Documentation and Knowledge Transfer

A. Cost Analysis

Table 3.45. Phase 6 — Hourly Rate Assumptions

Role	Hourly Rate (P)
Project Manager	₱725
System Analyst	₱550
Developer	₱480
UI/UX Designer	₱430
QA / Test Specialist	₱380
Technical Support Staff	₱300

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1. System Monitoring and Maintenance Cost

Table 3.46. Phase 6 — System Monitoring and Maintenance Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
System Performance Monitoring	Developer	1	₱480	5	$1 \times 480 \times 5$	₱2,400
Database Maintenance	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
Log Monitoring and Analysis	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Maintenance Planning	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
Maintenance Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱9,620

2. Technical Support and Issue Resolution Cost

Table 3.47. Phase 6 — Technical Support and Issue Resolution Cost

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
User Support Assistance	Tech Support Staff	1	₱300	6	$1 \times 300 \times 6$	₱1,800
Issue Diagnosis	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Bug Fix Implementation	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
System Verification	QA Specialist	1	₱380	3	$1 \times 380 \times 3$	₱1,140
Issue Resolution Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱8,710

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3. System Updates and Feature Enhancements Cost

Table 3.48. Phase 6 — System Updates and Feature Enhancements Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Enhancement Requirement Review	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
Feature Enhancement Development	Developer	1	₱480	6	$1 \times 480 \times 6$	₱2,880
Interface Improvements	UI/UX Designer	1	₱430	4	$1 \times 430 \times 4$	₱1,720
Enhancement Testing	QA Specialist	1	₱380	4	$1 \times 380 \times 4$	₱1,520
Update Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱9,970

4. Security Monitoring and Risk Management Cost

Table 3.49. Phase 6 — Security Monitoring and Risk Management Cost

Activity	Role	Persons	Rate (P)	Hours	Computation	Cost (P)
Security Monitoring	Developer	1	₱480	4	$1 \times 480 \times 4$	₱1,920
Vulnerability Assessment	QA Specialist	1	₱380	4	$1 \times 380 \times 4$	₱1,520
Risk Assessment	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Security Review Meeting	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450

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Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Security Report Documentation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650

Sub-Total: ₱8,190

5. System Documentation and Knowledge Transfer Cost

**Table 3.50. Phase 6 — System Documentation and Knowledge Transfer
Cost**

Activity	Role	Persons	Rate (₱)	Hours	Computation	Cost (₱)
Technical Documentation Update	System Analyst	1	₱550	4	$1 \times 550 \times 4$	₱2,200
System Maintenance Guide Creation	System Analyst	1	₱550	3	$1 \times 550 \times 3$	₱1,650
Knowledge Transfer Session	Project Manager	1	₱725	2	$1 \times 725 \times 2$	₱1,450
Staff Orientation	Tech Support Staff	2	₱300	3	$2 \times 300 \times 3$	₱1,800
Documentation Archiving	System Analyst	1	₱550	2	$1 \times 550 \times 2$	₱1,100

Sub-Total: ₱8,200

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B. Total Cost of Phase 6

Table 3.51. Phase 6 — Total Cost Summary

Activity	Cost (₱)
System Monitoring and Maintenance	₱9,620
Technical Support and Issue Resolution	₱8,710
System Updates and Feature Enhancements	₱9,970
Security Monitoring and Risk Management	₱8,190
System Documentation and Knowledge Transfer	₱8,200
TOTAL COST — PHASE 6	₱44,690

Agile Development Cost Summary

The table below consolidates the total costs across all six development phases of the proposed system. Phase 3 reflects the combined cost of two iterative Agile sprints: Sprint 1 covering core authentication, RBAC, and evaluation submission modules, and Sprint 2 covering AI integration, advanced dashboards, Bcrypt security implementation, and all remaining features.

Table 3.52. Overall Agile Development Cost Summary — All Phases

Phase	Description	Key Activities	Total Cost (₱)
Phase 1	Requirements Analysis and Planning	Stakeholder engagement, requirement elicitation, user stories, backlog, sprint planning	₱52,675
Phase 2	System Design and Prototype Development	Architecture design, database modeling, UI/UX	₱61,485

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Phase	Description	Key Activities	Total Cost (₱)
		wireframes, prototype build, design validation	
Phase 3 (Sprint 1)	Core System Module Development	Authentication + OTP, RBAC, evaluation forms, clearance module, core dashboards	₱84,945
Phase 3 (Sprint 2)	Advanced Module Development	AI module, PDF reports, audit trail, Bcrypt, email notifications, multi-campus support	₱96,085
Phase 4	System Testing, Deployment, and Maintenance	QA testing, UAT, server deployment, Cloudflare setup, user training, initial support	₱61,465
Phase 5	System Evaluation and Continuous Improvement	Performance evaluation, user feedback, enhancement planning, security review, improvements	₱45,965
Phase 6	Long-Term Maintenance and System Sustainability	System monitoring, tech support, updates, security risk management, documentation	₱44,690
	TOTAL OVERALL AGILE SYSTEM DEVELOPMENT COST		₱446,310

Benefit Analysis

This section identifies and quantifies the tangible and intangible benefits that the proposed evaluation system is expected to deliver to NAAP over its operational lifecycle. Tangible benefits are expressed as annual savings computed from the reduction of labor costs and operational expenses currently incurred through the manual evaluation process. Intangible benefits represent institutional

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improvements that are significant in value but not directly reducible to a single monetary figure.

Tangible Benefits (Annual)

The annual savings below are derived from the elimination or substantial reduction of manual activities that currently consume administrative staff time and institutional resources during each evaluation cycle. Each benefit is based on estimated staff-hours saved multiplied by the applicable average hourly rate.

Table 3.53. Annual Tangible Benefits — NAAP Evaluation System

Benefit Category	Description and Computation Basis	Annual Savings (P)
Reduced Administrative Labor	Automation of data encoding, form distribution, result consolidation, and report preparation eliminates an estimated 480 staff-hours per evaluation cycle at an average rate of P250/hr ($480 \times \text{P}250 = \text{P}120,000$)	P120,000
Faster Evaluation Processing	Automated submission tracking, real-time dashboards, and digital report generation reduce evaluation processing time from weeks to hours, saving an estimated 240 equivalent staff-hours per year ($240 \times \text{P}250 = \text{P}60,000$)	P60,000
Elimination of Paper and Printing Costs	Full digital submission removes recurring costs of printed evaluation forms, answer sheets, and physical report reproduction across all campuses (estimated lump-sum savings)	P25,000
Automated Clearance Verification	System-generated completion references replace manual per-student OSA cross-referencing during clearance season, saving an estimated 180 staff-hours per clearance period ($180 \times \text{P}250 = \text{P}45,000$)	P45,000

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Benefit Category	Description and Computation Basis	Annual Savings (P)
Total Annual Tangible Benefits		₱250,000

Intangible Benefits

The following benefits are not directly reducible to a single annual figure but represent substantial and lasting value for NAAP's institutional quality assurance and governance:

- Improved accuracy and integrity of evaluation data through enrollment-linked access control, automated duplicate prevention, and validated submission tracking
- Objective, consistent interpretation of qualitative student feedback through AI-assisted summarization (GPT-4.1 mini) combined with local frequency analysis, replacing subjective manual review
- Full regulatory compliance with CHED Memorandum Order No. 19, Series of 2025, through standardized questionnaire instruments and CHED-formatted PDF evaluation reports
- Compliance with the Data Privacy Act of 2012 (Republic Act No. 10173) through Bcrypt one-way password hashing, email OTP authentication, and anonymized evaluation reporting

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- Strengthened institutional accountability through a permanent, non-editable audit trail capturing all system actions, user changes, and AI summary edits with timestamps and user attribution
- Enhanced faculty trust and engagement in the evaluation process through transparent, consistently applied, and evaluator-anonymous procedures
- Multi-campus scalability allowing all NAAP campuses to be served from a single centralized deployment without proportional increases in administrative overhead
- Long-term data preservation through a soft-delete policy that retains all historical evaluation records, supporting trend analysis, academic quality reviews, and regulatory audits over multiple academic years

Total Benefits Over 3 Years

The total tangible benefit over the three-year assessment period is computed by multiplying the annual savings by the number of years:

$$\text{Total Tangible Benefits (3 Years)} = \text{Annual Savings} \times 3 = \text{₱}250,000 \times 3 = \text{₱}750,000$$

Net Benefit Computation

The net benefit is derived by subtracting the total project cost across all six phases from the total three-year tangible benefits. This represents the actual

financial gain the institution realizes after fully recovering the system's development, deployment, and maintenance investment:

$$\text{Net Benefit} = \text{Total Benefits (3 Years)} - \text{Total Project Cost (Phases 1-6)}$$

$$\text{Net Benefit} = \text{₱750,000} - \text{₱446,310} = \text{₱303,690}$$

The positive net benefit of ₱303,690 confirms that the system's financial returns exceed total costs across all six phases over the three-year period.

Return on Investment (ROI)

ROI measures the efficiency of the total project investment by expressing the net benefit as a percentage of the total cost. A higher ROI indicates that a greater proportion of the investment is recovered as net savings:

$$\text{ROI (\%)} = (\text{Net Benefit} \div \text{Total Project Cost}) \times 100$$

$$\text{ROI} = (\text{₱303,690} \div \text{₱446,310}) \times 100 = 68.04\%$$

An ROI of 68.04% means the institution recovers its full investment and gains an additional ₱0.68 in net savings for every ₱1.00 invested over the three-year period. This is a strong return that justifies the development and long-term operational expenditure.

Agile-Specific Cost Advantage Justification

The Agile SDLC model adopted for this project delivers economic advantages that strengthen the financial case for development. By validating features through stakeholder sprint reviews before subsequent development proceeds, rework costs from misaligned requirements are minimized. The two-sprint structure in Phase 3 is particularly advantageous because it allows core modules — authentication, RBAC, and evaluation submission — to be tested and confirmed functional before investment is made in the more complex AI and security features of Sprint 2. If stakeholder review during Sprint 1 identifies interface or logic issues, they are corrected before they propagate into the advanced modules, preventing exponentially more costly late-stage fixes.

Continuous integration testing throughout each sprint reduces the density and severity of defects reaching the production environment, directly reducing post-deployment support costs captured in Phases 5 and 6. Incremental delivery also allows NAAP to begin realizing operational savings from completed modules — such as digital evaluation submission and automated clearance checking — before the full system is deployed, accelerating the break-even point. The iterative nature of Agile development further protects the investment against obsolescence: as CHED updates evaluation requirements or NAAP expands its campuses, the system can be adapted within planned sprint cycles without requiring a complete rebuild.

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Cost-Benefit Summary

Table 3.54. Cost-Benefit Summary — AI-Enhanced CHED-Compliant Faculty Evaluation System

Metric	Value
Total Development Cost (Phases 1–4)	₱290,620
Total Evaluation and Improvement (Phase 5)	₱45,965
Total Long-Term Maintenance (Phase 6)	₱44,690
Overall Total Project Cost (Phases 1–6)	₱446,310
Annual Tangible Benefits	₱250,000
Total Tangible Benefits (3 Years)	₱750,000
Net Benefit (3 Years)	₱303,690
Return on Investment (ROI)	68.04%

The cost-benefit analysis confirms that the proposed AI-Enhanced CHED-Compliant Web-Based Faculty and Student Evaluation System represents a sound institutional investment for NAAP. The total three-year net benefit of ₱303,690 and a return on investment of 68.04 percent demonstrate that the financial returns from reduced administrative labor, eliminated paper costs, and automated clearance verification substantially exceed the total development, maintenance, and operational investment. When the system's intangible contributions — regulatory compliance, improved data governance, faculty performance transparency, and long-term audit capability — are considered alongside these figures, the overall institutional value of the proposed system is considerably greater than its direct financial return alone.

System Requirements Specification

This section formally defines the functional and non-functional requirements that the proposed system must satisfy. The requirements are derived from the gap analysis, stakeholder consultations, and the regulatory specifications of CHED Memorandum Order No. 19, Series of 2025.

Functional Requirements

The system shall provide secure, role-based access for seven distinct user groups: Student, Faculty, Supervisor/Dean, HR Officer, OSA Officer, VPAA, and System Administrator. Each role shall have access to a dedicated portal presenting only the functions and data relevant to that role's responsibilities.

Students shall be able to submit evaluation forms exclusively for faculty members assigned to their currently enrolled courses. The system shall automatically retrieve enrollment data to generate the list of evaluable faculty per student, preventing out-of-scope submissions. Students shall be able to save draft responses and resume evaluation completion within the active evaluation window.

The HR module shall allow authorized HR officers to configure evaluation periods, manage questionnaire instruments, publish AI-generated feedback summaries, generate CHED-compliant PDF evaluation reports, and monitor submission completion rates across all campuses in real time.

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The system shall implement an automated clearance verification function whereby the OSA dashboard displays, for each student, the completion status of required evaluations. The clearance reference shall be generated automatically once all required submissions are confirmed, eliminating the need for manual cross-referencing.

The AI-assisted feedback analysis module shall retrieve all open-ended student comments for a selected faculty member and evaluation period, transmit them to the OpenAI GPT-4.1 mini API with a structured analytical prompt, and return a thematic summary organized by commendations, constructive feedback, and recurring concerns. A locally executed frequency analysis module shall independently compute term frequencies from the same comment set and display results alongside the AI summary. The feedback analysis pipeline shall incorporate AI-Based Bias Detection to classify qualitative comments as Constructive, Neutral, or Biased prior to summarization, and shall display AI Transparency disclosures showing detected keywords, comment clusters, and analytic reasoning alongside every generated summary.

The system shall maintain a non-editable, append-only audit trail recording all significant actions — including login events, evaluation submissions, data modifications, AI summary generation, and administrative changes — with user identifiers, timestamps, and IP addresses.

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HR officers and VPAA users shall have access to comparative dashboards displaying faculty performance rankings, departmental evaluation averages, evaluation period trends, and campus-level summary statistics. All dashboard data shall be refreshed in real time from the live database. Dashboards shall additionally display longitudinal faculty performance trends across evaluation periods and Evaluation Credibility Scores for submitted evaluations, enabling administrators to distinguish between highly reliable and review-required evaluation inputs. Cross-source consistency alerts shall be surfaced in the HR dashboard when significant discrepancies are detected between student, peer, and supervisor evaluation results for the same faculty member and period.

The system shall support multi-campus data partitioning, ensuring that evaluation data from each NAAP campus is accessible only to authorized users assigned to that campus, while centralized administrative and VPAA views can access aggregated data across all campuses.

Email notifications shall be dispatched to students with pending evaluations at administrator-configured intervals during open evaluation periods. HR officers shall be able to configure notification frequency, timing, and message content through the administrative interface. Email delivery shall also be used for OTP distribution during first-time login verification and unrecognized device access events.

The questionnaire management module shall allow HR officers to create, edit, activate, and archive evaluation instruments conforming to the item format and rating scale requirements of CHED CMO No. 19, Series of 2025. Each questionnaire shall include configurable quantitative rating items and at least one open-ended comment field per instrument.

Analytics Framework and Advanced Feature Requirements

The proposed system integrates a multi-tier analytics framework organized into four complementary analytic categories: Descriptive, Diagnostic, Predictive, and Prescriptive Analytics. Each category serves a distinct purpose in transforming raw evaluation data into actionable institutional intelligence. The seven advanced features described in this section are distributed across these categories according to their analytical function and the type of insight they generate.

Descriptive Analytics

Descriptive analytics explains what has already happened by summarizing historical evaluation data in a form that is immediately understandable to administrators, faculty, and institutional decision-makers. The system applies descriptive analytics to produce consolidated performance summaries, period-over-period comparisons, and departmental aggregations from the accumulated evaluation record. Two advanced features are classified under this category:

Track Faculty Performance Over Time. The system stores all historical evaluation records for every faculty member across all completed evaluation periods. An AI-assisted trend detection module processes the accumulated data to identify improvement trends, decline patterns, and sustained consistency in faculty performance. Rather than presenting evaluation results solely on a per-semester basis, the system generates longitudinal performance views that allow HR officers and VPAA users to assess faculty development trajectories across multiple academic terms. Example outputs include computed percentage changes between evaluation periods (e.g., “Faculty improved by 18% over 3 semesters”) and graphical trend visualizations displayed on role-specific dashboards. Historical records are retained under the system’s soft-delete policy, ensuring that longitudinal data remains available for performance reviews, faculty development planning, and regulatory audits throughout the system’s operational life.

AI Transparency. The system surfaces the reasoning behind AI-generated feedback summaries rather than presenting only final conclusions. Instead of delivering a bare summary statement such as “Needs Improvement,” the AI Transparency feature displays the specific keywords detected in submitted comments, clusters of semantically related feedback, and the analytic reasoning applied to arrive at the summary classification. This design ensures that HR officers and faculty members can evaluate the basis of AI-generated assessments, identify cases where the model may have over-weighted isolated comments, and

exercise informed judgment before acting on AI outputs. Transparency disclosures are displayed alongside every AI-generated summary in the HR and faculty dashboards.

Diagnostic Analytics

Diagnostic analytics explains why a particular result occurred by examining the causes and patterns underlying the data. Where descriptive analytics reports what the numbers show, diagnostic analytics investigates whether those numbers are reliable and what factors produced them. The system applies diagnostic analytics through three advanced features that collectively address the integrity and validity of submitted evaluation data:

Evaluation Behavior Analysis. The system monitors and analyzes the behavioral patterns of evaluators during the submission process in order to detect responses that may reflect inattentiveness, strategic bias, or automated submission rather than genuine assessment. The analysis examines three behavioral dimensions: (1) time spent completing the evaluation form, where an unusually rapid completion time relative to the number of items may indicate that the respondent did not read the questions carefully; (2) response pattern uniformity, where a respondent who selects the same rating value (e.g., all 5 or all 1) across every item may not be providing differentiated assessments; and (3) repetitive response patterns across multiple submitted forms from the same

evaluator. Based on these signals, the system computes an Evaluation Behavior Score for each submitted evaluation. Submissions with low behavior scores are flagged for HR review rather than excluded automatically, preserving the integrity of the evaluation record while alerting administrators to potentially unreliable data points.

AI-Based Bias Detection. The system applies Natural Language Processing (NLP) to classify open-ended qualitative comments according to their constructiveness and neutrality before they are incorporated into AI-generated feedback summaries. The NLP module detects three categories of problematic content: emotional bias, characterized by language expressing personal animosity or strong negative sentiment unrelated to pedagogical performance (e.g., “I hate this prof”); non-constructive comments that provide no actionable information; and irrelevant feedback that does not address the faculty member’s professional conduct or teaching quality. Each detected comment is assigned one of three classification labels: Constructive (e.g., “Explains well but needs more examples”), Neutral (e.g., “Okay”), or Biased (e.g., “I hate this prof”). Comments classified as Biased are excluded from the AI summarization input or flagged with a reduced weighting, ensuring that AI-generated feedback summaries reflect genuine pedagogical assessments rather than personal sentiment. Classification labels and the rationale for each are displayed in the AI Transparency view accessible to HR officers.

Cross-Source Validation. The system performs automated cross-source consistency checks by comparing the evaluation results submitted through three distinct evaluator types: student evaluations, peer evaluations, and supervisor evaluations. Where the results from one evaluator group diverge substantially from the others for the same faculty member and evaluation period (e.g., student ratings are very low while supervisor ratings are very high), the system flags the inconsistency and presents a consistency alert in the HR dashboard. This cross-source diagnostic capability allows HR officers to investigate potential sources of rating discrepancy before final evaluation reports are published, supporting more reliable and defensible faculty performance determinations.

Predictive Analytics

Predictive analytics uses historical patterns and current data signals to estimate the likely reliability or outcome of future or current data points. In the context of the evaluation system, predictive analytics is applied not to forecast future faculty performance scores directly, but to estimate the credibility of submitted evaluations before they are weighted in performance computations. This is implemented through the following advanced feature:

Credibility Score Output. Each submitted evaluation is assigned an Evaluation Credibility Score on a scale of 0 to 100, computed by aggregating three component signals: the Evaluation Behavior Score derived from the Evaluation

Behavior Analysis module, the bias classification outcome from the AI-Based Bias Detection module, and the cross-source consistency result from the Cross-Source Validation module. The composite score provides a unified, data-driven estimate of how much confidence should be placed in a given evaluation submission. Credibility scores are interpreted according to three reliability bands: 90 to 100 indicates a highly reliable evaluation; 70 to 89 indicates an acceptable evaluation; and scores below 70 indicate that the evaluation requires HR review before inclusion in published performance reports. This scoring mechanism enables HR officers to apply differentiated scrutiny to evaluations proportional to their estimated reliability, rather than treating all submitted responses as equally valid inputs.

Prescriptive Analytics

Prescriptive analytics goes beyond reporting and diagnosis to recommend specific actions that should be taken in response to identified data conditions. It transforms analytical findings into operationally useful guidance, closing the loop between evaluation data and institutional improvement. The system implements prescriptive analytics through the following advanced feature:

Smart Recommendation Engine. The system implements an AI-assisted Smart Recommendation Engine that generates targeted faculty development suggestions based on the specific performance deficiencies identified in evaluation

results. The engine operates in two steps. In the first step, it identifies the evaluation dimensions where a faculty member's scores fall below defined thresholds (e.g., a low rating in the Clarity of Instruction domain). In the second step, it produces actionable improvement recommendations tailored to the identified weak areas (e.g., "Use more examples and visual aids to improve instructional clarity"). Recommendations are generated from a structured knowledge base aligned with common faculty development frameworks and are displayed in the faculty member's personal dashboard following each evaluation period. HR officers and deans also have access to aggregated recommendation outputs at the department level, enabling structured faculty development planning. The Recommendation Engine is designed exclusively for improvement support and does not generate disciplinary recommendations or influence performance ratings directly.

Non-Functional Requirements

The system shall satisfy software quality standards consistent with the ISO/IEC 25010 software product quality model. With respect to security, all personally identifiable data stored in the system database shall be protected in accordance with applicable data protection standards. User passwords shall be stored exclusively as one-way cryptographic hashes using Bcrypt with per-user salting, ensuring that plaintext credentials cannot be recovered by any means. An

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email-based one-time password (OTP) shall be issued on first-time login and on detection of unrecognized device access to verify user identity. A Cloudflare web application firewall shall provide perimeter-level protection against distributed denial-of-service attacks, brute-force login attempts, and common web application exploits.

With respect to usability, the system interface shall be role-specific, presenting each user group with only the navigation elements and functions relevant to their responsibilities. The interface shall be fully responsive, rendering correctly on desktop computers, laptop computers, tablet devices, and smartphones without requiring client-side plugins or additional software installation.

With respect to reliability, the system shall maintain continuous availability throughout active evaluation windows. Automated backup procedures shall execute at scheduled intervals, and backup integrity shall be verified through restoration testing on a periodic basis.

With respect to performance, the system shall process evaluation form submissions and dashboard data requests within acceptable response times under peak concurrent load conditions. Database queries shall be optimized through indexing and connection pooling to ensure that performance does not degrade significantly during high-traffic evaluation periods.

Proposed System Architecture and Design

The proposed system architecture and design provide a structured framework for the development and implementation of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines. The design emphasizes modularity, scalability, security, and usability, ensuring that the system can efficiently support diverse user roles, including students, faculty members, supervisors, the Vice President for Academic Affairs (VPAA), Office of Student Affairs (OSA), Human Resource personnel, and system administrators. The architecture follows a layered approach, integrating the presentation, application, and data management layers, while incorporating AI-assisted components to enhance the analysis of qualitative feedback. This structured design facilitates clear separation of responsibilities, supports centralized data management, and ensures secure, transparent, and reliable processing of evaluation data. The proposed architecture and design serve as the foundation for iterative development, testing, and deployment, enabling the system to meet institutional requirements, regulatory standards, and user expectations while maintaining flexibility for future enhancements.

System Architecture Overview

The proposed Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines is built on a web-based client-server architecture designed to facilitate centralized management of evaluation data, secure system access, and scalable deployment within the National Aviation Academy of the Philippines (NAAP). This architecture allows multiple stakeholders including students, faculty members, supervisors, the Vice President for Academic Affairs (VPAA), Human Resource personnel, the Office of Student Affairs (OSA), and system administrators to access and interact with the system through web-based interfaces, while ensuring centralized processing, storage, and management of evaluation data.

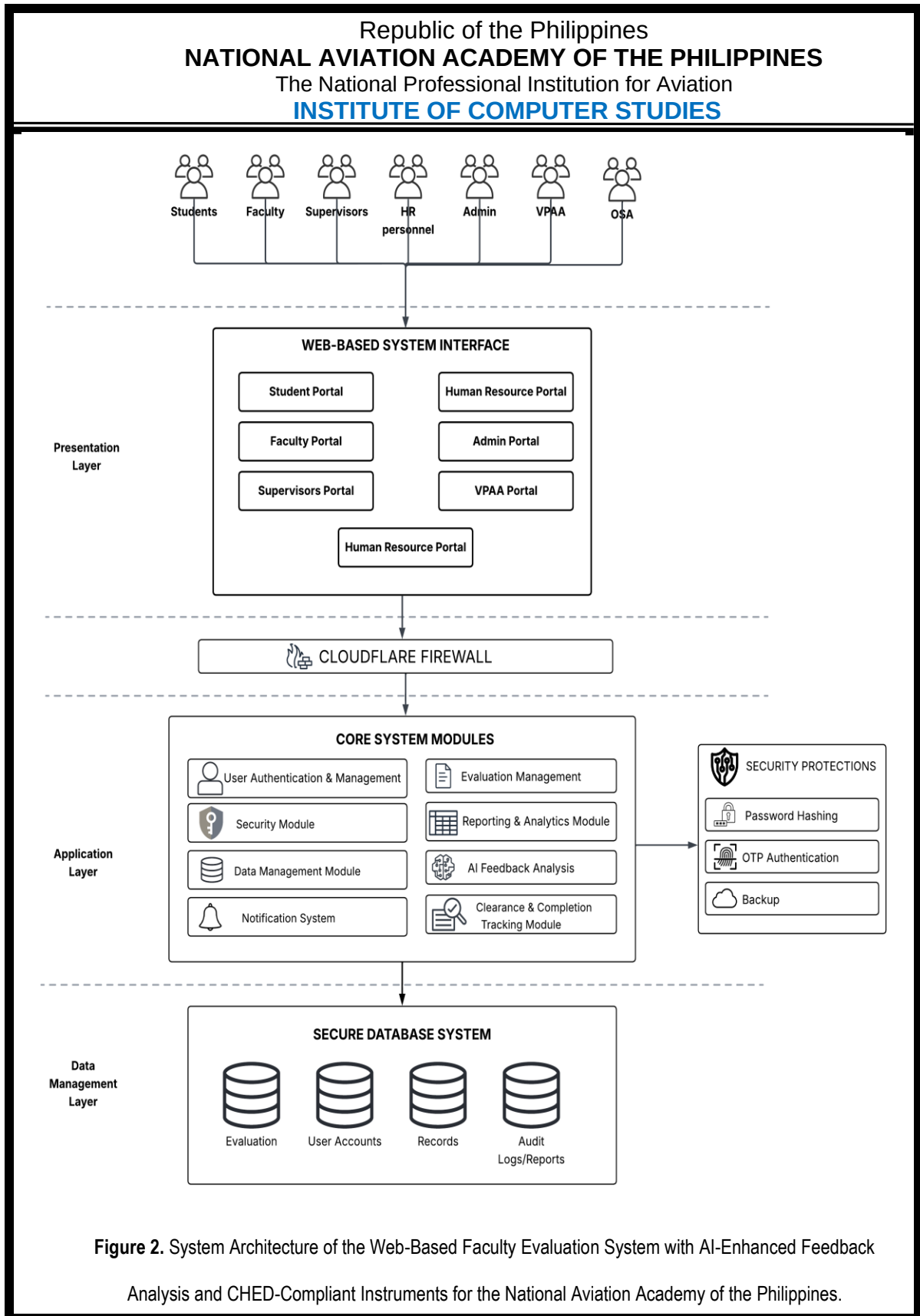


Figure 2. System Architecture of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines.

The system architecture is organized into three primary layers: the presentation layer, the application layer, and the data management layer. The presentation layer delivers web-based user interfaces that allow stakeholders—including students, faculty members, supervisors, the Vice President for Academic Affairs (VPAA), Human Resource personnel, the Office of Student Affairs (OSA), and system administrators to access evaluation forms, performance dashboards, administrative controls, and reporting tools via standard web browsers. The interface is designed to provide responsive interaction, ensuring accessibility across multiple devices while maintaining intuitive navigation and streamlined evaluation submission workflows.

The application layer functions as the system's core processing component, managing evaluation form processing, authentication and authorization, AI-assisted qualitative feedback analysis, report generation, and clearance verification functions. Role-based access control mechanisms are integrated within this layer to regulate system functionality according to user roles, ensuring data confidentiality and the integrity of evaluation processes. The AI module operates within the application layer, performing frequency-based text analysis and thematic summarization of qualitative evaluation responses. These analytical outputs support HR personnel in interpreting feedback patterns, facilitating informed decisions while preserving the authenticity of original evaluator responses.

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The data management layer provides centralized storage for evaluation records, user accounts, course assignments, and clearance verification data through a relational database management system. Centralized data storage ensures consistency, enables real-time reporting, and supports system scalability. Secure database access protocols and encrypted data transmission mechanisms protect sensitive evaluation information, ensuring compliance with institutional data privacy policies and national data protection regulations.

The client-server architecture allows efficient deployment and supports concurrent access during peak evaluation periods. The centralized server environment enables system administrators to configure evaluations, monitor system performance, and maintain security. Furthermore, the architecture is designed to support future system expansion and potential integration with institutional academic information systems, including enrollment and faculty assignment databases.

By combining web-based accessibility, AI-assisted analytical processing, and centralized data management, the proposed architecture delivers a scalable, secure, and efficient platform that enhances faculty evaluation management, streamlines administrative processes, and supports institutional quality assurance initiatives.

System Components and Modules

The Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments is structured into interconnected functional modules that collectively support secure, automated, and efficient evaluation processes at the National Aviation Academy of the Philippines (NAAP). The modular architecture promotes scalability, maintainability, and clear separation of system responsibilities.

At the center of the system is the Evaluation Management Module (Core Processing), which serves as the primary engine of the platform. This module manages the overall evaluation workflow, including the distribution of CHED-compliant evaluation forms, collection of responses from students, faculty, and supervisors, validation of submitted evaluations, and coordination with other modules for processing and storage. It ensures that all evaluation activities follow standardized procedures, maintaining consistency and reliability of collected data.

Connected to the core module is the Enrollment Validation Module, which verifies whether a student is officially enrolled under a specific instructor before allowing evaluation submission. This ensures that only authorized evaluators can participate, preventing invalid, duplicate, or unauthorized responses and strengthening data integrity.

The Authentication and Access Control Module works alongside the core system to regulate user access. It authenticates users through secure login

mechanisms and enforces role-based permissions for students, faculty, supervisors, and administrators. This ensures that users can only access functionalities relevant to their roles, protecting sensitive evaluation data.

The AI Feedback Analysis Module is integrated with the core processing module to enhance the interpretation of qualitative feedback. It performs automated text analysis using frequency analysis and AI-based summarization techniques, enabling Human Resource personnel and administrators to quickly identify recurring themes and generate meaningful insights from large volumes of comments.

On the administrative side, the System Administration Module supports the management and configuration of the system. It handles user account administration, evaluation setup, monitoring of system activity, and maintenance of evaluation instruments, ensuring smooth and reliable system operation.

The Reporting and Dashboard Module processes evaluation results and transforms them into visual and analytical outputs. It provides dashboards, reports, and performance summaries that allow administrators, supervisors, and decision-makers to monitor faculty performance, identify trends, and support data-driven decision-making.

The Notification Module facilitates system communication by sending alerts, reminders, and updates related to evaluation activities, deadlines, and

system actions. This ensures timely participation and improves overall system responsiveness.

The Clearance Monitoring and Tracking Module manages the verification of evaluation completion. It tracks whether students have fulfilled their evaluation requirements and generates clearance references necessary for academic processing. This module automates what was previously a manual process, reducing administrative workload and improving efficiency.

Finally, the Security Module ensures the protection of system data and user information. It implements essential security mechanisms such as OTP authentication and encryption handling, safeguarding sensitive data and ensuring compliance with data privacy standards.

Overall, the interaction of these modules creates a cohesive and efficient system where the Evaluation Management Module coordinates all operations, while supporting modules handle validation, security, analysis, reporting, and administration. This modular design enhances system reliability, simplifies maintenance, and allows future expansion of functionalities.

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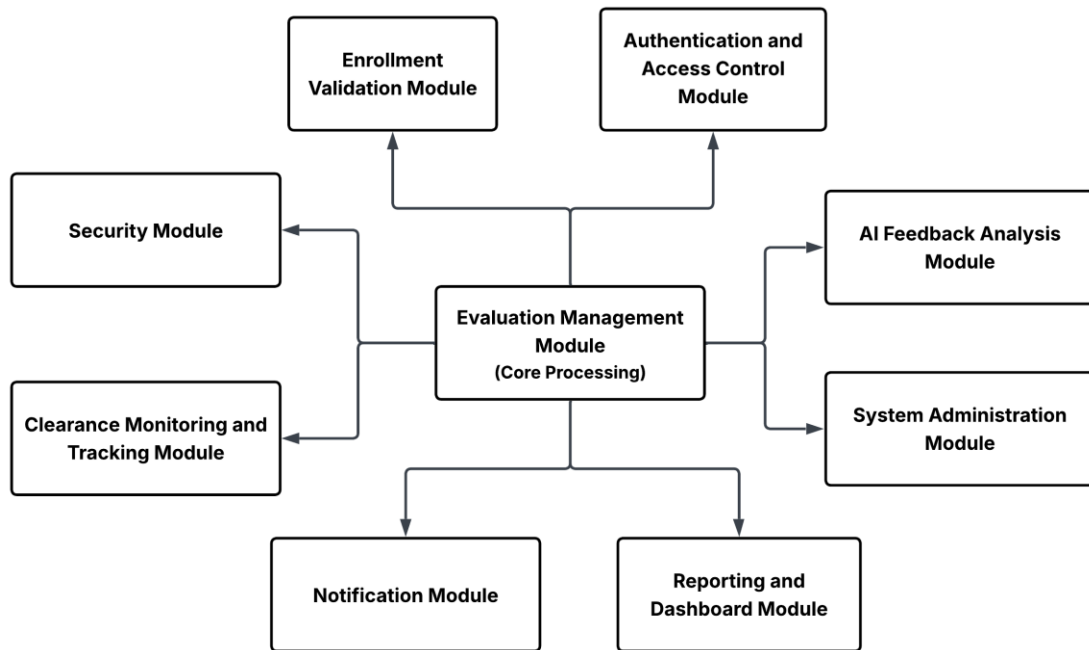


Figure 3. System Components and Modules of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines.

Figure 3 illustrates the modular architecture of the proposed Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines. At the core of the system is the Evaluation Management Module (Core Processing), which manages the overall evaluation workflow, including form distribution, response collection, validation, and coordination with other modules. Supporting modules include the Enrollment Validation Module, which verifies student–instructor relationships to ensure authorized evaluation participation; the

Authentication and Access Control Module, which enforces secure login and role-based access; and the Security Module, which implements OTP authentication and encryption mechanisms to protect sensitive data.

The system also integrates the AI Feedback Analysis Module, which processes qualitative responses using frequency analysis and AI-based summarization to generate meaningful insights. The Reporting and Dashboard Module provides visual summaries, performance analytics, and evaluation reports to support data-driven decision-making. The Clearance Monitoring and Tracking Module automates the verification of evaluation completion for student clearance, improving administrative efficiency. Additionally, the Notification Module facilitates communication through system alerts and reminders, while the System Administration Module manages user accounts, system configurations, and overall platform maintenance.

This modular architecture ensures efficient interaction among system components, enhances maintainability, and supports scalability for future improvements and expansion of system functionalities.

Process Modeling and System Workflow

Process modeling and system workflow provide a visual and structured representation of how the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments operates within

the National Aviation Academy of the Philippines (NAAP). This section illustrates the flow of information, interactions among stakeholders, and the sequence of operations required to conduct faculty evaluations efficiently and securely. Key components of the process modeling include the Context Diagram, which provides a high-level overview of the system and its interactions with external entities such as students, faculty, supervisors, the VPAA, the Office of Student Affairs (OSA), Human Resource personnel and system administrators. The Data Flow Diagrams (Level 0 and Level 1) further detail the movement of data between system modules, including evaluation form processing, AI-assisted feedback analysis, reporting, and clearance verification. Finally, the Procedural Workflow depicts step-by-step operational procedures, demonstrating how evaluation tasks are initiated, processed, monitored, and completed. Together, these modeling tools enable a clear understanding of system functionality, support effective system design and implementation, and provide a foundation for testing, validation, and future enhancements.

Context Diagram

The context diagram presents a high-level view of the Web-Based Student and Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments for the National Aviation Academy of the Philippines (NAAP). It illustrates the system as a centralized platform that interacts with

multiple stakeholders by exchanging evaluation data, reports, and system-related information.

Students interact with the system by receiving evaluation questionnaires, submitting evaluation responses, and monitoring their evaluation completion status. The system, in turn, provides completion tracking to ensure that students meet evaluation requirements necessary for clearance processing.

Faculty members engage with the system by submitting peer evaluations and receiving evaluation results, which allow them to review feedback on their teaching performance and professional interactions.

Supervisors and Deans participate by submitting formal evaluation assessments of faculty members. They also receive evaluation results generated by the system, enabling them to monitor faculty performance and support academic decision-making.

The Vice President for Academic Affairs (VPAA) accesses the system to review consolidated evaluation results, which serve as a basis for institutional planning, faculty assessment, and policy development.

Human Resource (HR) personnel interact with the system by receiving evaluation results and audit trails. These outputs support performance evaluation, employee monitoring, and documentation for faculty development and administrative review.

The Office of Student Affairs (OSA) utilizes the system to verify student evaluation completion status, which is required for clearance processing. This eliminates the need for manual verification and improves operational efficiency.

System Administrators (Admin) manage user accounts and receive audit trails from the system. They are responsible for maintaining system integrity, monitoring activities, and ensuring proper system configuration and operation.

Overall, the context diagram highlights the flow of data between external stakeholders and the evaluation system, including evaluation submissions, results, audit logs, and completion status. It defines the system boundary while abstracting internal processes, providing a clear overview of how different users interact with the system and how information is exchanged to support evaluation, monitoring, and administrative functions.

Figure 4 illustrates the context diagram showing the interaction between the evaluation system and external stakeholders.

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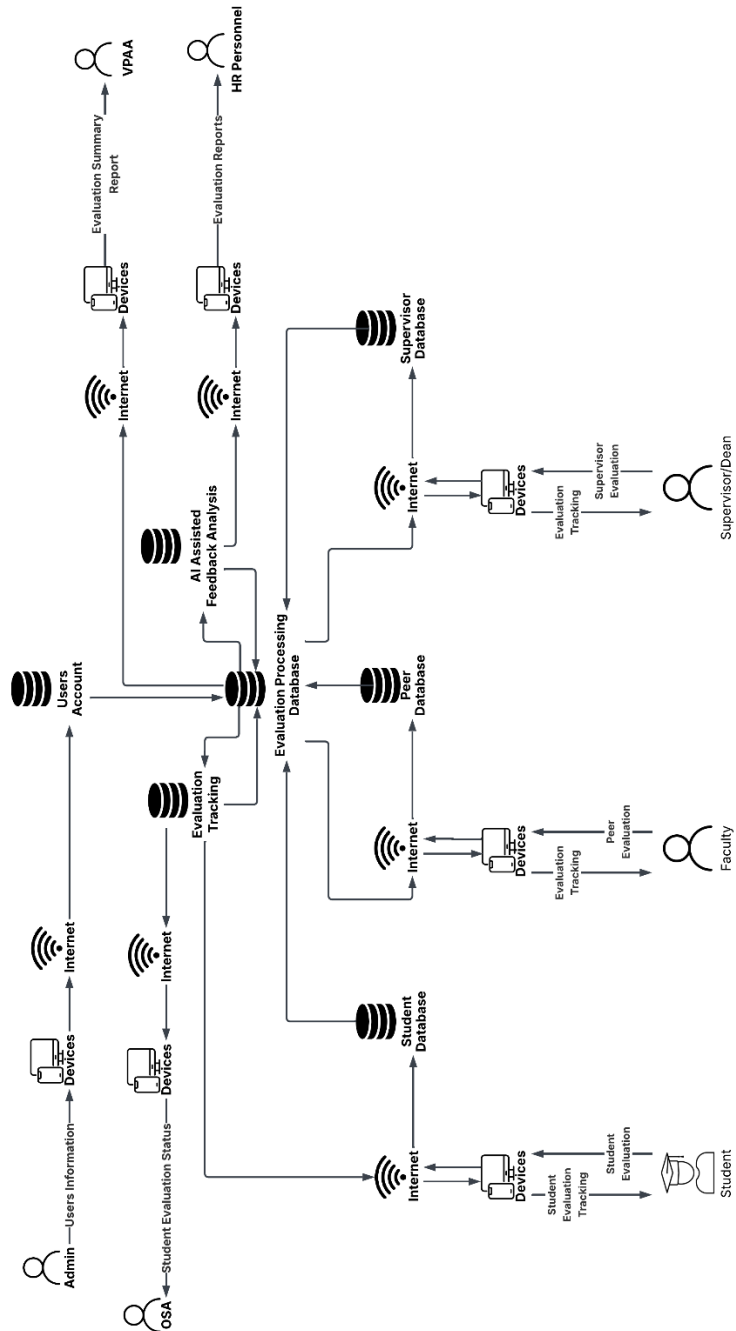


Figure 4. Context Diagram Showing External Stakeholder Interactions with the Web-Based Faculty Evaluation System

Figure 4 illustrates the context diagram of the proposed evaluation system. The diagram represents the system as a centralized process interacting with external stakeholders, including students, faculty, supervisors, the VPAA, the Office of Student Affairs (OSA), Human Resource personnel and system administrators. The diagram highlights the exchange of evaluation data, performance feedback, clearance verification information, and administrative control data between stakeholders and the system, establishing the system boundary and interaction scope.

Data Flow Diagram (Level 0)

The Level 0 Data Flow Diagram (DFD) illustrates the overall data flow of the Web-Based Faculty Evaluation System with AI-Assisted Feedback Analysis and CHED-Compliant Instruments at the National Aviation Academy of the Philippines (NAAP). In this diagram, the system is represented as a single central process (Process 0), which interacts with multiple external entities through defined data flows.

Students interact with the system by providing login credentials to gain access, accessing evaluation schedules, and completing student evaluation questionnaires. The system processes these inputs and generates student

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evaluation tracking information and AI-assisted evaluation results, allowing students to monitor their evaluation activities.

Faculty members participate in the system through peer evaluation and access evaluation schedules, evaluation tracking, and AI-assisted evaluation results. This enables them to review feedback and contribute to the evaluation of their colleagues.

Supervisors/Deans interact with the system by submitting supervisor evaluations, accessing evaluation schedules, and monitoring evaluation tracking. These functions support formal faculty assessment and oversight responsibilities.

Human Resource (HR) personnel access the system using login credentials and manage evaluation-related activities such as accessing evaluation questionnaires, monitoring evaluation schedules, and reviewing AI-assisted evaluation results. This allows HR to support personnel evaluation and performance monitoring.

The Office of Student Affairs (OSA) interacts with the system by accessing student evaluation status, evaluation schedules, and AI-assisted evaluation results. These data are used to verify student compliance and support administrative processes such as clearance.

The Vice President for Academic Affairs (VPAA) accesses the system through login credentials and receives evaluation summary reports, evaluation

schedules, and tracking information. These outputs support institutional planning, policy-making, and academic decision-making.

System Administrators (Admin) interact with the system by managing user information, configuring evaluation schedules, and maintaining evaluation tracking processes. They ensure proper system operation, access control, and overall functionality.

Overall, the diagram shows how data flows such as login credentials, evaluation questionnaires (student, peer, and supervisor), evaluation schedules, tracking information, AI-assisted evaluation results, user information, and summary reports are exchanged between external entities and the central system. The Level 0 DFD defines the system boundary while presenting a clear overview of stakeholder interaction without detailing internal processes. This highlights the system's role as a centralized platform for evaluation management, data processing, and information dissemination within the institution.

Figure 5 illustrates the Level 0 Data Flow Diagram showing the major system processes and data flow relationships.

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graph TD
    Student[Student] -- "Login Credentials" --> P0
    Student -- "Access" --> P0
    Student -- "Student Evaluation Tracking" --> P0
    Student -- "Student Evaluation Questionnaires" --> P0
    P0 -- "Student Evaluation" --> Student
    Faculty[Faculty] -- "Login Credentials" --> P0
    Faculty -- "Access" --> P0
    Faculty -- "Evaluation Schedule" --> P0
    Faculty -- "Peer Evaluation" --> P0
    P0 -- "AI Assisted Evaluation Results" --> Faculty
    P0 -- "Evaluation Tracking" --> Faculty
    OSA[OSA] -- "Login Credentials" --> P0
    OSA -- "Access" --> P0
    OSA -- "Student Evaluation Status" --> P0
    P0 -- "Evaluation Schedule" --> OSA
    Supervisor[Supervisor/Dean] -- "Login Credentials" --> P0
    Supervisor -- "Access" --> P0
    Supervisor -- "Supervisor Evaluation" --> P0
    P0 -- "AI Assisted Evaluation Results" --> Supervisor
    P0 -- "Evaluation Schedule" --> Supervisor
    P0 -- "Evaluation Tracking" --> Supervisor
    HR[HR] -- "Access" --> P0
    HR -- "Supervisor Evaluation Questionnaires" --> P0
    HR -- "Evaluation Schedule" --> P0
    P0 -- "AI Assisted Evaluation Results" --> HR
    P0 -- "Evaluation Tracking" --> HR
    VPAA[VPAA] -- "Access" --> P0
    VPAA -- "Evaluation Schedule" --> P0
    VPAA -- "Login Credentials" --> P0
    P0 -- "Evaluation Tracking" --> VPAA
    P0 -- "Evaluation Summary Reports" --> VPAA
    Admin[Admin] -- "Users Information" --> P0
    Admin -- "Access" --> P0
    Admin -- "Evaluation Schedule" --> P0
    Admin -- "Login Credentials" --> P0
    Admin -- "Evaluation Tracking" --> P0
    
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Figure 5. DFD level 0 diagram Showing External Stakeholder Interactions with the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments.

Figure 5 illustrates the Level 0 Data Flow Diagram of the proposed evaluation system. The diagram represents the system as a single centralized process (Process 0) interacting with external stakeholders, including students, faculty, supervisors/deans, the Vice President for Academic Affairs (VPAA), the Office of Student Affairs (OSA), Human Resource (HR) personnel, and system administrators. It highlights the exchange of key data flows such as login

Figure 5. DFD level 0 diagram Showing External Stakeholder Interactions with the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments.

Figure 5 illustrates the Level 0 Data Flow Diagram of the proposed evaluation system. The diagram represents the system as a single centralized process (Process 0) interacting with external stakeholders, including students, faculty, supervisors/deans, the Vice President for Academic Affairs (VPAA), the Office of Student Affairs (OSA), Human Resource (HR) personnel, and system administrators. It highlights the exchange of key data flows such as login

credentials, evaluation submissions (student and supervisor submissions), evaluation schedules and tracking information, evaluation results, student evaluation status, and evaluation summary reports. The diagram establishes the system boundary and clearly defines how stakeholders interact with the system for evaluation processing, monitoring, reporting, and administrative control.

Data Flow Diagram (Level 1)

The Level 1 Data Flow Diagram (DFD) presents a detailed decomposition of the central process shown in the Level 0 DFD of the Web-Based Faculty Evaluation System with AI-Assisted Feedback Analysis and CHED-Compliant Instruments at the National Aviation Academy of the Philippines (NAAP). The system is broken down into eight major processes, namely: (1.0) User Authentication and Access, (2.0) Evaluation Schedule Management, (3.0) Student Evaluation Processing, (4.0) Peer Evaluation Processing, (5.0) Supervisor/Dean Evaluation Processing, (6.0) AI-Assisted Evaluation Results Generation, (7.0) Evaluation Tracking, and (8.0) User Management. These processes interact with data stores such as Evaluation Schedule, Student Evaluation, Peer Evaluation, Supervisor Evaluation, and Users Account, as well as external entities including students, faculty, supervisors/deans, HR personnel, VPAA, OSA, and administrators.

The User Authentication and Access Process (1.0) handles login credentials submitted by all external entities. The system verifies user information from the Users Account data store and grants access privileges based on user roles, ensuring secure and authorized system usage.

The Evaluation Schedule Management Process (2.0) manages the creation, configuration, and distribution of evaluation schedules. Administrators and HR personnel configure evaluation schedules, which are stored in the Evaluation Schedule data store. These schedules are then accessed by students, faculty, supervisors/deans, VPAA, and other authorized users to guide evaluation activities.

The Student Evaluation Processing (3.0), Peer Evaluation Processing (4.0), and Supervisor/Dean Evaluation Processing (5.0) handle the submission and storage of evaluation data. Students submit student evaluation questionnaires, faculty members perform peer evaluations, and supervisors/deans submit their respective evaluations. These inputs are stored in their corresponding data stores: Student Evaluation, Peer Evaluation, and Supervisor Evaluation.

The AI-Assisted Evaluation Results Generation Process (6.0) processes the collected evaluation data from all sources. It performs statistical analysis and AI-assisted summarization to generate evaluation results. These results are made available to faculty members, supervisors/deans, and HR personnel, supporting performance assessment and feedback interpretation.

The Evaluation Tracking Process (7.0) monitors the completion and status of evaluation activities. It generates evaluation tracking information accessible to students, faculty, supervisors/deans, VPAA, and administrators. Additionally, it produces student evaluation status, which is specifically accessed by the Office of Student Affairs (OSA) for clearance verification purposes.

The User Management Process (8.0) handles the maintenance of user accounts and user information. Administrators manage user data, ensuring proper role assignment, access control, and system integrity.

Overall, the Level 1 DFD illustrates how data flows such as login credentials, evaluation schedules, student/peer/supervisor evaluation questionnaires, AI-assisted evaluation results, evaluation tracking information, student evaluation status, and user information are processed within the system. The interaction between processes, data stores, and external entities demonstrates a structured and efficient workflow that supports secure access, accurate evaluation processing, effective monitoring, and data-driven decision-making across the institution.

Figure 6 to 12 illustrates the Level 1 Data Flow Diagram showing detailed system sub processes and internal data flow relationships.

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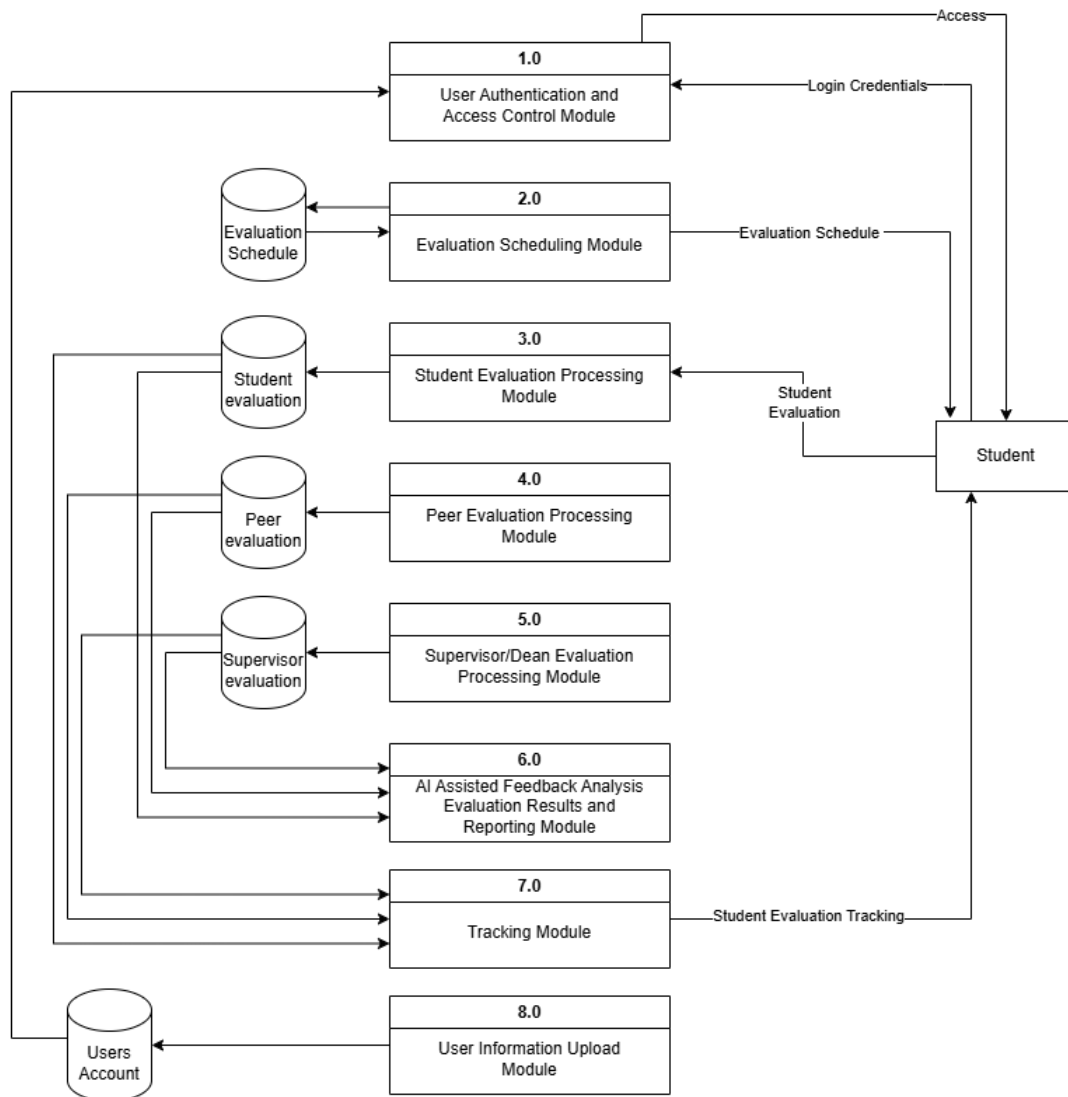


Figure 6. Level 1 Data Flow Diagram for Student of the Web-Based Faculty Evaluation System

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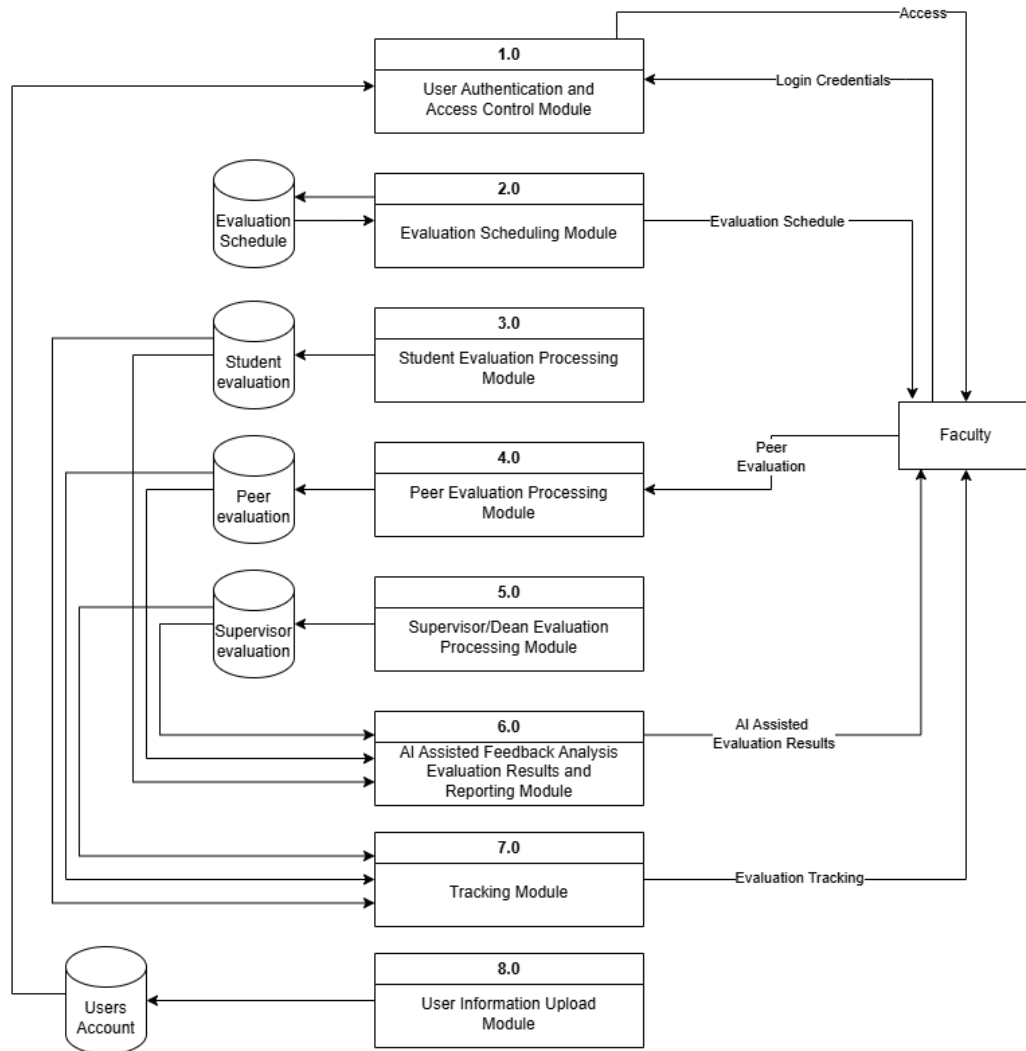


Figure 7. Level 1 Data Flow Diagram for Faculty of the Web-Based Faculty Evaluation System

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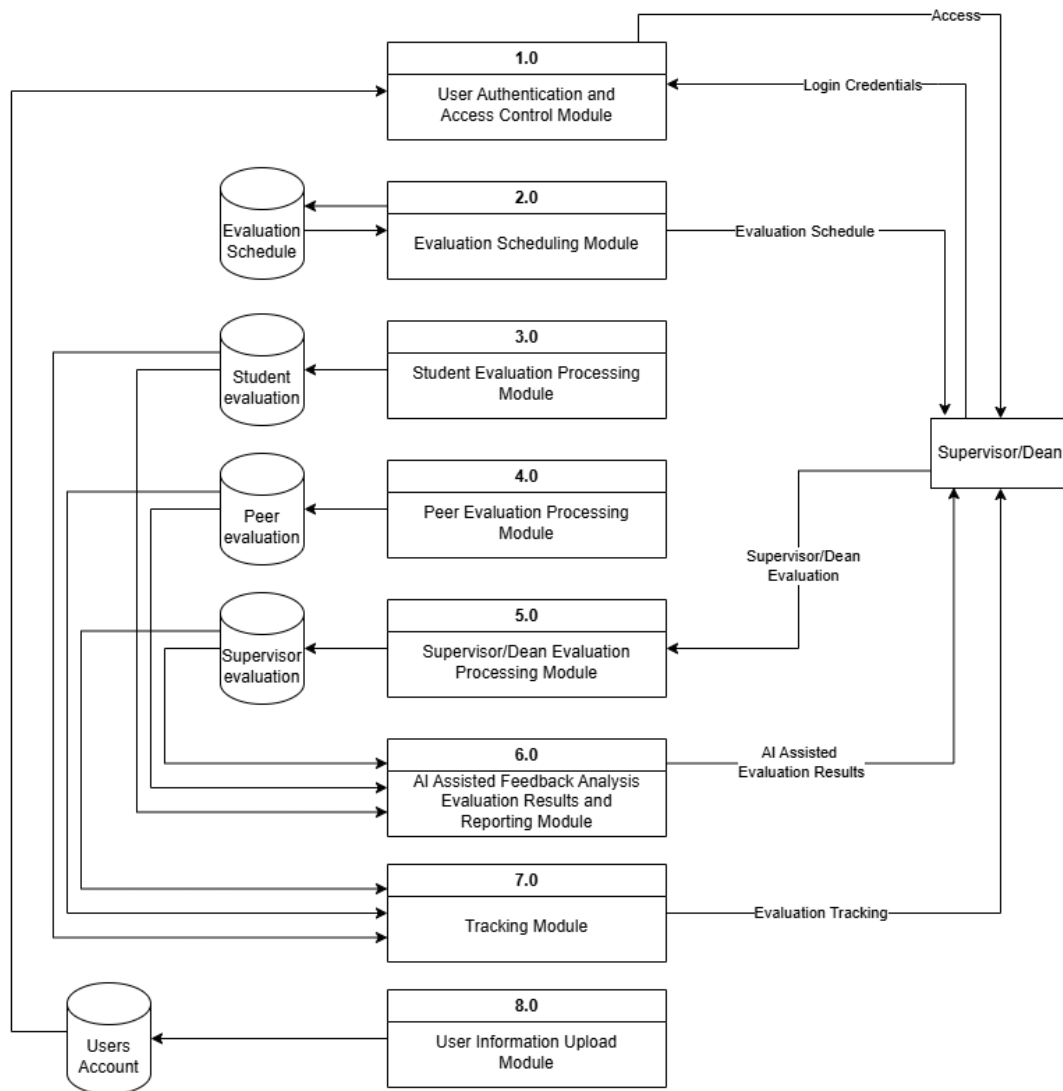


Figure 8. Level 1 Data Flow Diagram for Supervisor/Dean of the Web-Based Faculty Evaluation System

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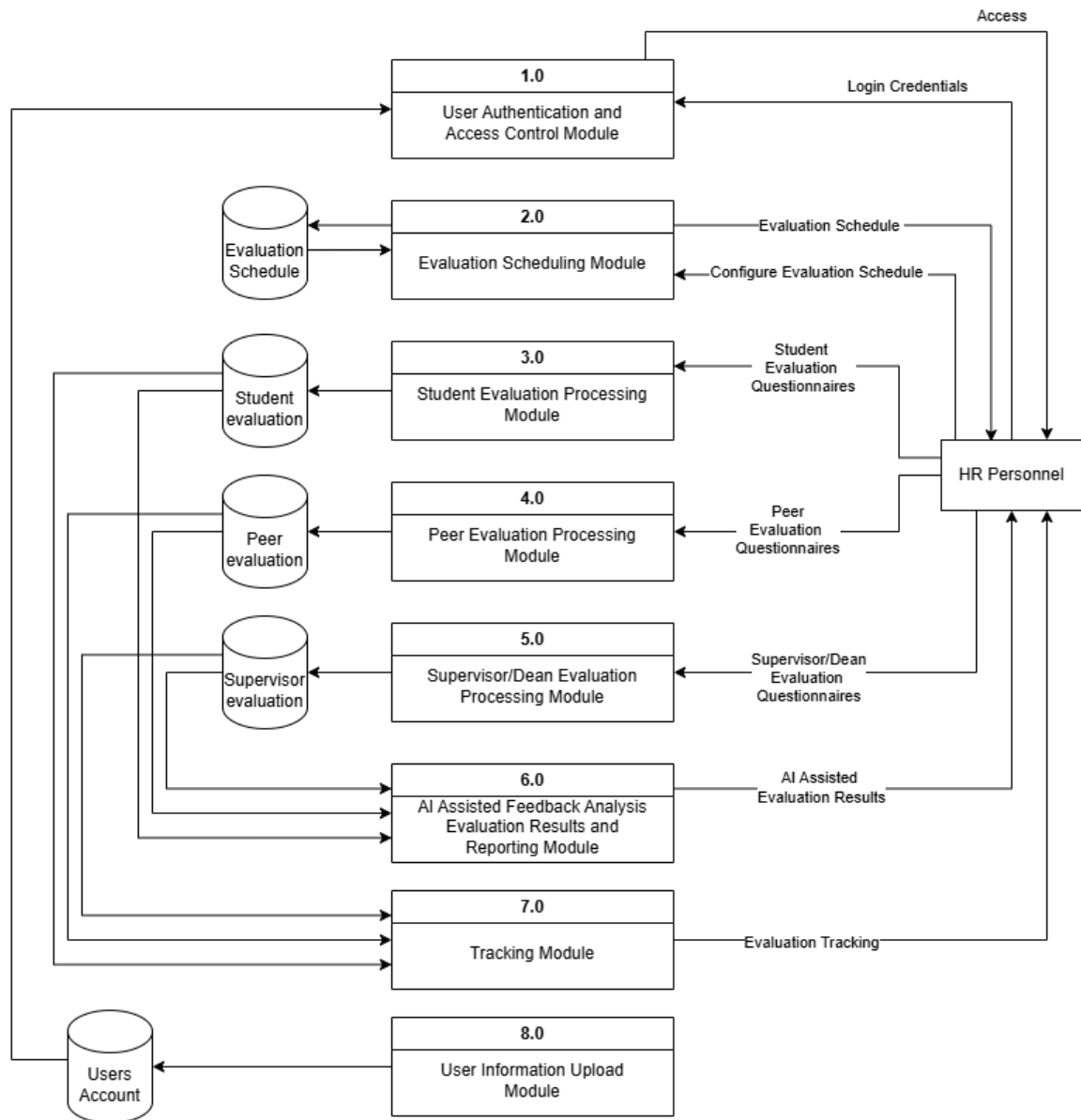


Figure 9. Level 1 Data Flow Diagram for HR personnel of the Web-Based Faculty Evaluation System

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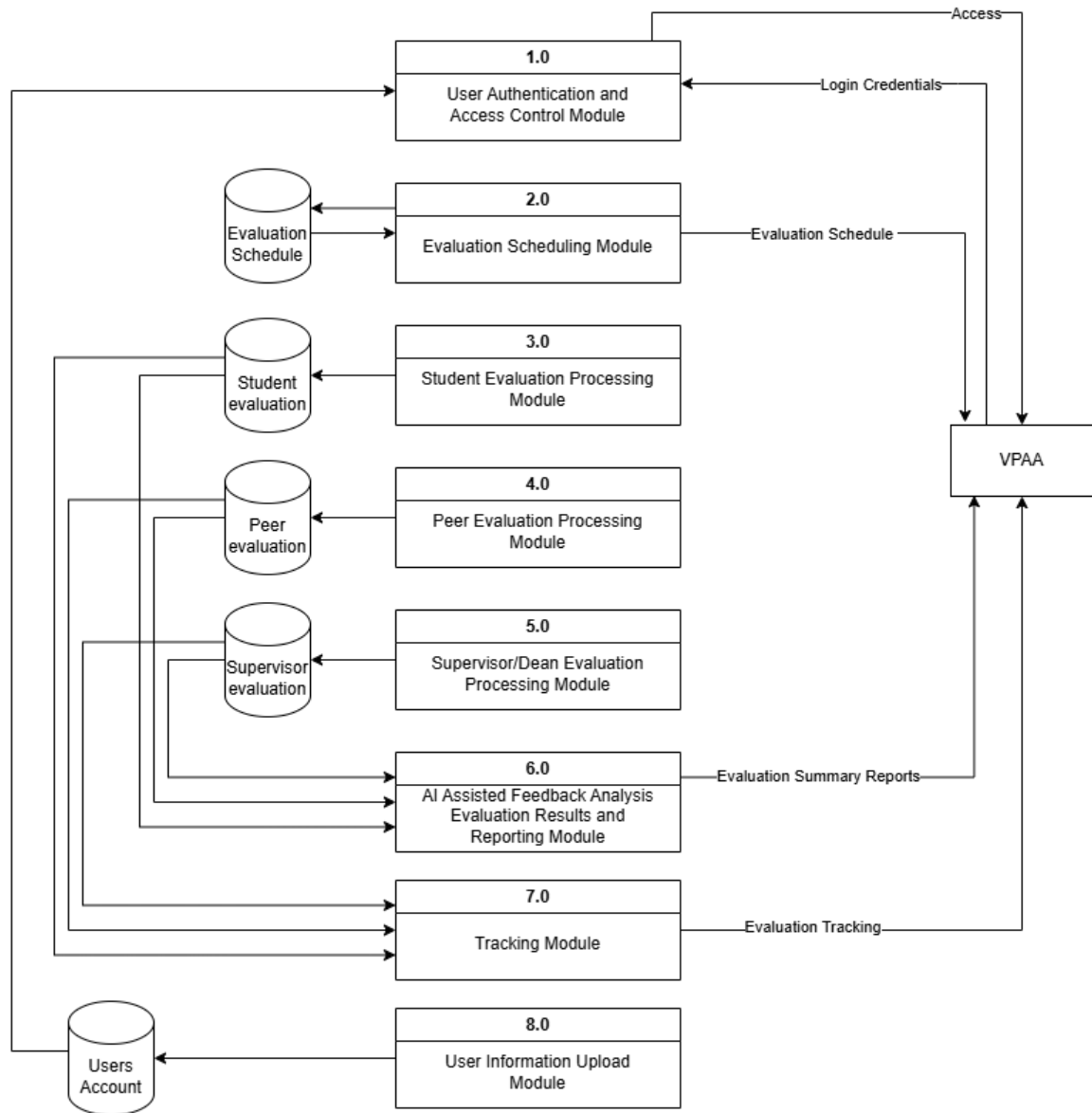


Figure 10. Level 1 Data Flow Diagram for VPAA of the Web-Based Faculty Evaluation System

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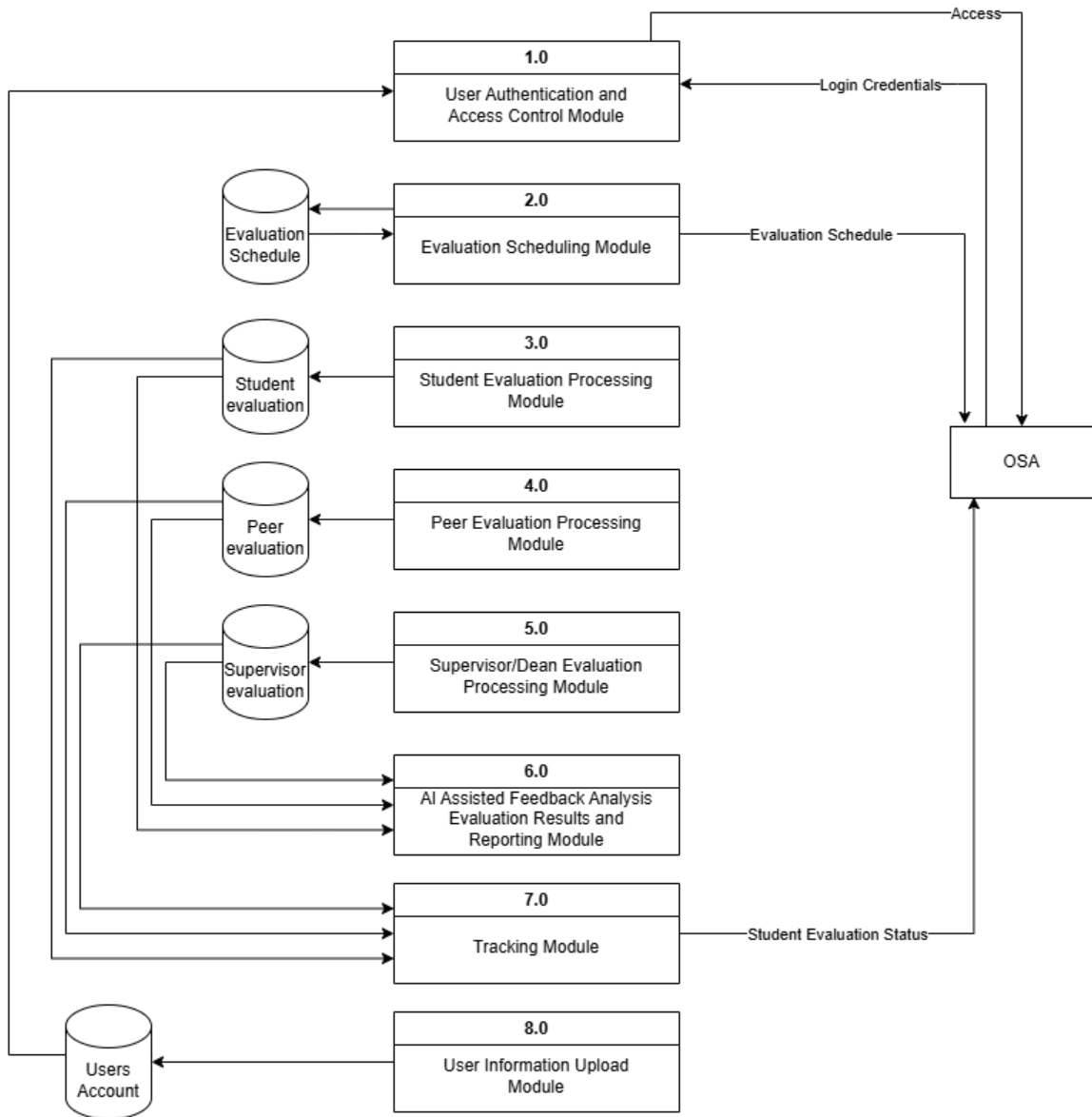


Figure 11. Level 1 Data Flow Diagram for OSA of the Web-Based Faculty Evaluation System

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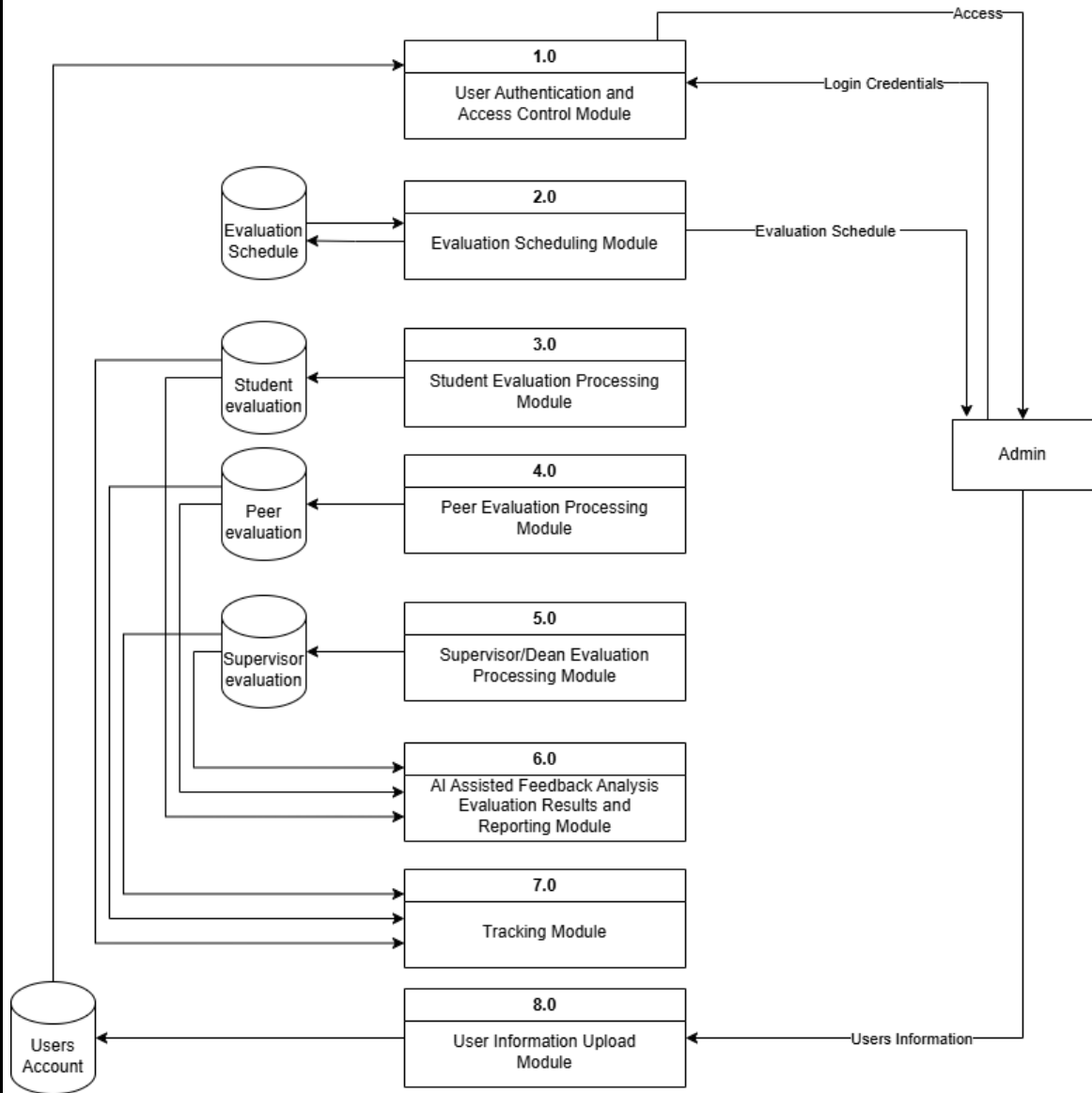


Figure 12. Level 1 Data Flow Diagram for Admin of the Web-Based Faculty Evaluation System

Procedural Workflow

The procedural workflow illustrates the sequence of user interactions and system processes in the operation of the Web-Based Faculty Evaluation System with AI-Assisted Feedback Analysis and CHED-Compliant Instruments at the National Aviation Academy of the Philippines (NAAP). It presents how different stakeholders interact with the system from login authentication up to evaluation submission, processing, reporting, and system updates.

The workflow begins when a user accesses the system and proceeds to login authentication. The system performs credential verification and assigns role-based access depending on the user type, which may include student, faculty, supervisor/dean, HR personnel, VPAA, OSA, or administrator. This role-based mechanism ensures secure access and directs users to their respective dashboards.

Once authenticated, users are redirected to their designated dashboards. Students access the student dashboard, where they view assigned evaluation tasks, complete evaluation forms, and submit their responses. The system then stores the submitted data in the database, updates the dashboard, and records completion for clearance verification and system updates.

Faculty members access the faculty dashboard to perform peer evaluations. They view assigned peer evaluation tasks, complete evaluation forms, and submit responses. The system stores the data, performs evaluation

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processing and AI-assisted analysis, generates reports, updates the dashboard, and executes system updates.

Supervisors/Deans access their dashboard to manage and conduct supervisor evaluations. They create assignment rooms for peer evaluation, distribute evaluation tasks manually or automatically, and complete both peer and supervisor evaluation forms. Submitted data are stored in the database and processed through evaluation analysis and AI-assisted summarization, followed by report generation, dashboard updates, and system updates.

HR personnel access their dashboard to monitor and manage evaluation processes. They display trend charts, set evaluation periods, and configure CHED-compliant questionnaires. HR personnel can also view professors' evaluation results and AI-assisted feedback summaries, supporting performance monitoring and administrative decision-making.

The VPAA accesses the VPAA dashboard to view trend charts and professors' evaluation results, which provide insights for institutional planning and academic decision-making.

The Office of Student Affairs (OSA) accesses the system to monitor student evaluation completion through the student evaluation monitoring feature. This enables verification of student compliance for clearance purposes.

Administrators access the admin dashboard to manage system configuration and monitor audit logs. They also handle user and system management functions to ensure system integrity, security, and proper operation.

Throughout the workflow, the system performs internal processes such as data storage, validation, evaluation processing, AI-assisted feedback analysis, report generation, and dashboard updates. These processes ensure that collected data are transformed into accurate, meaningful, and actionable information.

Overall, the procedural workflow demonstrates a structured and role-based process that integrates authentication, evaluation submission, AI-assisted analysis, reporting, monitoring, and system management. This ensures an efficient, secure, and centralized evaluation system that supports data-driven decision-making and enhances faculty evaluation processes within the institution.

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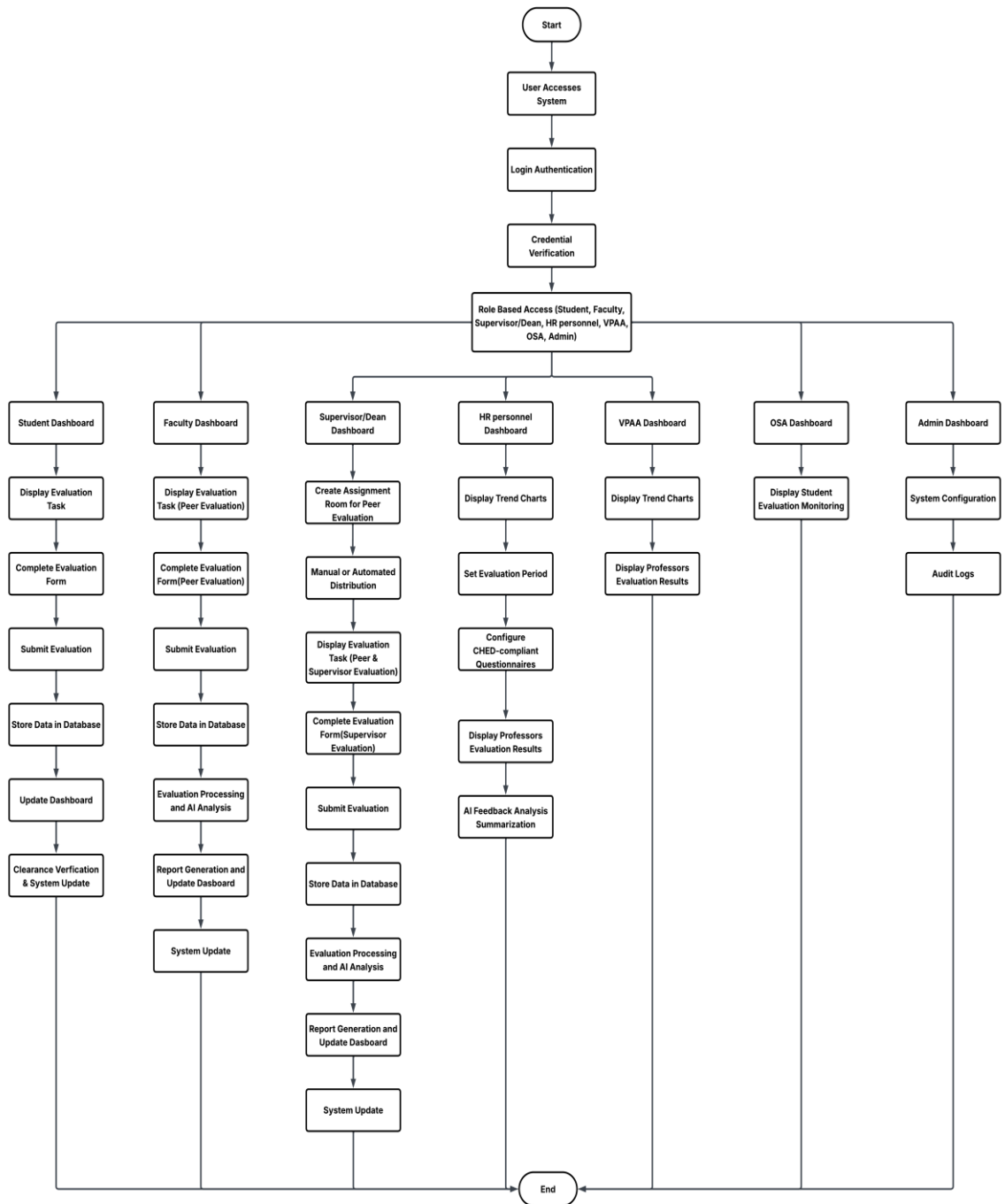


Figure 13. Procedural Workflow Diagram of the Web-Based Faculty Evaluation System

Data Modeling and Database Design

The Data Modeling and Database Design section outlines the structured approach used to organize, represent, and implement the data requirements of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments at the National Aviation Academy of the Philippines (NAAP). This section begins with the Conceptual Data Model, which identifies the key entities, their attributes, and the relationships between them, providing a high-level view of the system's data requirements. It is followed by the Entity-Relationship Diagram (ERD), which visually represents the entities, relationships, and cardinalities to clarify how data interacts within the system. Finally, the Logical Database Design translates these conceptual and relational models into a structured database schema, detailing tables, primary and foreign keys to support efficient storage, retrieval, and management of evaluation data. Together, these models provide a foundation for building a secure, consistent, and scalable database that supports the system's operational and analytical requirements.

Conceptual Data Model

The conceptual data model defines the high-level data structure of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments at the National Aviation Academy of the Philippines

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(NAAP) by identifying key entities and their relationships within the evaluation management environment. This model emphasizes essential evaluation data components without detailing technical database implementation. The Users entity represents all system stakeholders, including students, faculty members, supervisors, Human Resource personnel, and administrators. Each user is assigned specific roles and access privileges that determine their evaluation responsibilities and interactions with the system. The Evaluation Forms entity represents standardized CHED-compliant evaluation instruments linked to specific courses and faculty members. These forms include structured evaluation criteria and rating scales used to assess teaching performance, professional behavior, and collaboration effectiveness. The Evaluation Responses entity captures evaluation data submitted by authorized users. It stores both quantitative ratings and qualitative feedback, linking each response to the evaluator, the evaluated faculty member, and the corresponding evaluation form to ensure traceability and data integrity. The Evaluation Results entity represents processed outcomes generated by aggregating evaluation responses. It stores consolidated scores, AI-generated summaries of qualitative feedback, and performance trend analyses, which supervisors and Human Resource personnel use for faculty monitoring and development planning. The Clearance Records entity tracks students' evaluation completion status and generates system-issued verification reference numbers for

clearance processing. This ensures compliance with institutional evaluation requirements and supports automated clearance validation.

Relationships among entities support structured data processing: users submit evaluation responses via evaluation forms, which are processed to produce evaluation results. Clearance records are generated based on evaluation completion, linking student compliance to institutional clearance procedures.

Overall, the conceptual data model organizes evaluation information into structured entities and logical relationships, promoting data consistency, evaluation transparency, and effective performance monitoring. It forms the foundation for the subsequent logical and physical database design stages.

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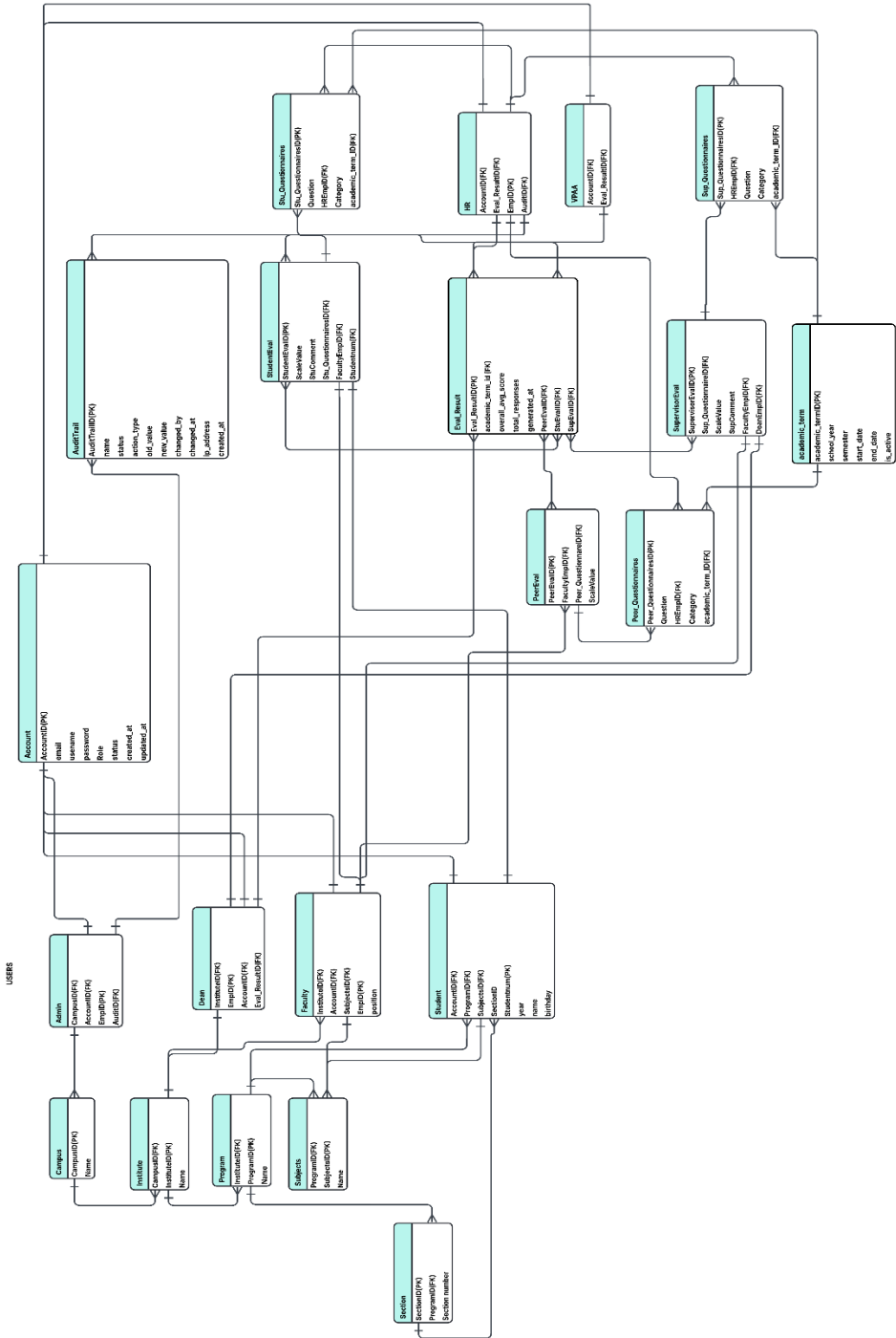


Figure 14. Conceptual Data Model of the Web-Based Faculty Evaluation System

Figure 14 illustrates the Conceptual Data Model of the Web-Based Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments of the National Aviation Academy of the Philippines (NAAP). The diagram presents the major data entities of the system and shows how these entities are logically related to support the faculty evaluation process.

Entity Relationship Diagram

The Entity Relationship Diagram (ERD) illustrates the logical structure and relationships among the major data entities of the Web-Based Faculty Evaluation System with AI-Assisted Feedback Analysis and CHED-Compliant Instruments at the National Aviation Academy of the Philippines (NAAP). It defines how data is organized, stored, and interconnected to support accurate, secure, and efficient evaluation processing.

At the core of the system is the Account entity, which represents all system users. It stores essential user credentials such as email, username, password, role, status, and timestamps. The Account entity enables role-based access and is associated with role-specific entities including Admin, Dean, Faculty, Student, HR, and VPAA, allowing the system to manage user identities and responsibilities. System activities are further monitored through the AuditTrail entity, which records actions such as changes made, timestamps, and user-related activity logs.

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The academic structure of the institution is defined through the entities Campus, Institute, Program, Subjects, and Section. These entities are hierarchically related, where a campus contains institutes, institutes contain programs, and programs are associated with subjects and sections. The Student and Faculty entities are linked to these structures through foreign keys, ensuring that users are properly assigned to their respective academic units, subjects, and sections.

The evaluation process is represented by three main entities: StudentEval, PeerEval, and SupervisorEval. These entities store evaluation data submitted by students, faculty (peer evaluation), and supervisors/deans, respectively. Each evaluation record contains scale values and is linked to both the evaluator and the evaluated faculty member. These entities are also associated with their respective questionnaire entities: Stu_Questionnaires, Peer_Questionnaires, and Sup_Questionnaires, which define the standardized CHED-compliant questions, categories, and academic term context used during evaluation.

All evaluation data are consolidated in the Eval_Result entity, which stores computed results such as overall average scores, total responses, and evaluation dates. This entity integrates results from student, peer, and supervisor evaluations, providing a comprehensive assessment of faculty performance.

These results are accessed by HR and VPAA entities for monitoring, reporting, and decision-making purposes.

The `academic_term` entity defines the timeframe of evaluations, including school year, semester, start and end dates, and active status. All questionnaire and evaluation-related entities are linked to this entity to ensure that evaluation data are properly organized and managed within a specific academic period.

Additionally, HR personnel are responsible for configuring evaluation questionnaires, as shown by their relationship with the questionnaire entities. The Admin entity manages user accounts and system control, ensuring proper configuration, maintenance, and access management across the system.

Overall, the ERD enforces data integrity through well-defined primary and foreign key relationships among entities. These relationships ensure that evaluation data are consistently linked to valid users, academic structures, questionnaires, and academic terms. This structured design supports efficient data storage, accurate processing, reliable reporting, and comprehensive evaluation analysis, enabling the system to effectively support institutional quality assurance and decision-making.

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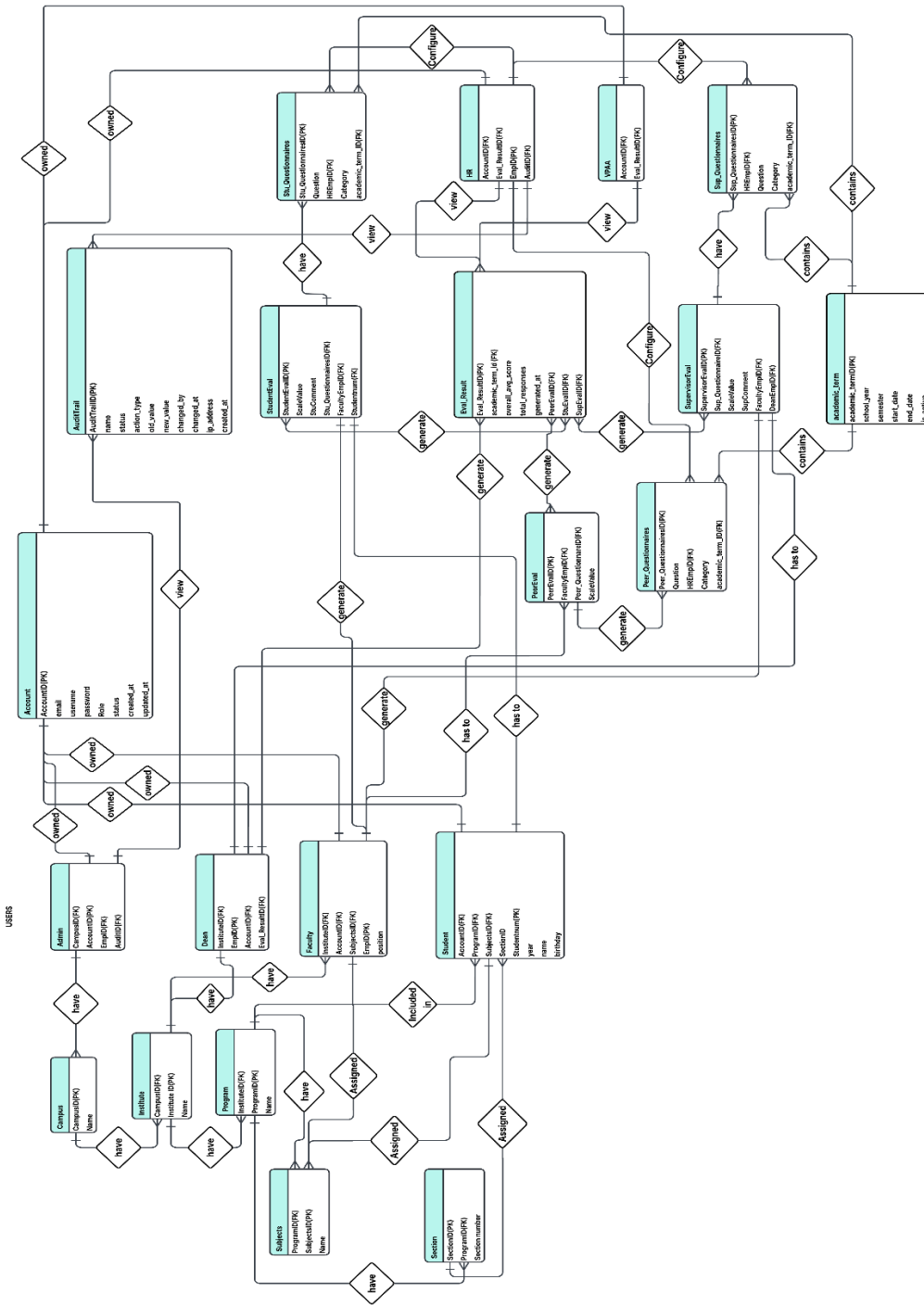


Figure 15. ERD Model of the Web-Based Faculty Evaluation System

Logical Database Design

The logical database design defines how data is structured and managed in the Web-Based Student and Faculty Evaluation System with AI-Enhanced Feedback Analysis and CHED-Compliant Instruments at the National Aviation Academy of the Philippines (NAAP). It translates system requirements into a relational database model composed of interconnected tables that support secure storage, efficient retrieval, and accurate processing of evaluation data.

At the core of the database is the Account table, which stores user credentials such as email, username, password, role, and status. This table enables role-based access control for different users, including students, faculty, deans, HR personnel, VPAA, and administrators. It is linked to role-specific tables such as Student, Faculty, Dean, HR, VPAA, and Admin, allowing the system to manage user-specific information and responsibilities.

The academic structure is defined through tables such as Campus, Institute, Program, Subjects, and Section, which establish hierarchical relationships within the institution. These tables ensure that students and faculty are properly assigned to their respective academic units, enabling accurate mapping of evaluations to subjects, programs, and sections.

Evaluation instruments are stored in dedicated tables, including Stu_Questionnaires, Peer_Questionnaires, and Sup_Questionnaires, which contain CHED-compliant questions, categories, and references to the academic

term. This structure allows the system to dynamically assign appropriate evaluation forms based on user roles and evaluation type.

Evaluation transactions are captured in the StudentEval, PeerEval, and SupervisorEval tables. These tables record evaluation responses submitted by students, faculty (peer evaluation), and supervisors/deans, respectively. Each record includes scale values and qualitative comments, along with references to evaluators, evaluated faculty members, and corresponding questionnaires. This ensures traceability and integrity of evaluation data.

Processed evaluation results are consolidated in the Eval_Result table, which stores computed values such as overall average scores, total responses, and generated timestamps. This table integrates data from multiple evaluation sources and serves as the basis for reporting and performance analysis.

The academic_term table defines the time frame for evaluations, including school year, semester, start and end dates, and active status. All evaluation-related tables are linked to this entity to ensure that data is organized and retrieved based on the correct academic period.

Additional tables such as HR and VPAA are linked to evaluation results and filtered outputs, enabling access to reports and analytical insights for monitoring, evaluation, and institutional planning. The Admin table supports system administration by managing user accounts and system configuration.

To maintain data consistency and efficiency, the database design follows normalization principles, reducing redundancy and ensuring logical separation of data. Relationships between tables are enforced through primary and foreign key constraints, guaranteeing referential integrity and preventing invalid or duplicate records.

Overall, this logical database design supports accurate evaluation processing, efficient reporting, AI-assisted feedback analysis, and reliable monitoring of faculty performance. It provides a scalable and maintainable foundation for the system, ensuring long-term usability and alignment with institutional and regulatory requirements.

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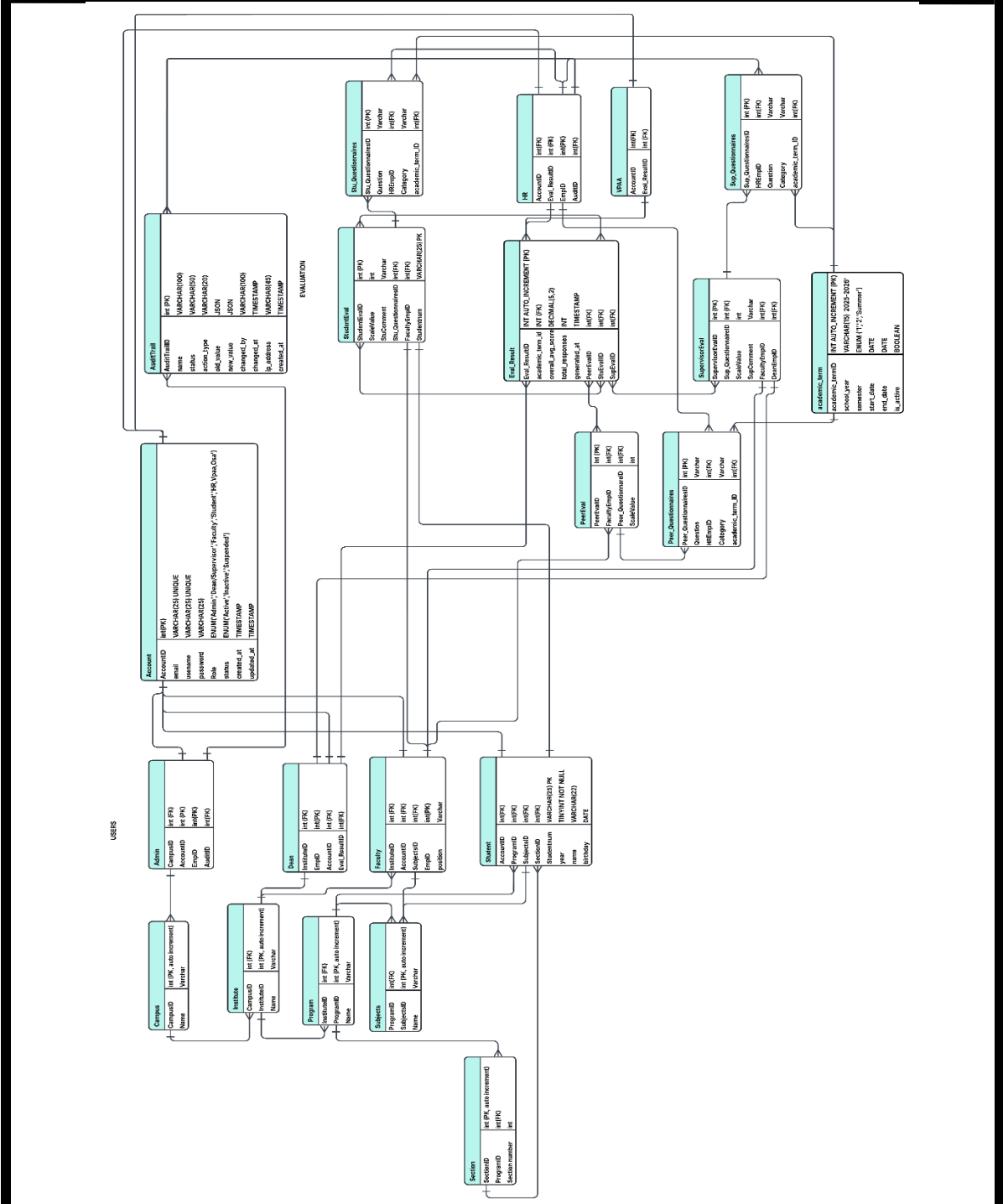


Figure 16. Logical Database Design Model of the Web-Based Faculty Evaluation System

System Security and Data Privacy Design

The protection of evaluation data and the personal information of all system users is a foundational design principle of the proposed system. The security architecture is composed of multiple independent layers: role-based access control enforced at the application layer, Bcrypt password hashing, risk-adaptive OTP authentication via institutional email, HTTPS data transmission with TLS, Cloudflare perimeter firewall protection, and scheduled encrypted database backups. Each layer addresses a distinct class of threat, and the combination ensures that no single point of failure can compromise the confidentiality, integrity, or availability of evaluation data.

Access Control and Authentication Mechanisms

Access to the system is governed by a role-based access control (RBAC) framework that assigns each registered user to one of seven defined roles: Student, Faculty, Supervisor/Dean, HR Officer, OSA Officer, VPAA, or System Administrator. The system determines the authenticated user's role at login and constructs the session context accordingly, restricting access to only the modules, data records, and operations that the assigned role is authorized to perform.

Authentication employs a risk-adaptive OTP mechanism. On first-time login, the system issues a one-time password to the user's registered institutional email

address for identity verification — since the email account was provisioned by the administrator, successful OTP verification confirms that the user is the legitimate owner of that address. Subsequent logins from recognized devices and sessions do not require OTP entry. However, if the system detects a login attempt from an unrecognized device or identifies anomalous session characteristics, it re-triggers OTP verification to confirm the legitimacy of the access attempt before granting entry. The OTP is time-limited and single-use, ensuring that even a compromised password credential cannot grant system access without simultaneous control of the user's institutional email inbox.

Session management controls are applied throughout all authenticated interactions. Session tokens are invalidated upon logout, upon expiry of the configured session duration, and upon detection of concurrent login attempts from different IP addresses. All session-related events — including successful logins, failed login attempts, OTP generation, and session terminations — are captured in the audit trail with the user identifier, timestamp, and IP address of the access attempt, creating a comprehensive record that supports security review and incident investigation.

Data Encryption and Protection Measures

User passwords are the most security-sensitive credentials stored in the system database. The system uses Bcrypt, an adaptive cryptographic hashing algorithm, to store all user passwords. Bcrypt is specifically designed for password storage: it is computationally slow by design, making brute-force and dictionary attacks significantly more costly for an attacker even if the database is compromised. Each password hash is generated with a unique per-user salt, preventing precomputed rainbow table attacks and ensuring that identical passwords produce different stored hash values for different users. Bcrypt's work factor (cost parameter) can be increased over time as hardware capabilities improve, ensuring that the security of stored password hashes remains robust throughout the system's operational life.

All data transmitted between client devices and the institutional server is protected using HTTPS with current TLS protocols, preventing interception, tampering, or eavesdropping during evaluation submission, dashboard access, and report download operations. At the network perimeter, a Cloudflare web application firewall is positioned in front of the system's public domain, filtering all incoming traffic before it reaches the application server. This configuration provides active defense against distributed denial-of-service attacks, automated credential stuffing attempts, SQL injection, cross-site scripting, and other common

web application threats without requiring changes to the server hardware configuration.

Automated backup procedures run on a scheduled basis, creating encrypted backup archives written to a secondary storage location isolated from the primary database. Backup integrity is verified through scheduled restoration tests. This regime ensures that complete, current evaluation records can be recovered in the event of hardware failure, data corruption, or a successful attack on the primary database server.

Data Privacy and Ethical Considerations

The proposed system is designed in full compliance with Republic Act No. 10173, the Data Privacy Act of 2012, which governs the collection, storage, processing, and sharing of personal information in the Philippines. The system collects only the personal data fields required for evaluation processing and user authentication: name, institutional email address, role assignment, course enrollment, and evaluation responses. No additional personal data is collected beyond what is operationally necessary.

Student evaluation submissions are processed in a manner that protects evaluator anonymity. Evaluation responses are dissociated from student identifiers before being presented to faculty members or included in HR reports. The AI-

assisted summarization process operates on aggregated comment text without attributing individual comments to specific students in the output. Faculty members and deans have no access to individual submission records; they receive only aggregated ratings and thematic summaries.

The soft-delete policy enforced throughout the system serves both data integrity and privacy compliance objectives. Records flagged for deletion are retained in a deactivated state rather than permanently removed, preserving the integrity of historical evaluation data for audit purposes while allowing records to be excluded from active operational views. This approach ensures that evaluation history remains available for regulatory review without exposing deactivated personal data in routine system operations.

Should a data subject formally exercise their right to erasure or correction under RA 10173, the system administrator can flag the relevant records for anonymization or update. The audit trail records all data subject requests and the administrative actions taken in response, providing a documented compliance record.

Audit Logging and Activity Monitoring

The system maintains a comprehensive, non-editable audit trail that captures all significant actions performed within the platform. Logged events

include user authentication events, evaluation form submissions and draft saves, HR administrative actions (evaluation period creation, questionnaire configuration, report generation), AI summary generation and edits, administrative user management actions, and all data modification events across all modules. Each audit record includes the user identifier, the action performed, the affected data record or module, the timestamp, and the source IP address.

The audit trail is accessible to system administrators through a dedicated administrative view that supports filtering by user, action type, module, and date range. The non-editable property of the audit trail is enforced at the database layer: audit records are inserted through a write-only procedure and the database role assigned to the application layer does not include update or delete permissions on the audit table. This design ensures that the audit trail cannot be altered by any application-level action, including actions performed by users with the System Administrator role.

At the system level, activity monitoring tools provide administrators with real-time and historical visibility into server resource utilization, application error rates, backup job completion status, and response time metrics. Automated alerts notify administrators when monitored metrics exceed configured thresholds, enabling proactive identification and resolution of performance anomalies or security incidents before they escalate into service disruptions.

Implementation Environment and Technology Stack

This section describes the software environment, hardware infrastructure, development toolset, and deployment configuration that together constitute the implementation environment for the proposed system.

Software Requirements

The system requires a web server software environment capable of hosting a server-side web application framework, a relational database management system, and supporting services for email delivery and external API communication. The server must be configured to handle HTTPS requests routed through the Cloudflare proxy layer, support environment variable management for API credentials, and execute scheduled background tasks for backup procedures and email notifications. External dependencies include the OpenAI GPT-4.1 mini API for AI-assisted feedback analysis and an SMTP service for automated email delivery, including OTP distribution.

A relational database management system (RDBMS) is required to store and manage all evaluation data, user account records, questionnaire configurations, audit log entries, and session information. The database must support ACID-compliant transaction processing to ensure data consistency across concurrent evaluation submissions, and must be configured with access controls

that restrict database connections to authenticated application-layer processes only.

The system must be accessible through standard modern web browsers on all major operating systems without requiring the installation of additional client-side software or browser extensions. This requirement ensures that students accessing the system from personal devices, university computer laboratories, or mobile devices can interact with the evaluation platform using whatever browser environment is available to them. Client-side functionality is implemented using standard HTML5, CSS3, and JavaScript, with responsive design principles applied to ensure usability across screen sizes ranging from desktop monitors to smartphone displays.

Hardware Requirements

The system is designed for deployment on NAAP's institutional server infrastructure, which must meet minimum 2025-standard hardware specifications to ensure reliable performance during peak concurrent evaluation windows. The hardware recommendations below reflect current market availability and are aligned with the system's operational demands: multi-role concurrent access, real-time dashboard queries, AI API call processing, PDF report rendering, and scheduled database backup execution.

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Server Hardware Requirements

Table 3.55. Recommended Server Hardware Specifications — 2025

Standards

Component	Minimum Specification	Recommended Specification
Processor (CPU)	Intel Xeon E-2300 Series (8-core) or AMD EPYC 7003 Series (8-core), 3.0 GHz base clock	Intel Xeon Gold 6300 Series or AMD EPYC 9004 Series, 12–16 cores, 3.5 GHz+, with hardware virtualization support
RAM	16 GB DDR4 ECC RAM (Error-Correcting Code)	32 GB DDR5 ECC RAM for improved concurrent session handling and query caching
Primary Storage	512 GB NVMe SSD (M.2 or U.2 form factor, PCIe 4.0)	1 TB NVMe SSD RAID-1 array for redundancy; <100µs read latency for database queries
Backup / Secondary Storage	1 TB SATA SSD or HDD, physically or logically isolated from primary	2 TB NVMe or enterprise SATA SSD on isolated volume, with offsite or cloud backup target
Network Interface	1 Gbps Ethernet NIC	Dual 1 Gbps or single 10 Gbps Ethernet NIC with failover support
Operating System	64-bit Linux Server (Ubuntu 22.04 LTS / Debian 12 or equivalent)	Ubuntu 24.04 LTS with automatic security patch management enabled
Power Supply	Redundant 500W PSU (80+ Gold certified)	Redundant 750W PSU (80+ Platinum), UPS-backed for protection against power interruptions

The recommended server configuration — a 12- to 16-core processor, 32 GB of DDR5 ECC RAM, and NVMe SSD storage — is sufficient to handle peak

concurrent load during evaluation windows when large numbers of students, faculty, and supervisors may simultaneously access the system. ECC RAM is specified to protect against memory errors that could corrupt database operations or session data during high-load periods. NVMe SSD storage is specified to ensure sub-millisecond I/O response times for database queries, which directly impacts the perceived responsiveness of the evaluation submission and dashboard access functions.

Storage must be provisioned with sufficient headroom for several years of data growth beyond initial deployment. Each evaluation period generates records proportional to the number of enrolled students, faculty assignments, and open-ended comments submitted. Backup storage must be physically or logically isolated from the primary database storage volume so that a single hardware failure does not simultaneously destroy both the live database and its backup copies. Administrators should configure automated off-site backup replication to a cloud storage target or a separate physical location for disaster recovery purposes.

Client-Side Hardware Requirements

Client-side hardware requirements are intentionally minimal by design. Any device capable of running a contemporary web browser — including desktop computers, laptop computers, tablet computers, and smartphones released within the last five years — is sufficient for accessing the evaluation system. This broad

device compatibility ensures that all students and faculty can fulfill their evaluation obligations from any location and any personal device without requiring institutional hardware provisioning or software installation. The responsive interface design ensures full functionality on screen resolutions as low as 375px width (standard smartphone portrait mode).

Table 3.56. Minimum Client-Side Hardware Requirements

Device Type	Minimum Requirement
Desktop / Laptop Computer	Any x86-64 system with 4 GB RAM, running Windows 10/11, macOS 12+, or Ubuntu 20.04+, with a modern browser (Chrome 110+, Firefox 115+, Edge 110+, or Safari 16+)
Tablet Computer	Any ARM or x86-64 tablet with 3 GB RAM running iOS 15+, iPadOS 15+, or Android 12+, with a current browser
Smartphone	Any smartphone with 2 GB RAM running iOS 15+ or Android 10+, with a current browser; 375px minimum screen width supported
Network Connectivity	Stable internet connection of at least 1 Mbps for evaluation submission; 5 Mbps recommended for dashboard and PDF report access

Development Tools and Programming Technologies

The system is developed using a modern server-side web application framework that supports structured application routing, server-side form processing, session management, database interaction, and integration with external APIs. The framework's component-based architecture supports the modular system design, allowing individual modules — such as the evaluation

submission engine, the AI analysis module, the reporting generator, and the administrative management interface — to be developed, tested, and updated independently.

Database development and management is performed using standard relational database tools supporting schema design, migration management, query optimization, and performance monitoring. The database schema is normalized to the Third Normal Form (3NF) to eliminate data redundancy and ensure efficient, consistent data retrieval across all system modules. Primary and foreign key constraints are defined throughout the schema to enforce referential integrity and prevent invalid data relationships from forming during the system's operational life.

The AI-assisted feedback analysis feature is implemented through integration with the OpenAI GPT-4.1 mini API. When an authorized user initiates comment summarization for a faculty member's evaluation period, the system retrieves all submitted open-ended comments, structures them into an appropriately formatted API request with a system-defined analytical prompt, and returns a concise thematic summary organized around commendations, constructive feedback, and recurring concerns. This API-based summarization is complemented by a locally executed frequency analysis module that independently tokenizes submitted comments, removes common stop words, and computes term frequency distributions to identify the most commonly mentioned

words and phrases. The frequency analysis results serve as an independent validation layer for the AI-generated summary and are displayed alongside it to assist HR in assessing summary completeness.

Security implementation tools include the Bcrypt library for cryptographic password hashing with per-user salt generation, an OTP generation library for time-limited single-use authentication codes and expiry management, and HTTPS/TLS configuration at the web server level. The Cloudflare firewall is configured at the DNS level by routing the system's public domain through Cloudflare's proxy infrastructure, requiring no modifications to the server hardware. PDF report generation is handled by a server-side rendering library capable of producing documents conforming to the layout requirements specified by CHED CMO No. 19, Series of 2025. Version control, sprint task management, and collaborative development are supported through standard Agile development tooling.

Deployment Environment

The system is deployed on NAAP's institutional server, configured as a web server environment that handles incoming HTTPS requests and routes them to the appropriate application components. The Cloudflare firewall sits at the DNS level in front of the institutional domain, intercepting and filtering all traffic before it is

forwarded to the server. This configuration is activated by pointing the system's DNS records through Cloudflare's proxy infrastructure, and it requires no modifications to the physical server hardware or operating system configuration.

The application and database layers may be co-located on a single server or separated across two servers within the institutional network, depending on available infrastructure at the time of deployment. If co-located, database access is restricted to authenticated local connections from the application layer only — the database port is not exposed to external network traffic. Database queries are executed through the application's data access layer, which enforces parameterized query construction throughout, preventing SQL injection vulnerabilities from arising at the data access boundary.

Automated backup jobs are scheduled to execute at defined intervals, writing encrypted backup archives to a secondary storage location. Administrators receive automated alerts if a backup job fails to complete successfully. System monitoring tools track server resource utilization, application error rates, and response times, alerting administrators when metrics exceed defined thresholds. These monitoring capabilities allow performance issues or security anomalies to be identified and investigated before they escalate into service interruptions.

The modular application architecture ensures that individual components can be updated, patched, or replaced without requiring a full system redeployment.

Updates to the AI analysis module, for instance, can be deployed and tested independently of the evaluation submission engine or the administrative management module. This design approach reduces the operational risk associated with system maintenance and ensures that the platform can be kept current with evolving security requirements and institutional needs throughout its operational lifecycle.

Summary of Technology Stack

Table 3.57. Summary of Technology Stack — AI-Enhanced CHED-Compliant Evaluation System

Layer	Component	Technology / Approach
Network Security	Web Application Firewall	Cloudflare (Cloud-Based WAF and DDoS Protection)
Transport	Data Transmission Security	HTTPS with TLS Encryption
Application	Server-Side Framework	Modern Web Application Framework
Application	Client-Side Interface	HTML5, CSS3, JavaScript (Responsive Design)
AI and Analytics	Feedback Summarization	OpenAI GPT-4.1 mini API
AI and Analytics	Frequency Analysis Module	Local Text Processing and Term Frequency Analysis
AI and Analytics	Evaluation Behavior Analysis	Rule-Based Behavioral Pattern Scoring (Time, Uniformity, Repetition Detection)
AI and Analytics	AI-Based Bias Detection	NLP Comment Classification Engine (Constructive / Neutral / Biased Labels)
AI and Analytics	Cross-Source Validation	Multi-Evaluator Consistency Checker (Student, Peer, Supervisor Comparison)

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Layer	Component	Technology / Approach
AI and Analytics	Credibility Score Engine	Composite Evaluation Reliability Scoring (0–100 Scale, Three Reliability Bands)
AI and Analytics	Smart Recommendation Engine	AI-Assisted Faculty Development Recommender (Weak-Area Detection and Targeted Suggestions)
AI and Analytics	Longitudinal Performance Tracker	Historical Evaluation Trend Analysis (Multi-Semester Improvement and Decline Detection)
AI and Analytics	AI Transparency Module	Keyword Cluster and Reasoning Disclosure Layer (Displayed Alongside Every AI Summary)
Data Management	Database System	Relational DBMS — ACID-Compliant with 3NF Schema
Data Protection	Password Storage	Bcrypt One-Way Hashing with Per-User Salt
Authentication	Multi-Factor Authentication	Email-Based One-Time Password (OTP)
Access Control	Authorization Model	Role-Based Access Control (RBAC) — 7 Roles
Reporting	Evaluation Report Generation	CHED-Compliant PDF Rendering Engine
Communication	Transactional Email Service	SMTP-backed delivery for OTP and password-reset email workflows.
Resilience	Data Backup and Recovery	Scheduled Encrypted Backup with Restoration Testing

The technology decisions summarized above reflect a consistent set of design principles: preference for proven, widely supported technologies that minimize implementation risk; use of open standards and frameworks that reduce long-term maintenance dependency on proprietary vendors; and deliberate layering of security controls so that no single point of failure can expose sensitive evaluation data. Each component was selected with the specific operational

context of NAAP in mind, balancing analytical capability, institutional IT capacity, regulatory compliance requirements, and the scalability needed to accommodate growth in users, campuses, and evaluation data volume throughout the system's intended operational life. The adoption of Bcrypt for password hashing reflects current best practices in credential security, offering adaptive work factors that can be tuned to maintain resistance to brute-force attacks as computing hardware capabilities improve over time.

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